

GRADE 5 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)

71 question sets • 1930 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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Evaluate expressions that use parentheses

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting multi-digit numbers	4.NBT.B.4
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FORWARD 2	Evaluate expressions with negative numbers	7.NS.A.3

SET 1 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q1/5

Evaluate.

$$(35 - 11) \div 4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q2/5

Evaluate.

$$50 - (6 \times 7)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q3/5

Evaluate.

$$3 + (24 \div 6) \times 5$$

- A. 35
- B. 135
- C. 27
- D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q4/5

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

In $4 \times (2 + 5)$, you add $2 + 5$ first.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q1/5

Evaluate.

$$(6 + 4) \times 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q2/5

Which step comes first in $18 - (3 \times 4)$?

- A.** Subtract $18 - 4$
- B.** Multiply 3×4
- C.** Subtract $18 - 3$
- D.** Add $3 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Multiply 3×4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q3/5

Evaluate.

$$20 - (5 + 5)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q4/5

True or false?

$$(10 - 4) + 3 = 10 - 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$7 \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q1/5

21

Divide.

$$24 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q2/5

True or false?

$$9 \times 7 = 56$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q3/5

$$48 \div 8 = ?$$

- A.** 8
- B.** 5
- C.** 7
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q4/5

Multiply.

$$3 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 5.OA.A.1

BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$348 + 275$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

623

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q1/5

27

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$502 - 168$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

334

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q2/5

28

True or false?

$$1,250 - 375 = 985$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q3/5

$$609 + 194 = ?$$

- A.** 793
- B.** 703
- C.** 813
- D.** 803

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 803

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$1,000 - 432$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

568

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 1 • BACKWARD 3 • Q5/5

31

SET 1 • 5.OA.A.1

FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4(x + 3)$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q1/5

Evaluate $20 - 2y$ when $y = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q2/5

Evaluate $(a + b) \div 3$ when $a = 10$ and $b = 8$.

- A.** 4
- B.** 9
- C.** 54
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q3/5

Evaluate $3m + 2$ when $m = 7$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Short answer

CORRECT ANSWER

23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q4/5

True or false?

When $n = 4$, the value of $5(n - 1)$ is 19.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 1 • FORWARD 1 • Q5/5

SET 1 • 5.OA.A.1

FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q1/5

Find the value.

$$(-12 + 4) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q2/5

Find the value.

$$6 - (2 - 10)$$

- A.** -2
- B.** 2
- C.** 14
- D.** 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q3/5

Find the value.

$$(-5) \times (-2) + (-7)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q4/5

True or false?

$$(-20) \div (3 + 2) = 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 1 • FORWARD 2 • Q5/5

Evaluate expressions with brackets and braces

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting within 100	2.NBT.B.5
FORWARD 1	Evaluate expressions with whole-number exponents	6.EE.A.1
FORWARD 2	Grouping symbols with negative numbers	7.NS.A.3

SET 2 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Short answer

CORRECT ANSWER

34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q1/5

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q2/5

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

- A. 36
- B. 20
- C. 44
- D. 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q3/5

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q4/5

Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q5/5

50

QUESTION ID

S2-M-Q5

TYPE

Short answer

CORRECT ANSWER

17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q1/5

True or false?

In $(8 + 2) \div 5$, you divide $2 \div 5$ first.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q2/5

Evaluate.

$$4 \times (9 - 6)$$

- A.** 33
- B.** 10
- C.** 30
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q3/5

Evaluate.

$$30 \div (2 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q4/5

True or false?

$$(3 + 5) \times 2 = 16$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$8 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q1/5

Divide.

$$56 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q2/5

True or false?

$$36 \div 4 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q3/5

$$7 \times 9 = ?$$

- A.** 63
- B.** 56
- C.** 72
- D.** 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q4/5

Divide.

$$81 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 5.OA.A.1

BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$46 + 37$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q1/5

64

QUESTION ID

S2-B3-Q1

TYPE

Short answer

CORRECT ANSWER

83

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$72 - 28$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q2/5

65

QUESTION ID

S2-B3-Q2

TYPE

Short answer

CORRECT ANSWER

44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q2/5

65

True or false?

$$55 + 29 = 74$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q3/5

66

QUESTION ID

S2-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q3/5

$$90 - 47 = ?$$

- A.** 47
- B.** 43
- C.** 53
- D.** 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q4/5

67

QUESTION ID

S2-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$38 + 54$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q5/5

68

QUESTION ID

S2-B3-Q5

TYPE

Short answer

CORRECT ANSWER

92

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 2 • BACKWARD 3 • Q5/5

68

SET 2 • 5.OA.A.1

FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q1/5

70

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q1/5

Evaluate.

$$[10 - 2^3] \times 5$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q2/5

71

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q2/5

Evaluate.

$$4^2 - (6 + 2)$$

- A.** 8
- B.** 2
- C.** 0
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q3/5

72

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q3/5

Evaluate.

$$(5^2 - 1) \div 6$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q4/5

73

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q4/5

True or false?

$$3 \times 2^2 = 36$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q5/5

74

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 5.OA.A.1

FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q1/5

76

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q1/5

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q2/5

77

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q2/5

Find the value.

$$-4 \times [5 - (-1)]$$

- A.** 16
- B.** -16
- C.** -24
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q3/5

78

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. -24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q3/5

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q4/5

79

QUESTION ID

S2-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q4/5

True or false?

$$[-8 - (-2)] \times 3 = -30$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q5/5

80

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 2 • FORWARD 2 • Q5/5

Place grouping symbols to make an equation true

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	When regrouping keeps the same value	3.OA.B.5
BACKWARD 3	Multiplication and division facts	3.OA.C.7
FORWARD 1	Use parentheses to write equivalent expressions	6.EE.A.3
FORWARD 2	Expand and factor expressions with rational coefficients	7.EE.A.1

SET 3 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q1/5

83

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

$$(4 + 6) \times 3 = 30$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q1/5

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q2/5

84

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

$$20 - (8 - 2) = 14$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q2/5

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

A. $2 \times (5 + 3) \times 4 = 52$ **B.** $(2 \times 5 + 3) \times 4 = 52$

C. $2 \times (5 + 3 \times 4) = 52$ **D.** $2 \times 5 + (3 \times 4) = 52$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q3/5

85

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$(2 \times 5 + 3) \times 4 = 52$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q3/5

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q4/5

86

QUESTION ID

S3-M-Q4

TYPE

Short answer

CORRECT ANSWER

$$36 \div (6 - 2) = 9$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q4/5

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q5/5

87

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

$$2 \times [9 - (3 + 1)] \\ = 10$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 3 • MAIN • Q5/5

SET 3 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q1/5

89

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q1/5

Evaluate.

$$4 + (6 \times 3)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q2/5

90

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q2/5

True or false?

$$(20 - 8) - 2 = 20 - (8 - 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q3/5

91

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q3/5

Evaluate.

$$36 \div (6 - 2)$$

- A. 4
- B. 34
- C. 9
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q4/5

92

QUESTION ID

S3-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q4/5

Evaluate.

$$(36 \div 6) - 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q5/5

93

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 5.OA.A.1

BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

95

QUESTION ID

S3-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q1/5

True or false?

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

96

QUESTION ID

S3-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q2/5

Which equation is true?

- A. $(5 + 2) + 3 = 5 + (2 + 3)$
- B. $(9 - 4) - 1 = 9 - (4 - 1)$
- C. $(16 \div 4) \div 2 = 16 \div (4 \div 2)$
- D. $(7 - 3) - 2 = 7 - (3 - 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5

97

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

$(5 + 2) + 3 = 5 + (2 + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

98

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q4/5

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

99

QUESTION ID

S3-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 5.OA.A.1

BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q1/5

101

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q1/5

Divide.

$$42 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q2/5

102

QUESTION ID

S3-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q2/5

True or false?

$$8 \times 7 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q3/5

103

QUESTION ID

S3-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q3/5

$$64 \div 8 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q4/5

104

QUESTION ID

S3-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q4/5

Multiply.

$$6 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q5/5

105

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • 5.OA.A.1

FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q1/5

107

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q1/5

Expand.

$$4(2y - 1)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q2/5

108

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$8y - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q2/5

Which expression is equivalent to $6(a + 2)$?

- A. $6a + 2$
- B. $a + 12$
- C. $6a + 12$
- D. $8a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q3/5

109

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $6a + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q3/5

Factor.

$$5m + 20$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q4/5

110

QUESTION ID

S3-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5(m + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q4/5

True or false?

$$2(x + 7) = 2x + 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q5/5

111

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 5.OA.A.1

FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •
Expressions & Equations • 7.EE.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Expand.

$$-2(3x - 4)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q1/5

113

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$-6x + 8$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q1/5

113

Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q2/5

114

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3y + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q2/5

Which expression is equivalent to $-5(n + 2)$?

- A. $-5n + 2$
- B. $-5n - 10$
- C. $-5n + 10$
- D. $5n - 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q3/5

115

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-5n - 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q3/5

Factor out 3.

$$3x - 12$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q4/5

116

QUESTION ID

S3-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$3(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q4/5

True or false?

$$0.5(4a - 8) = 2a - 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q5/5

117

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 3 • FORWARD 2 • Q5/5

Write numerical expressions from words

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Quotients as equal shares	3.OA.A.2
BACKWARD 3	Sums and differences	2.OA.A.1
FORWARD 1	Write expressions with variables	6.EE.A.2.a
FORWARD 2	Write and solve equations from words	7.EE.B.4.a

SET 4 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q1/5

120

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

$(9 + 6) \times 4$

GRADING NOTE $4 \times (9 + 6)$ is also correct. Parentheses are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q1/5

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q2/5

121

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(30 - 5) \div 5$

GRADING NOTE Parentheses are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q2/5

Which expression shows "triple the sum of 12 and 8"?

- A. $12 + 8 \times 3$
- B. $3 \times 12 + 8$
- C. $3 \times (12 + 8)$
- D. $(3 + 12) \times 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q3/5

122

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \times (12 + 8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q3/5

Write an expression for "divide 40 by 8, then add 7."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q4/5

123

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

$(40 \div 8) + 7$

GRADING NOTE $40 \div 8 + 7$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q4/5

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q5/5

124

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

$6 \times (15 - 9)$

GRADING NOTE $(15 - 9) \times 6$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 5.OA.A.2

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

True or false?

5×7 can mean 5 groups of 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q1/5

126

QUESTION ID

S4-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q1/5

Which expression shows 4 groups of 6?

- A. 46
- B. $6 - 4$
- C. $4 + 6$
- D. 4×6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q2/5

127

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 4×6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q2/5

Write a multiplication expression for 8 groups of 3.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q3/5

128

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

8 × 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q3/5

True or false?

$$3 + 3 + 3 + 3 = 3 \times 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q4/5

129

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q4/5

What is the product of 7 and 5?

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q5/5

130

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 5.OA.A.2

BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

True or false?

15 ÷ 3 can mean 15 shared equally into 3 groups.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q1/5

132

QUESTION ID

S4-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q1/5

Which expression shows 20 apples shared equally among 4 bags?

- A. 20×4
- B. $4 \div 20$
- C. $20 \div 4$
- D. $20 - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q2/5

133

QUESTION ID

S4-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $20 \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q2/5

What is the quotient of 18 and 6?

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q3/5

134

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q3/5

Write a division expression for 32 split into groups of 8.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q4/5

135

QUESTION ID

S4-B2-Q4

TYPE

Short answer

CORRECT ANSWER

32 ÷ 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q4/5

True or false?

The quotient of 12 and 2 is 24.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q5/5

136

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 5.OA.A.2

BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

What is the sum of 14 and 9?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q1/5

138

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q1/5

What is the difference of 20 and 7?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q2/5

139

QUESTION ID

S4-B3-Q2

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q2/5

True or false?

The difference of 15 and 6 is 21.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q3/5

140

QUESTION ID

S4-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q3/5

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q4/5

141

QUESTION ID

S4-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q4/5

Write an expression for "7 more than 25."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q5/5

142

QUESTION ID

S4-B3-Q5

TYPE

Short answer

CORRECT ANSWER

25 + 7

GRADING NOTE 7 + 25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • 5.OA.A.2

FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for "the sum of n and 8, times 3."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q1/5

144

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3(n + 8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q1/5

Write an expression for "y divided by 5, then minus 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q2/5

145

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$y \div 5 - 2$

GRADING NOTE $y/5 - 2$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q2/5

Which expression means "4 less than the product of 2 and x "?

- A. $4 - 2x$
- B. $2x - 4$
- C. $2(x - 4)$
- D. $2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q3/5

146

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $2x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q3/5

Write an expression for "twice the difference of m and 6."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q4/5

147

QUESTION ID

S4-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$2(m - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q4/5

True or false?

"9 more than k " can be written as $k + 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q5/5

148

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 4 • FORWARD 1 • Q5/5

SET 4 • 5.OA.A.2

FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Three times the sum of a number and 4 is 27.
Write and solve an equation to find the number.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q1/5

150

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3(n + 4) = 27; n = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q1/5

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q2/5

151

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$$(1/2)n - 6 = 10;$$
$$n = 32$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q2/5

Which equation represents "2 less than 5 times a number is 33"?

- A. $5n + 2 = 33$
- B. $5n - 2 = 33$
- C. $2 - 5n = 33$
- D. $5(n - 2) = 33$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q3/5

152

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5n - 2 = 33$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q3/5

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q4/5

153

QUESTION ID

S4-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$n \div 4 + 3 = 9; n = 24$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q4/5

True or false?

The solution of $2(x + 3) = 18$ is $x = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q5/5

154

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 4 • FORWARD 2 • Q5/5

Interpret numerical expressions without evaluating them

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Groups of a sum	3.OA.A.1
BACKWARD 3	Equal shares	3.OA.A.2
FORWARD 1	Identify parts of an expression	6.EE.A.2.b
FORWARD 2	Rewrite expressions to show how quantities are related	7.EE.A.2

SET 5 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does $4 \times (563 + 218)$ compare to $563 + 218$?

- A. It is 4 more.
- B. It is 4 times as large.
- C. It is 4 less.
- D. It is one-fourth as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q1/5

157

QUESTION ID

S5-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. It is 4 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q1/5

True or false?

$(7,215 - 1,005) \div 5$ is one-fifth as large as $7,215 - 1,005$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q2/5

158

QUESTION ID

S5-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q2/5

Without calculating, how many times as large is
 $9 \times (48 + 27)$ as $48 + 27$?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q3/5

159

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q3/5

Which expression is twice as large as 356×12 ?

- A. $(356 + 2) \times 12$
- B. $356 \times 12 + 2$
- C. 356×14
- D. $2 \times (356 \times 12)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q4/5

160

QUESTION ID

S5-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $2 \times (356 \times 12)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q4/5

True or false?

$6 \times 25 + 4$ is 6 times as large as $25 + 4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q5/5

161

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 5 • MAIN • Q5/5

SET 5 • 5.OA.A.2

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

35 is 5 times as many as 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

163

QUESTION ID

S5-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q1/5

Fill in the blank.

48 is ____ times as many as 6.

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

164

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q2/5

Which equation shows "36 is 4 times as many as 9"?

A. $9 = 4 \times 36$

B. $36 = 4 + 9$

C. $36 - 9 = 4$

D. $36 = 4 \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

165

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $36 = 4 \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

30 is 3 times as many as 27.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

166

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q4/5

Tom has 6 cards. Ana has 3 times as many cards.

How many cards does Ana have?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

167

QUESTION ID

S5-B1-Q5

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 5.OA.A.2

BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 1)$ means 3 groups of $(4 + 1)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q1/5

169

QUESTION ID

S5-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q1/5

Which expression means 6 groups of $(10 + 2)$?

- A. $6 \times (10 + 2)$
- B. $(6 + 10) \times 2$
- C. $6 \times 10 + 2$
- D. $6 + (10 + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q2/5

170

QUESTION ID

S5-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $6 \times (10 + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q2/5

How many groups of $(5 + 4)$ are in $7 \times (5 + 4)$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q3/5

171

QUESTION ID

S5-B2-Q3

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$$2 \times (8 + 1) = (8 + 1) + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q4/5

172

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q4/5

Write 4 groups of 9 as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q5/5

173

QUESTION ID

S5-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 × 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 5.OA.A.2

BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

True or false?

12 shared equally among 3 people is 4 each.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q1/5

175

QUESTION ID

S5-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q1/5

A 20-inch ribbon is cut into 5 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q2/5

176

QUESTION ID

S5-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q2/5

Which expression finds one of 4 equal shares of 28?

- A. $28 - 4$
- B. $28 \div 4$
- C. 28×4
- D. $4 \div 28$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q3/5

177

QUESTION ID

S5-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $28 \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q3/5

True or false?

Dividing a number by 2 splits it into 2 equal shares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q4/5

178

QUESTION ID

S5-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q4/5

30 crayons are shared equally among 6 children.

How many crayons does each child get?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q5/5

179

QUESTION ID

S5-B3-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 5.OA.A.2

FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •
6.EE.A.2.b

In $5(x + 2)$, what is 5?

- A.** A term of $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. A factor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q1/5

How many terms are in $4a + 7b - 1$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q2/5

What is the coefficient of m in $9m + 4$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q3/5

True or false?

In $3(y + 6)$, the quantity $(y + 6)$ is a sum of two terms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q4/5

What is the expression $12 \div (n + 1)$ called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. A quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 5 • FORWARD 1 • Q5/5

SET 5 • 5.OA.A.2

FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p is increased by 5%.

Which expression shows the new price?

- A. $0.05p$
- B. $1.05p$
- C. $5p$
- D. $p + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $1.05p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q1/5

True or false?

$$x + 0.3x = 0.4x$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q2/5

Write $n - 0.4n$ as one term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.6n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q3/5

Which expression is equivalent to $2(3t)$?

- A. $5t$
- B. $2t + 3t$
- C. $6t$
- D. $3t + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $6t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q4/5

A sale price is $0.75p$, where p is the original price.

What percent of the original price is taken off?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 5 • FORWARD 2 • Q5/5

Generate two numerical patterns and compare their terms

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Equations relating dependent and independent variables	6.EE.C.9
FORWARD 2	Equations for proportional relationships	7.RP.A.2.c

SET 6 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6.

Pattern B starts at 0 and adds 18.

Write the first 5 terms of each pattern.

Pattern A: _____

Pattern B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

Pattern A: 0, 6, 12, 18, 24; Pattern B: 0, 18, 36, 54, 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q1/5

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

- A.** 8 more **B.** 12 more
C. 4 times as large **D.** 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q2/5

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: _____

Pattern B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Pattern A: 25;
Pattern B: 50**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q3/5

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q4/5

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 6 • MAIN • Q5/5

SET 6 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 6." The pattern starts at 0.

Write the first five terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 6, 12, 18, 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q1/5

True or false?

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q2/5

What is the rule for the pattern?

0, 9, 18, 27, 36

- A.** Add 9
- B.** Add 18
- C.** Multiply by 9
- D.** Add 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Add 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q3/5

The rule is "add 11." The pattern starts at 0.

What is the 4th term?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q4/5

What comes next?

0, 15, 30, 45, ____

Next term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 5.OA.B.3

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q1/5

True or false?

$$6 \times 8 = 42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q2/5

What number times 3 equals 24?

- A.** 8
- B.** 7
- C.** 21
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q3/5

Multiply.

$$5 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q4/5

What number times 6 equals 42?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 5.OA.B.3

FORWARD 1

Equations relating dependent and independent variables • Grade 6 •
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans c in b boxes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q1/5

212

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 6b$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q1/5

In $c = 6b$, which variable is the dependent variable?

- A. b
- B. c
- C. 6
- D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q2/5

213

QUESTION ID

S6-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. c

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q2/5

Write an equation for y in terms of x .

x	1	2	3	4
y	5	10	15	20

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q3/5

214

QUESTION ID

S6-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$y = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q3/5

True or false?

In $y = x + 4$, y is the independent variable.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q4/5

215

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q4/5

A plant grows 2 cm each week.

Write an equation for its growth g after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q5/5

216

QUESTION ID

S6-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$$g = 2w$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 5.OA.B.3

FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost \$0.35 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q1/5

218

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 0.35n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x .

x	4	6	10
y	10	15	25

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q2/5

219

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 2.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q2/5

Which equation represents a proportional relationship?

- A. $y = 1.5$
- B. $y = x + 1.5$
- C. $y = 1.5x + 2$
- D. $y = 1.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q3/5

220

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 1.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q3/5

A train travels 150 miles in 2 hours at a constant speed.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q4/5

221

QUESTION ID

S6-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$d = 75t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$y = \frac{3}{4}x$ represents a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q5/5

222

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 6 • FORWARD 2 • Q5/5

Explain why two patterns have the relationship they do

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Notice features of a pattern that the rule does not state	4.OA.C.5
BACKWARD 2	Explain patterns in the multiplication table	3.OA.D.9
FORWARD 1	Use an equation to describe how two quantities change together	6.EE.C.9
FORWARD 2	Identify the constant of proportionality	7.RP.A.2.b

SET 58 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Each term in Pattern B is twice the matching term in Pattern A.

Explain why.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q1/5

225

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

Both patterns start at 0 and take the same number of steps, and B adds 8, which is 2×4 , at every step, so B has always added twice as much as A.

GRADING NOTE Must connect the rule amounts (8 is twice 4) to the same start (0) and the same number of steps. Only restating "B is double A" or checking a few terms is not a full explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q1/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 15.

Why is each term in Pattern B 5 times the matching term in Pattern A?

- A.** Both start at 0, and B adds 5 times as much as A at every step.
- B.** Pattern B starts 5 higher than Pattern A.
- C.** Pattern B has 5 more terms than Pattern A.
- D.** Each term in B is 12 more than the matching term in A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q2/5

226

QUESTION ID

S58-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. Both start at 0,
and B adds 5 times
as much as A at
every step.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q2/5

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6. Each term in B is 3 times the matching term in A because B adds 3 times as much at every step.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q3/5

227

QUESTION ID

S58-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q3/5

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 1 and adds 4. Each term in Pattern B is twice the matching term in Pattern A.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q4/5

228

QUESTION ID

S58-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q4/5

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 20.

Without listing the terms, how does the 10th term of B compare with the 10th term of A? Explain.

Comparison: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q5/5

229

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

B's 10th term is 4 times A's 10th term (180 and 45).

GRADING NOTE Explanation: both start at 0 and reach the 10th term after the same 9 steps; B adds 20 and A adds 5 each step, and 20 is 4×5 , so B has added 4 times as much. The values 180 and 45 are not required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 58 • MAIN • Q5/5

SET 58 • 5.OA.B.3

BACKWARD 1

Notice features of a pattern that the rule does not state • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

True or false?

The rule is "add 2," starting at 1. Every term is odd.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q1/5

231

QUESTION ID

S58-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q1/5

The rule is "add 4," starting at 0. Which is true of every term?

- A. It is a multiple of 4.
- B. It is odd.
- C. It ends in 4.
- D. It is a multiple of 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q2/5

232

QUESTION ID

S58-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. It is a multiple of 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q2/5

The rule is "add 10," starting at 5.

What do you notice about the ones digit of every term?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q3/5

233

QUESTION ID

S58-B1-Q3

TYPE

Short answer

CORRECT ANSWER

**The ones digit is
always 5.**

GRADING NOTE Accept any statement that every term ends in 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q3/5

True or false?

The rule is "add 5," starting at 0. The terms alternate between ending in 0 and ending in 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q4/5

234

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q4/5

The rule is "add 3," starting at 1: 1, 4, 7, 10, 13.

Which describes the terms?

- A. They alternate odd, even, odd, even.
- B. They are all odd.
- C. They are all even.
- D. They are all multiples of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q5/5

235

QUESTION ID

S58-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. They alternate
odd, even, odd,
even.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 5.OA.B.3

BACKWARD 2

Explain patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

True or false?

Every product in the 4s row of the multiplication table is even.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q1/5

237

QUESTION ID

S58-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q1/5

Why is each product in the 6s row twice the product in the 3s row, in the same column?

- A. 6 is 2×3 , so $6 \times n$ is $2 \times (3 \times n)$.
- B. 6 is 3 more than 3.
- C. The 6s row has twice as many products.
- D. Products in later rows are always even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q2/5

238

QUESTION ID

S58-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. 6 is 2×3 , so
 $6 \times n$ is $2 \times (3 \times n)$.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q2/5

True or false?

Every product in the 5s row ends in 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q3/5

239

QUESTION ID

S58-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q3/5

The 2s row is 2, 4, 6, 8, 10, ...

What is the same about every product?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q4/5

240

QUESTION ID

S58-B2-Q4

TYPE

Short answer

CORRECT ANSWER

They are all even.

GRADING NOTE Accept "even", "multiples of 2", or "end in 0, 2, 4, 6 or 8".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q4/5

True or false?

In the 9s row, 9×4 is 9 more than 9×3 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q5/5

241

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 5.OA.B.3

FORWARD 1

Use an equation to describe how two quantities change together • Grade 6 •
Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x .

Explain how the table shows it.

x	1	2	3	4
y	7	14	21	28

Equation: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q1/5

243

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$y = 7x$$

GRADING NOTE Explanation: each y -value is 7 times its x -value (or y increases by 7 when x increases by 1, starting from 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q1/5

Each time x increases by 1, y increases by 5, and y is 0 when x is 0.

Which equation fits?

A. $y = 5x$

B. $y = 5$

C. $y = x + 5$

D. $x = 5y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q2/5

244

QUESTION ID

S58-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q2/5

True or false?

In $y = x + 3$, each y-value is 3 times its x-value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q3/5

245

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q3/5

A gym charges \$9 for each class.

Write an equation for the cost c of n classes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q4/5

246

QUESTION ID

S58-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$c = 9n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q4/5

True or false?

For $y = 4x$, when x doubles, y also doubles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q5/5

247

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 5.OA.B.3

FORWARD 2

Identify the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality?

x	2	5	8
y	6	15	24

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q1/5

249

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q1/5

What is the constant of proportionality in $y = 0.6x$?

- A. 0.6
- B. 0
- C. 6
- D. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q2/5

250

QUESTION ID

S58-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q2/5

True or false?

In the table x: 4, 6, 10 and y: 2, 3, 5, the constant of proportionality is 2.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q3/5

251

QUESTION ID

S58-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q3/5

A car travels 50 miles on every 2 gallons of gas.

What is the constant of proportionality, in miles per gallon?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q4/5

252

QUESTION ID

S58-F2-Q4

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q4/5

True or false?

A proportional relationship graphs as a line through $(0, 0)$ and $(1, 7)$. Its constant of proportionality is 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q5/5

253

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 58 • FORWARD 2 • Q5/5

Form ordered pairs from two patterns and graph them

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 3	Read a scaled bar graph	3.MD.B.3
BACKWARD 4	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Plot pairs of values from a ratio table	6.RP.A.3.a
FORWARD 2	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a

SET 7 • 5.OA.B.3

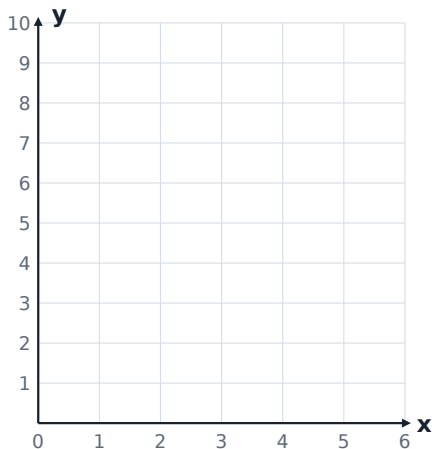
MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q1/5

256

QUESTION ID

S7-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (0, 0), (1, 2), (2, 4), (3, 6), (4, 8)

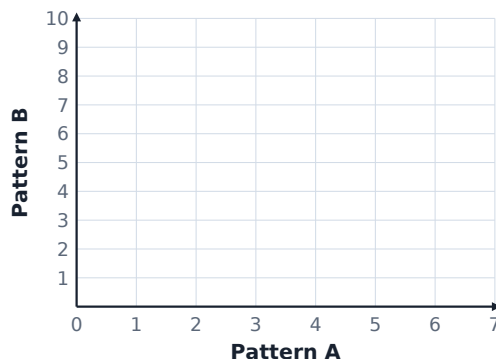
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q1/5

Graph the ordered pairs from the two patterns.

Pattern A	0	2	4	6
Pattern B	0	3	6	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q2/5

257

QUESTION ID

S7-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (0, 0), (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q2/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

Write the first four ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q3/5

258

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

**(0, 0), (3, 9), (6, 18),
(9, 27)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

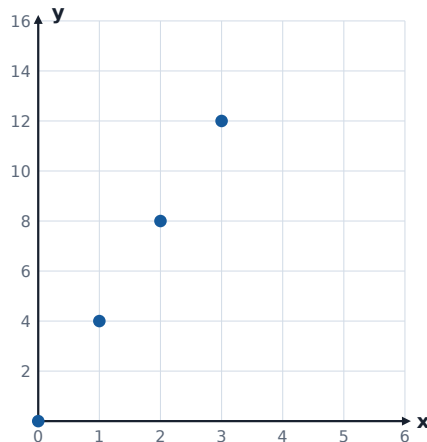
ID S7-M-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q3/5

The points come from two patterns.

Which statement is true?

- A.** The y-values increase by 1 each time. **B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value. **D.** Each y-value is 4 more than its x-value.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q4/5

259

QUESTION ID

S7-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Each y-value is 4 times its x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

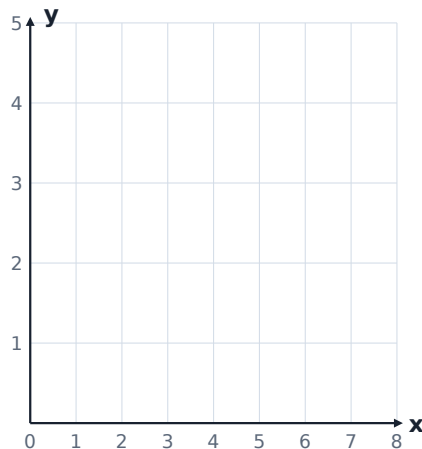
ID S7-M-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q4/5

Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q5/5

260

QUESTION ID

S7-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (0, 0), (2, 1), (4, 2), (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 7 • MAIN • Q5/5

SET 7 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 3." The pattern starts at 0.

What is the 5th term?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q1/5

262

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q1/5

True or false?

The pattern 0, 5, 10, 15 follows the rule "add 5."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q2/5

263

QUESTION ID

S7-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q2/5

Which pattern follows the rule "add 2," starting at 0?

- A.** 0, 2, 4, 6
- B.** 2, 4, 6, 8
- C.** 0, 1, 2, 3
- D.** 0, 2, 4, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q3/5

264

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0, 2, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q3/5

What is the rule for the pattern?

0, 8, 16, 24

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q4/5

265

QUESTION ID

S7-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Add 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q4/5

The rule is "add 10." The pattern starts at 0.

What is the 6th term?

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q5/5

266

QUESTION ID

S7-B1-Q5

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

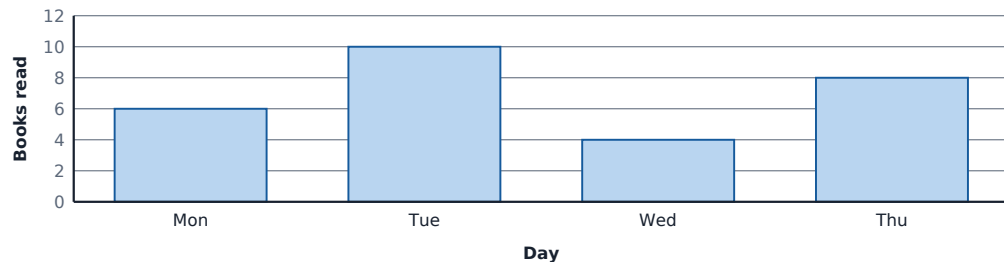
Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 5.OA.B.3

BACKWARD 3

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many books does each grid line on the scale stand for?



- A.** 4 **B.** 1 **C.** 12 **D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

268

QUESTION ID

S7-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

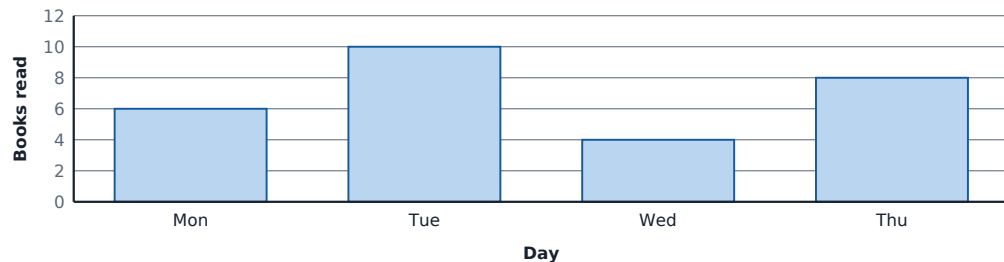
D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

How many books were read on Tuesday?



Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q2/5

269

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

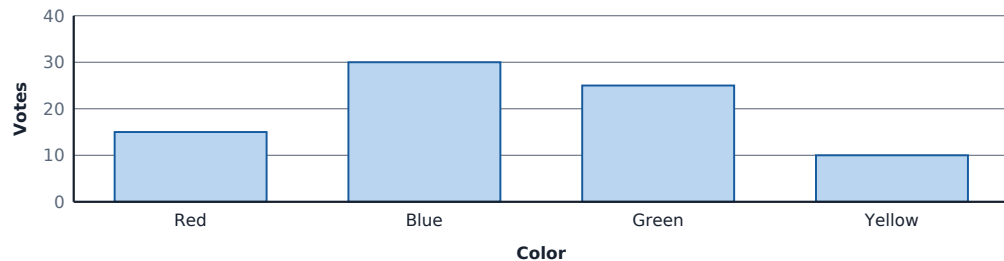
10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q2/5

How many votes did Green get?



Votes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q3/5

270

QUESTION ID

S7-B3-Q3

TYPE

Short answer

CORRECT ANSWER

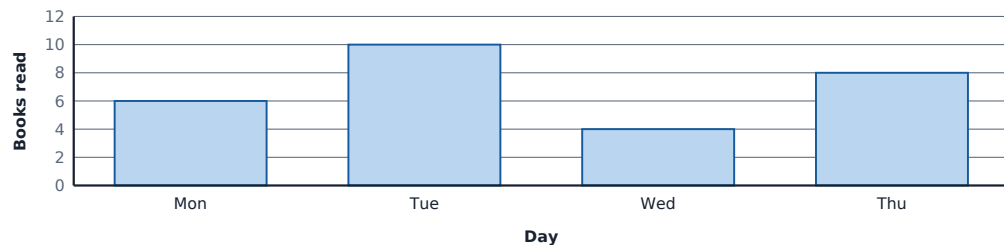
25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q3/5

How many more books were read on Tuesday than on Wednesday?



Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q4/5

271

QUESTION ID

S7-B3-Q4

TYPE

Short answer

CORRECT ANSWER

6

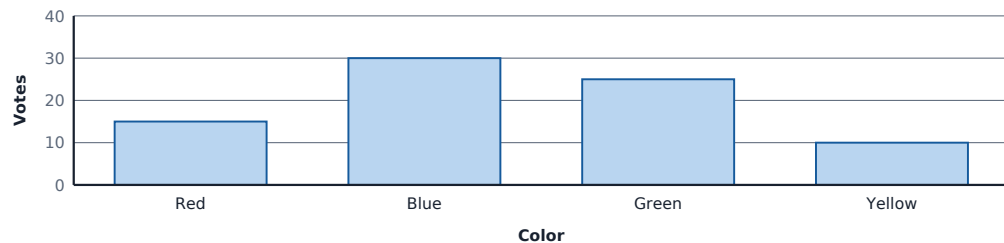
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Red got 5 more votes than Yellow.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q5/5

272

QUESTION ID

S7-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • 5.OA.B.3

BACKWARD 4

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q1/5

274

QUESTION ID

S7-B4-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q1/5

True or false?

From 0 to 4 on this number line there are 4 equal spaces.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q2/5

275

QUESTION ID

S7-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q2/5

The spaces on a number line are equal and each stands for 1.

Which number is 3 spaces to the right of 5?

A. 8

B. 2

C. 53

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q3/5

276

QUESTION ID

S7-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q4/5

277

QUESTION ID

S7-B4-Q4

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q4/5

True or false?

On a number line, 9 is to the left of 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q5/5

278

QUESTION ID

S7-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B4-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 7 • BACKWARD 4 • Q5/5

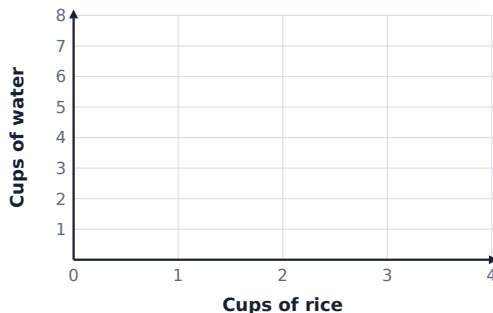
SET 7 • 5.OA.B.3

FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

Cups of rice	1	2	3
Cups of water	2	4	6



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q1/5

280

QUESTION ID

S7-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 2), (2, 4), (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q1/5

The table shows equivalent ratios.

Find the missing value.

Laps	2	4	6	8
Minutes	5	10	?	20

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q2/5

281

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q2/5

Which point comes from the ratio table?

Boxes	1	2	3
Pens	6	12	18

- A. (2, 6) B. (12, 2)
C. (2, 12) D. (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q3/5

282

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 12)

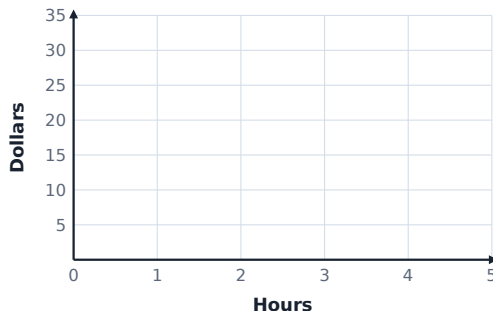
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q3/5

Plot the pairs of values from the ratio table on the coordinate plane.

Hours	1	2	3	4
Dollars	8	16	24	32



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q4/5

283

QUESTION ID

S7-F1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 8), (2, 16), (3, 24), (4, 32)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q4/5

True or false?

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q5/5

284

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • 5.OA.B.3

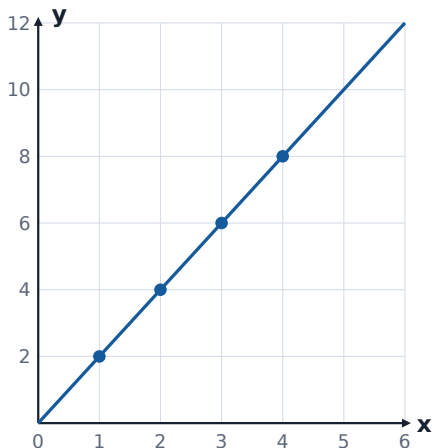
FORWARD 2

Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The points lie on a straight line through the origin, so the relationship is proportional.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q1/5

286

QUESTION ID

S7-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

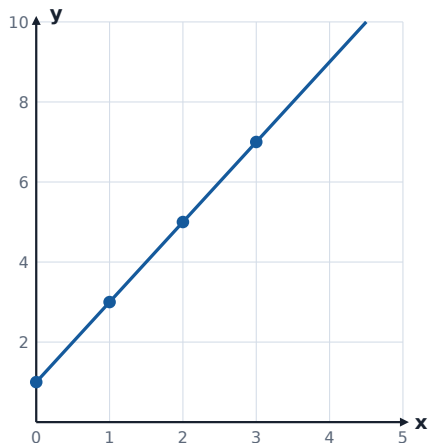
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q1/5

Which statement about the graph is true?

- A. It is not proportional because the line does not pass through $(0, 0)$.
- B. It is not proportional because the points are not on a line.
- C. It is proportional because the points are on a line.
- D. It is proportional because y increases by 2.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q2/5

287

QUESTION ID

S7-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. It is not
proportional
because the line
does not pass
through $(0, 0)$.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

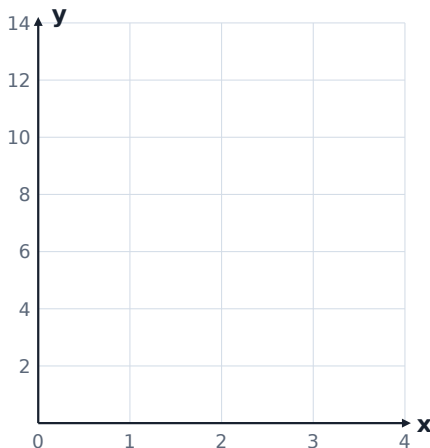
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q2/5

Graph the pairs (1, 4), (2, 8), and (3, 12).

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q3/5

288

QUESTION ID

S7-F2-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at (1, 4), (2, 8) and (3, 12)

Written: Yes; the points lie on a straight line through (0, 0)

GRADING NOTE Grade the drawing and the written answer. Drawing: all three points at the correct places. Written: "yes" (proportional) with a reason from the graph: the points are on a straight line that passes through the origin (or each y is 4 times its x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q3/5

True or false?

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q4/5

289

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q4/5

Which set of points shows a proportional relationship?

- A. (1, 1), (2, 4), (3, 9)
- B. (1, 2), (2, 3), (3, 4)
- C. (0, 3), (1, 6), (2, 9)
- D. (2, 5), (4, 10), (6, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q5/5

290

QUESTION ID

S7-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5), (4, 10), (6, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 7 • FORWARD 2 • Q5/5

A digit is worth 10 times as much as the same digit one place to its right

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	Each place is worth 10 times the place to its right	4.NBT.A.1
BACKWARD 2	Tenths and hundredths as decimals	4.NF.C.6
BACKWARD 3	Tenths as hundredths	4.NF.C.5
FORWARD 1	Compute with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 8 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A. It is 100 times as much.
- B. It is 10 times as much.
- C. It is the same.
- D. It is $\frac{1}{10}$ as much.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q1/5

293

QUESTION ID

S8-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. It is 10 times as much.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q1/5

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q2/5

294

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q2/5

True or false?

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q3/5

295

QUESTION ID

S8-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q3/5

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: _____

4 in 6,421.8: _____

Times as much: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q4/5

296

QUESTION ID

S8-M-Q4

TYPE

Short answer

CORRECT ANSWER

40; 400; 10 times as much

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q4/5

In which number is the 6 worth 10 times as much as the 6 in 3.6?

- A.** 0.36
- B.** 3.06
- C.** 46.1
- D.** 60.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q5/5

297

QUESTION ID

S8-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 46.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 8 • MAIN • Q5/5

SET 8 • 5.NBT.A.1

BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

299

QUESTION ID

S8-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q1/5

What is the value of the 7 in 2,740?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

300

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q2/5

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

301

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q3/5

How many tens are in 1 hundred?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

302

QUESTION ID

S8-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q4/5

True or false?

70 is 10 times as much as 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

303

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 5.NBT.A.1

BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{3}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q1/5

305

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q1/5

Write $\frac{47}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q2/5

306

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q2/5

True or false?

$$0.6 = \frac{6}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q3/5

307

QUESTION ID

S8-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q3/5

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q4/5

308

QUESTION ID

S8-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.05

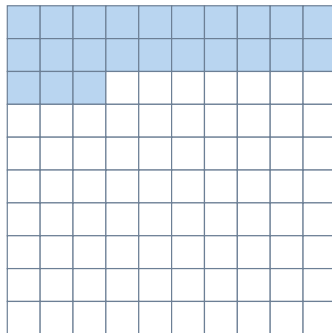
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q5/5

309

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 5.NBT.A.1

BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

True or false?

$$\frac{1}{10} = \frac{10}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q1/5

311

QUESTION ID

S8-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q1/5

How many hundredths are in 1 tenth?

Hundredths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q2/5

312

QUESTION ID

S8-B3-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q2/5

Which fraction is equal to $\frac{4}{10}$?

A. $\frac{4}{100}$

B. $\frac{400}{100}$

C. $\frac{14}{100}$

D. $\frac{40}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q3/5

313

QUESTION ID

S8-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 40/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q3/5

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q4/5

314

QUESTION ID

S8-B3-Q4

TYPE

Short answer

CORRECT ANSWER

70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q4/5

True or false?

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q5/5

315

QUESTION ID

S8-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • 5.NBT.A.1

FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$27.6 + 4.38$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q1/5

317

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

31.98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q1/5

317

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$50.2 - 7.65$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q2/5

318

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

42.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q2/5

318

Multiply.

$$1.2 \times 0.4$$

- A.** 4.8
- B.** 0.48
- C.** 0.048
- D.** 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q3/5

319

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.5 \div 2.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q4/5

320

QUESTION ID

S8-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q4/5

320

True or false?

$$0.3 \times 0.3 = 0.9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q5/5

321

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 5.NBT.A.1

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver at -4.5 m descends 2.25 m.

What is the diver's new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q1/5

323

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-6.75 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q1/5

The temperature was 3.5°F . It fell 6.2°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q2/5

324

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-2.7°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q2/5

A stock's value changed by $-\$0.75$ each day for 4 days.

What was the total change?

- A. $-\$0.30$
- B. $-\$4.75$
- C. $\$3.00$
- D. $-\$3.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q3/5

325

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $-\$3.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q3/5

A bank balance of $-\$12.40$ increases by $\$20.00$.

What is the new balance?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q4/5

326

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$\$7.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q4/5

True or false?

$$-1.5 \times 4 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q5/5

327

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 8 • FORWARD 2 • Q5/5

A digit is worth $\frac{1}{10}$ of the same digit one place to its left

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	One place to the right is worth $\frac{1}{10}$ as much	4.NBT.A.1
BACKWARD 2	Unit fractions	3.NF.A.1
BACKWARD 3	Decimals for tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Write fractions as decimals using long division	7.NS.A.2.d

SET 9 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

True or false?

In 44.4, the 4 in the tenths place is worth $\frac{1}{10}$ of the 4 in the ones place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q1/5

330

QUESTION ID

S9-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q1/5

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

- A. 10 times as much
- B. $\frac{1}{10}$
- C. The same amount
- D. $\frac{1}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q2/5

331

QUESTION ID

S9-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q2/5

What is $\frac{1}{10}$ of 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q3/5

332

QUESTION ID

S9-M-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is $\frac{1}{10}$ of 0.8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q4/5

333

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q4/5

333

In which number is the 9 worth $\frac{1}{10}$ of the 9 in 19.4?

- A. 5.92
- B. 3.009
- C. 9.01
- D. 90.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q5/5

334

QUESTION ID

S9-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 5.92

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.1 • Set 9 • MAIN • Q5/5

SET 9 • 5.NBT.A.1

BACKWARD 1

One place to the right is worth $\frac{1}{10}$ as much • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

True or false?

In 330, the 3 in the tens place is worth $\frac{1}{10}$ of the 3 in the hundreds place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q1/5

336

QUESTION ID

S9-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$400 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q2/5

337

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q2/5

337

Which number is $\frac{1}{10}$ of 3,000?

- A. 3
- B. 300
- C. 30,000
- D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q3/5

338

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q3/5

How many hundreds are in 1 thousand?

Hundreds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q4/5

339

QUESTION ID

S9-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$$60 \div 10 = 600$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q5/5

340

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 5.NBT.A.1

BACKWARD 2

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{8}$ means 1 of 8 equal parts of a whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q1/5

342

QUESTION ID

S9-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q1/5

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q2/5

343

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q2/5

Which fraction is greater?

A. $\frac{1}{8}$

B. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q3/5

344

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q3/5

True or false?

A whole cut into 4 equal parts has parts that are each $\frac{1}{3}$ of the whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q4/5

345

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q4/5

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q5/5

346

QUESTION ID

S9-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 5.NBT.A.1

BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{9}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q1/5

348

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q1/5

Write 0.8 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q2/5

349

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8/10

GRADING NOTE 4/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q2/5

True or false?

0.4 and 0.04 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q3/5

350

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q3/5

Which decimal is equal to $\frac{6}{10}$?

- A. 6.0
- B. 0.06
- C. 6.10
- D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q4/5

351

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.6

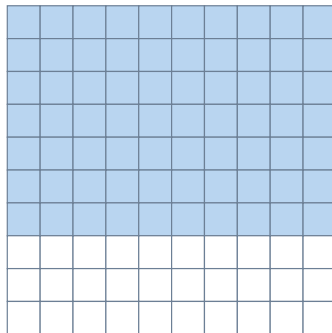
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q5/5

352

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.70

GRADING NOTE 0.7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • 5.NBT.A.1

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.8 \div 0.4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q1/5

354

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q1/5

354

Divide.

$$9.36 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q2/5

355

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q2/5

Divide.

$$0.56 \div 0.7$$

- A.** 80
- B.** 0.08
- C.** 8
- D.** 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q3/5

356

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$15 \div 0.25$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q4/5

357

QUESTION ID

S9-F1-Q4

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q4/5

357

True or false?

$$6.3 \div 0.9 = 0.7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q5/5

358

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 5.NBT.A.1

FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System
• 7.NS.A.2.d

Use long division to write $\frac{7}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q1/5

360

QUESTION ID

S9-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.875

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q1/5

Use long division to write $\frac{2}{9}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q2/5

361

QUESTION ID

S9-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.222... (the 2 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q2/5

Which fraction has a terminating decimal?

A. $\frac{5}{6}$

B. $\frac{4}{7}$

C. $\frac{1}{3}$

D. $\frac{3}{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q3/5

362

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q3/5

Use long division to write $\frac{9}{20}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q4/5

363

QUESTION ID

S9-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

0.45

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q4/5

True or false?

The decimal form of $\frac{1}{11}$ repeats.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q5/5

364

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 9 • FORWARD 2 • Q5/5

Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Multiplying and dividing by 10	4.NBT.A.1
BACKWARD 2	Multiplying by multiples of 10	3.NBT.A.3
FORWARD 1	Write and evaluate whole-number exponents	6.EE.A.1
FORWARD 2	Multiply and divide negative numbers by powers of 10 — nearest...	7.NS.A.2.c

SET 10 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$37 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q1/5

367

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

37,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q1/5

367

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$45,000 \div 100$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q2/5

368

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q2/5

368

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q3/5

369

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

10^4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q3/5

$$6 \times 10^3 = ?$$

- A.** 60,000
- B.** 6,000
- C.** 18
- D.** 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q4/5

370

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q4/5

True or false?

$$820 \div 10^2 = 82$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q5/5

371

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • MAIN • Q5/5

SET 10 • 5.NBT.A.2

BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q1/5

373

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

250

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q1/5

373

True or false?

300 is 10 times as much as 30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q2/5

374

QUESTION ID

S10-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q2/5

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q3/5

375

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q3/5

How many tens are in 800?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q4/5

376

QUESTION ID

S10-B1-Q4

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7,000 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q5/5

377

QUESTION ID

S10-B1-Q5

TYPE

Short answer

CORRECT ANSWER

700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 10 • BACKWARD 1 • Q5/5

377

SET 10 • 5.NBT.A.2

BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q1/5

379

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

240

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q1/5

379

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q2/5

380

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q2/5

True or false?

$$7 \times 30 = 2,100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q3/5

381

QUESTION ID

S10-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q3/5

$$5 \times 90 = ?$$

- A.** 4,500
- B.** 450
- C.** 45
- D.** 95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q4/5

382

QUESTION ID

S10-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 70$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q5/5

383

QUESTION ID

S10-B2-Q5

TYPE

Short answer

CORRECT ANSWER

560

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 3 • Number & Operations in Base Ten • 3.NBT.A.3 • Set 10 • BACKWARD 2 • Q5/5

383

SET 10 • 5.NBT.A.2

FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

3^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q1/5

385

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q1/5

Write $5 \times 5 \times 5$ using an exponent.

Exponent form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q2/5

386

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q2/5

$$2^5 = ?$$

- A.** 25
- B.** 10
- C.** 64
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q3/5

387

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q3/5

Evaluate.

$$10^3 + 2^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q4/5

388

QUESTION ID

S10-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1,004

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

$$4^3 = 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q5/5

389

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 10 • FORWARD 1 • Q5/5

FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$-3.5 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q1/5 • NEAREST RELATED

391

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q1/5 • NEAREST RELATED

391

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$-420 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q2/5 • NEAREST RELATED

392

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q2/5 • NEAREST RELATED

392

$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q3/5 • NEAREST RELATED

393

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q3/5 • NEAREST RELATED

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.2 \div (-10)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q4/5 • NEAREST RELATED

394

QUESTION ID

S10-F2-Q4

TYPE

Short answer

CORRECT ANSWER

−0.72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q4/5 • NEAREST RELATED

394

True or false?

$$(-10) \times (-10) \times (-10) = -1,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q5/5 • NEAREST RELATED

395

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 10 • FORWARD 2 • Q5/5 • NEAREST RELATED

Place the decimal point when multiplying or dividing a decimal by a power of 10

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Decimal notation for fractions	4.NF.C.6
BACKWARD 2	Place value when multiplying by 10 or 100	4.NBT.A.1
FORWARD 1	Compute fluently with multi-digit decimals	6.NS.B.3
FORWARD 2	Powers of 10 with negative rational numbers — nearest related (n...	7.NS.A.2.c

SET 11 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.82 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q1/5

398

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

382

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q1/5

398

Divide.

$$56.4 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q2/5

399

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

5.64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.075 \times 10^2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q3/5

400

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q3/5

400

Divide.

$$2.9 \div 10^3$$

- A. 0.029
- B. 0.29
- C. 0.0029
- D. 2,900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q4/5

401

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.0029

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q4/5

True or false?

When you multiply a number by 1,000, each digit moves 3 places to the left.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q5/5

402

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • MAIN • Q5/5

SET 11 • 5.NBT.A.2

BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

Write $\frac{35}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q1/5

404

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q1/5

True or false?

$$0.7 = \frac{7}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q2/5

405

QUESTION ID

S11-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q2/5

Which decimal is $2\frac{4}{10}$?

- A. 24
- B. 2.4
- C. 0.24
- D. 2.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q3/5

406

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q3/5

Write 0.09 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q4/5

407

QUESTION ID

S11-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q4/5

True or false?

$$1.5 = 1\frac{5}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q5/5

408

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 5.NBT.A.2

BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

When a whole number is multiplied by 10, each digit moves one place to the left.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q1/5

410

QUESTION ID

S11-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$64 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q2/5

411

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q2/5

411

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q3/5

412

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$9,000 \div 100$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q4/5

413

QUESTION ID

S11-B2-Q4

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q4/5

413

True or false?

$$500 \div 10 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q5/5

414

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 5.NBT.A.2

FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2.4 \times 0.35$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q1/5

416

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q1/5

416

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$1.44 \div 0.12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q2/5

417

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q2/5

417

Multiply.

$$12.5 \times 0.8$$

- A.** 100
- B.** 10
- C.** 0.1
- D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q3/5

418

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q3/5

Add.

$$0.875 + 12.4$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q4/5

419

QUESTION ID

S11-F1-Q4

TYPE

Short answer

CORRECT ANSWER

13.275

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q4/5

True or false?

$$4.5 \div 0.05 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q5/5

420

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 11 • FORWARD 1 • Q5/5

FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Find the value.

$$-0.45 \times 100$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

422

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q1/5 • NEAREST RELATED

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$-63.5 \div 1,000$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

423

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−0.0635

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q2/5 • NEAREST RELATED

423

Find the value.

$$(-2.7) \times (-10)$$

- A. -0.27
- B. -27
- C. 0.27
- D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

424

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q3/5 • NEAREST RELATED

Find the value.

$$0.8 \times (-100)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

425

QUESTION ID

S11-F2-Q4

TYPE

Short answer

CORRECT ANSWER

−80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

$$-5.2 \div 100 = -0.052$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

426

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 11 • FORWARD 2 • Q5/5 • NEAREST RELATED

Explain patterns in the number of zeros and the placement of the decimal point

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	A digit is worth 10 times as much as in the place to its right	4.NBT.A.1
BACKWARD 2	Tenths as hundredths	4.NF.C.5
FORWARD 1	Explain decimal placement when multiplying and dividing decima...	6.NS.B.3
FORWARD 2	Write fractions as decimals and explain why they terminate or rep...	7.NS.A.2.d

SET 59 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Explain why $46 \times 1,000 = 46,000$ has three more zeros than 46.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q1/5

429

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

Multiplying by 1,000 makes each digit worth 1,000 times as much, so each digit moves 3 places to the left; the 3 empty places (hundreds, tens, ones) are filled with zeros.

GRADING NOTE Must use place value: each factor of 10 moves every digit one place to the left ($1,000 = 10 \times 10 \times 10$, so three places) and zeros fill the empty places. "Just add three zeros" without the place-value reason is not enough.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q1/5

Why is $5.7 \times 100 = 570$?

- A. You write two zeros after 5.7.
- B. Each digit becomes worth 100 times as much, so it moves 2 places to the left.
- C. Each digit becomes worth $\frac{1}{100}$ as much.
- D. You move the decimal point 2 places to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q2/5

430

QUESTION ID

S59-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Each digit becomes worth 100 times as much, so it moves 2 places to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q2/5

True or false?

$38 \div 10 = 3.8$ because dividing by 10 makes each digit worth $\frac{1}{10}$ as much, so each digit moves one place to the right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q3/5

431

QUESTION ID

S59-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q3/5

Mia says $2.4 \times 10 = 2.40$ because "multiplying by 10 adds a zero."

Give the correct product and explain her mistake.

Correct product: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q4/5

432

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Explanation: 2.40 has the same value as 2.4, so writing a zero after the decimal does not multiply by 10. Multiplying by 10 makes each digit worth 10 times as much (2 ones become 2 tens, 4 tenths become 4 ones), so the product is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q4/5

$$10^3 = 1,000.$$

Explain how the exponent tells you the number of zeros.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q5/5

433

QUESTION ID

S59-M-Q5

TYPE

Short answer

CORRECT ANSWER

The exponent 3 means 10 is used as a factor 3 times ($10 \times 10 \times 10$); each factor of 10 moves the 1 one place to the left, so 1,000 has 3 zeros.

GRADING NOTE Must connect each factor of 10 to one place-value shift (one more zero).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 59 • MAIN • Q5/5

SET 59 • 5.NBT.A.2

BACKWARD 1

A digit is worth 10 times as much as in the place to its right • Grade 4 •
Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 770, the 7 in the hundreds place is worth 10 times the 7 in the tens place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q1/5

435

QUESTION ID

S59-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q1/5

Why is $30 \times 10 = 300$?

- A. The 3 tens become 3 hundreds.
- B. The 3 moves one place to the right.
- C. Each digit becomes worth less.
- D. You add 10 to 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q2/5

436

QUESTION ID

S59-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. The 3 tens
become 3 hundreds.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q2/5

How many tens are equal to 6 hundreds?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q3/5

437

QUESTION ID

S59-B1-Q3

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q3/5

True or false?

In 5,505, the 5 in the thousands place is worth 10 times the 5 in the ones place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q4/5

438

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q4/5

Write the number that is 10 times as much as 408.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q5/5

439

QUESTION ID

S59-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4,080

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 5.NBT.A.2

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Write $\frac{3}{10}$ as a fraction with denominator 100.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q1/5

441

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q1/5

True or false?

$$\frac{6}{10} = \frac{60}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q2/5

442

QUESTION ID

S59-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q2/5

$$\frac{40}{100} = ?$$

A. $\frac{40}{10}$

B. $\frac{400}{10}$

C. $\frac{4}{10}$

D. $\frac{4}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q3/5

443

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{2}{10} + \frac{5}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q4/5

444

QUESTION ID

S59-B2-Q4

TYPE

Short answer

CORRECT ANSWER

25/100**GRADING NOTE** 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q4/5

True or false?

$$\frac{7}{10} = \frac{7}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q5/5

445

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 5.NBT.A.2

FORWARD 1

Explain decimal placement when multiplying and dividing decimals • Grade 6 •
The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.3 \times 0.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q1/5

447

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q1/5

447

Why does $0.4 \times 0.03 = 0.012$ have three decimal places?

- A. Because $4 \times 3 = 12$.
- B. Tenths times hundredths gives thousandths.
- C. Because 0.03 is greater than 0.4.
- D. Because the product is greater than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q2/5

448

QUESTION ID

S59-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

**B. Tenths times
hundredths gives
thousandths.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.2 \div 0.08$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q3/5

449

QUESTION ID

S59-F1-Q3

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q3/5

449

True or false?

$$1.25 \times 0.4 = 0.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q4/5

450

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q4/5

Explain why $4.8 \div 0.6$ has the same quotient as $48 \div 6$.

Quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q5/5

451

QUESTION ID

S59-F1-Q5

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Both numbers were multiplied by 10 (4.8 becomes 48 and 0.6 becomes 6), which does not change the quotient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 59 • FORWARD 1 • Q5/5

FORWARD 2

Write fractions as decimals and explain why they terminate or repeat • Grade 7

• The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

453

QUESTION ID

S59-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.375

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

True or false?

$\frac{3}{40}$ can be written as a decimal that ends.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

454

QUESTION ID

S59-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which decimal equals $\frac{7}{25}$?

- A. 0.35
- B. 0.28
- C. 2.8
- D. 0.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

455

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

456

QUESTION ID

S59-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

0.41666... (the 6 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

$$\frac{2}{3} = 0.67 \text{ exactly.}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

457

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

Read and write decimals to thousandths using numerals and words

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Decimals for tenths and hundredths	4.NF.C.6
BACKWARD 2	Reading and writing whole numbers	4.NBT.A.2
BACKWARD 3	Naming tenths and hundredths	4.NF.C.5
FORWARD 1	Find and position rational numbers on a number line	6.NS.C.6.c
FORWARD 2	Write fractions as decimals	7.NS.A.2.d

SET 12 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q1/5

460

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q1/5

Write 3.508 in words.

Words: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q2/5

461

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

**three and five
hundred eight
thousandths**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q2/5

Which number is nine and forty-one thousandths?

- A.** 9.410
- B.** 9.41
- C.** 9.0041
- D.** 9.041

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q3/5

462

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9.041

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q3/5

Write the number in standard form.

seventeen and three tenths

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q4/5

463

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

17.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q4/5

True or false?

0.072 is read "seventy-two hundredths."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q5/5

464

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 12 • MAIN • Q5/5

SET 12 • 5.NBT.A.3.a

BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q1/5

466

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q1/5

True or false?

$$0.5 = \frac{5}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q2/5

467

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{12}{100}$?

- A. 1.2
- B. 12.0
- C. 0.12
- D. 0.012

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q3/5

468

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q3/5

Write $3\frac{7}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q4/5

469

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3.7

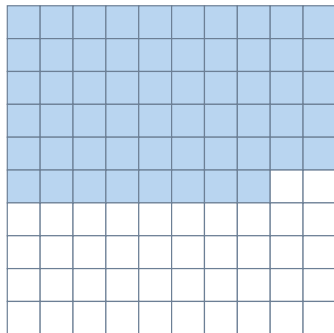
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q4/5

What decimal does the shaded part of the grid show?

Decimal: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q5/5

470

QUESTION ID

S12-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0.58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 5.NBT.A.3.a

BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write the number in standard form.

two thousand, four hundred six

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

472

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2,406

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

True or false?

5,030 is read "five thousand three hundred."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

473

QUESTION ID

S12-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

Which number is eighty thousand, nine hundred?

- A.** 80,900
- B.** 8,900
- C.** 89,000
- D.** 80,090

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

474

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 80,900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

Write 607 in words.

Words: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

475

QUESTION ID

S12-B2-Q4

TYPE

Short answer

CORRECT ANSWER

six hundred seven

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

Write the number in standard form.

twelve thousand, fifty

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

476

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

12,050

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 5.NBT.A.3.a

BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

True or false?

Three tenths is equal to thirty hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q1/5

478

QUESTION ID

S12-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q1/5

How many hundredths are in 4 tenths?

Hundredths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q2/5

479

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q2/5

Which fraction is "five hundredths"?

A. $\frac{100}{5}$

B. $\frac{5}{10}$

C. $\frac{50}{100}$

D. $\frac{5}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q3/5

480

QUESTION ID

S12-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q3/5

Write "nine tenths" as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q4/5

481

QUESTION ID

S12-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q4/5

True or false?

$$\frac{60}{100} = \frac{6}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q5/5

482

QUESTION ID

S12-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

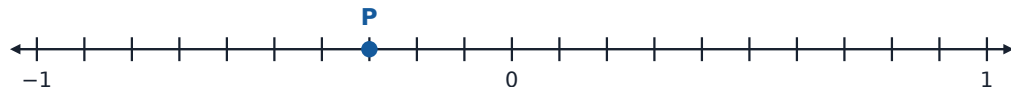
Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • 5.NBT.A.3.a

FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q1/5

484

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q1/5

Plot and label point Q at 0.45 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q2/5

485

QUESTION ID

S12-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

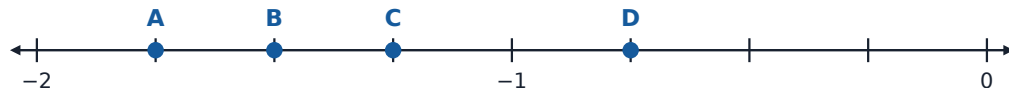
Point Q plotted at 0.45, the fifth tick after 0.4 (halfway between 0.4 and 0.5), and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q2/5

Which point is at -1.25 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q3/5

486

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

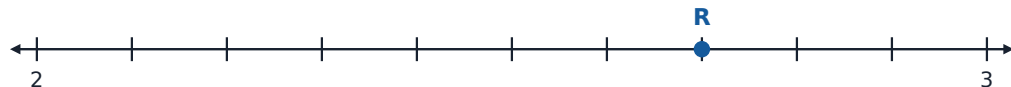
C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q3/5

What number does point R represent?



R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q4/5

487

QUESTION ID

S12-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q4/5

True or false?

On a number line, -0.6 is to the left of -0.5 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q5/5

488

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 12 • FORWARD 1 • Q5/5

SET 12 • 5.NBT.A.3.a

FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q1/5

490

QUESTION ID

S12-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

0.375

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q1/5

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q2/5

491

QUESTION ID

S12-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.41666... (the 6 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q2/5

$$\frac{11}{20} = ?$$

- A.** 0.55
- B.** 5.5
- C.** 0.2
- D.** 0.11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q3/5

492

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q3/5

Use long division to write $\frac{4}{9}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q4/5

493

QUESTION ID

S12-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

0.444... (the 4 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q4/5

True or false?

$$\frac{7}{25} = 0.32$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q5/5

494

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 12 • FORWARD 2 • Q5/5

Write decimals to thousandths in expanded form

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Expanded form of whole numbers	4.NBT.A.2
BACKWARD 2	Decimals and fractions with denominators 10 and 100	4.NF.C.6
FORWARD 1	Write expanded form with exponents	6.EE.A.1
FORWARD 2	Decimal expansions of fractions — nearest related (no direct Gra...	7.NS.A.2.d

SET 13 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q1/5

497

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

$$\begin{aligned} &3 \times 100 + 5 \times 10 \\ &+ 2 \times 1 + 6 \times (1/10) \\ &+ 8 \times (1/100) \end{aligned}$$

GRADING NOTE 300 + 50 + 2 + 0.6 + 0.08 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q1/5

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q2/5

498

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

47.309

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q2/5

Which is the expanded form of 6.045?

- A. $6 + 45$
- B. $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$
- C. $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$
- D. $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q3/5

499

QUESTION ID

S13-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.
 $6 \times 1 + 4 \times (1/100)$
 $+ 5 \times (1/1000)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q3/5

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q4/5

500

QUESTION ID

S13-M-Q4

TYPE

Short answer

CORRECT ANSWER

800.26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q4/5

True or false?

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q5/5

501

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 13 • MAIN • Q5/5

SET 13 • 5.NBT.A.3.a

BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q1/5

503

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000 + 300 + 70 + 1

GRADING NOTE $4 \times 1,000 + 3 \times 100 + 7 \times 10 + 1 \times 1$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q1/5

True or false?

$$6,052 = 6,000 + 500 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q2/5

504

QUESTION ID

S13-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q2/5

Which number is $7,000 + 400 + 9$?

- A. 7,409
- B. 7,490
- C. 74,009
- D. 7,049

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q3/5

505

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7,409

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q3/5

Write $3 \times 1,000 + 8 \times 100 + 5 \times 1$ in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q4/5

506

QUESTION ID

S13-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3,805

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$$900 + 20 + 4 = 9,024$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q5/5

507

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 5.NBT.A.3.a

BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q1/5

509

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6/10

GRADING NOTE 3/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q1/5

True or false?

$$\frac{4}{100} = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q2/5

510

QUESTION ID

S13-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q2/5

Which fraction is equal to 0.25?

A. $\frac{1}{25}$

B. $\frac{25}{100}$

C. $\frac{25}{10}$

D. $\frac{2}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q3/5

511

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q3/5

Write $2 + \frac{3}{10} + \frac{7}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q4/5

512

QUESTION ID

S13-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2.37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q4/5

True or false?

$$\frac{9}{10} = 0.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q5/5

513

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 5.NBT.A.3.a

FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •
6.EE.A.1

Write $5,000 + 300 + 20 + 7$ using powers of 10.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q1/5

515

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$5 \times 10^3 + 3 \times 10^2 \\ + 2 \times 10^1 + 7$$

GRADING NOTE Writing the last term as 7×10^0 or 7×1 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q1/5

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q2/5

516

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4,060

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q2/5

Which number is equal to 10^4 ?

- A.** 40
- B.** 100,000
- C.** 1,000
- D.** 10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q3/5

517

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q3/5

Write $600 + 90 + 1$ using powers of 10.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q4/5

518

QUESTION ID

S13-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$6 \times 10^2 + 9 \times 10^1 + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q4/5

True or false?

$$2 \times 10^2 + 5 = 250$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q5/5

519

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 13 • FORWARD 1 • Q5/5

FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Write $\frac{3}{4}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q1/5 • NEAREST RELATED

521

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q1/5 • NEAREST RELATED

Use long division to write $\frac{1}{6}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q2/5 • NEAREST RELATED

522

QUESTION ID

S13-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.1666... (the 6 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which fraction is equal to 0.666... (the 6 repeats)?

A. $\frac{3}{5}$

B. $\frac{6}{10}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q3/5 • NEAREST RELATED

523

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q3/5 • NEAREST RELATED

Use long division to write $\frac{5}{8}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q4/5 • NEAREST RELATED

524

QUESTION ID

S13-F2-Q4

TYPE

Answer with required work

CORRECT ANSWER

0.625

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

$$\frac{2}{5} = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q5/5 • NEAREST RELATED

525

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 13 • FORWARD 2 • Q5/5 • NEAREST RELATED

Compare decimals to thousandths

MAIN	Grade 5 main question	5.NBT.A.3.b
BACKWARD 1	Comparing decimals to hundredths	4.NF.C.7
BACKWARD 2	The symbols $>$, $<$, and $=$	1.NBT.B.3
FORWARD 1	Order rational numbers	6.NS.C.7.b
FORWARD 2	Compare fractions and decimals using decimal expansions	7.NS.A.2.d

SET 14 • 5.NBT.A.3.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Write $>$, $<$, or $=$ to compare.

0.47 ____ 0.405

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q1/5

528

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q1/5

Write $>$, $<$, or $=$ to compare.

3.06 ____ 3.060

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q2/5

529

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q2/5

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q3/5

530

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q3/5

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q4/5

531

QUESTION ID

S14-M-Q4

TYPE

Short answer

CORRECT ANSWER

**2.13, 2.3, 2.301,
2.31**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q4/5

True or false?

$$0.8 < 0.79$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q5/5

532

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 14 • MAIN • Q5/5

SET 14 • 5.NBT.A.3.b

BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions
• 4.NF.C.7

Write $>$, $<$, or $=$ to compare.

0.6 ____ 0.58

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q1/5

534

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q1/5

True or false?

$$0.3 = 0.03$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q2/5

535

QUESTION ID

S14-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q2/5

Which number is less?

A. 0.09

B. 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q3/5

536

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q3/5

Write $>$, $<$, or $=$ to compare.

0.75 ____ 0.8

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q4/5

537

QUESTION ID

S14-B1-Q4

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q4/5

True or false?

$0.42 > 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q5/5

538

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.7 • Set 14 • BACKWARD 1 • Q5/5

BACKWARD 2

The symbols $>$, $<$, and $=$ • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q1/5

540

QUESTION ID

S14-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q1/5

True or false?

$$9 > 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q2/5

541

QUESTION ID

S14-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

52 ____ 25

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q3/5

542

QUESTION ID

S14-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q3/5

542

True or false?

The symbol $>$ means "is less than."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q4/5

543

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q4/5

Which statement is true?

- A.** $14 < 41$
- B.** $14 = 41$
- C.** $14 > 41$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q5/5

544

QUESTION ID

S14-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $14 < 41$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 5.NBT.A.3.b

FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write $>$, $<$, or $=$ to compare.

-0.4 ____ -0.45

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q1/5

546

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q1/5

Which number is least?

- A.** -2
- B.** -2.5
- C.** 0.5
- D.** -2.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q2/5

547

QUESTION ID

S14-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. -2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q2/5

Order from least to greatest.

-1.2 , 0.3 , -1.25 , 0

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q3/5

548

QUESTION ID

S14-F1-Q3

TYPE

Short answer

CORRECT ANSWER

-1.25 , -1.2 , 0 , 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q3/5

True or false?

$$-3.7 > -3.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q4/5

549

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q4/5

A temperature of -2.5°C is colder than a temperature of -1.8°C .

Write an inequality to show this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q5/5

550

QUESTION ID

S14-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$-2.5 < -1.8$

GRADING NOTE $-1.8 > -2.5$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 14 • FORWARD 1 • Q5/5

SET 14 • 5.NBT.A.3.b

FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Is $\frac{5}{8}$ greater than or less than 0.6?

Decimal: _____

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q1/5

552

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**0.625; greater than
0.6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q1/5

Which number is greatest?

- A. $\frac{4}{5}$
- B. 0.7
- C. $\frac{3}{4}$
- D. 0.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q2/5

553

QUESTION ID

S14-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q2/5

Order from least to greatest.

$\frac{2}{3}$, 0.6, $\frac{5}{8}$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q3/5

554

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.6, 5/8, 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q3/5

True or false?

$$\frac{1}{3} > 0.33$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q4/5

555

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q4/5

Write $>$, $<$, or $=$ to compare.

$$\frac{7}{20} \text{ ____ } 0.4$$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q5/5

556

QUESTION ID

S14-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 14 • FORWARD 2 • Q5/5

Round decimals to any place

MAIN

Grade 5 main question

5.NBT.A.4

BACKWARD 1

Rounding whole numbers

4.NBT.A.3

BACKWARD 2

Decimals on a number line

4.NF.C.6

FORWARD 1

Estimate to check computations with decimals

6.NS.B.3

FORWARD 2

Use estimation to judge whether an answer is reasonable

7.EE.B.3

SET 15 • 5.NBT.A.4

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q1/5

559

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q1/5

Round 12.749 to the nearest hundredth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q2/5

560

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

12.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q2/5

Round 0.865 to the nearest whole number.

- A.** 0.9
- B.** 1
- C.** 0
- D.** 0.87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q3/5

561

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q3/5

Round 39.96 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q4/5

562

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

40.0

GRADING NOTE 40 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q4/5

True or false?

4.351 rounded to the nearest hundredth is 4.36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q5/5

563

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 15 • MAIN • Q5/5

SET 15 • 5.NBT.A.4

BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q1/5

565

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,700

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q1/5

True or false?

350 rounded to the nearest hundred is 300.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q2/5

566

QUESTION ID

S15-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q2/5

Round 7,249 to the nearest thousand.

- A. 7,200
- B. 7,000
- C. 7,250
- D. 8,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q3/5

567

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q3/5

Round 85 to the nearest ten.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q4/5

568

QUESTION ID

S15-B1-Q4

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q4/5

True or false?

1,349 rounded to the nearest hundred is 1,400.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q5/5

569

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 5.NBT.A.4

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q1/5

571

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q1/5

True or false?

0.7 is closer to 1 than to 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q2/5

572

QUESTION ID

S15-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q2/5

Is 0.43 closer to 0.4 or to 0.5?

A. 0.4

B. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q3/5

573

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q3/5

What decimal is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q4/5

574

QUESTION ID

S15-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q4/5

True or false?

2.5 is halfway between 2 and 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q5/5

575

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 5.NBT.A.4

FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q1/5

577

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE $19 + 6 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q1/5

Which is the best estimate of 4.9×6.1 ?

- A. 36
- B. 30
- C. 24
- D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q2/5

578

QUESTION ID

S15-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q2/5

Estimate $31.8 \div 7.9$ by rounding each number to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q3/5

579

QUESTION ID

S15-F1-Q3

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE $32 \div 8 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q3/5

True or false?

29.6 – 10.2 is about 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q4/5

580

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q4/5

Kim says $3.9 \times 2.1 = 81.9$.

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q5/5

581

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE 3.9×2.1 is about $4 \times 2 = 8$, so 81.9 is far too large (the exact product is 8.19).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 5.NBT.A.4

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Round the price to the nearest dollar. Then use the rounded price to estimate the sale price.

Rounded price: _____

Estimated sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q1/5

583

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

\$25; \$20

GRADING NOTE $\$25 - 20\% \text{ of } \$25 = \$25 - \$5 = \$20$ (or $0.8 \times \$25$). Both parts are required. The exact sale price, \$19.80, is not the estimate asked for.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q1/5

A worker earns \$19.85 per hour for 9.75 hours.

Which is the best estimate of her pay?

- A. \$20
- B. \$2,000
- C. \$200
- D. \$100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q2/5

584

QUESTION ID

S15-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. \$200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q2/5

True or false?

A 6.2% tax on \$49.50 is about \$3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q3/5

585

QUESTION ID

S15-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q3/5

Nina says $\frac{3}{4} \times 39.8$ is about 40.

Round 39.8 to the nearest whole number and estimate

$\frac{3}{4} \times 39.8$. Is her estimate reasonable?

Your estimate: _____

Reasonable? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q4/5

586

QUESTION ID

S15-F2-Q4

TYPE

Short answer

CORRECT ANSWER

30; no

GRADING NOTE $\frac{3}{4} \times 40 = 30$, so 40 is not reasonable (the exact value is 29.85). Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q4/5

A board is $8\frac{1}{4}$ feet long. It is cut into pieces that are $1\frac{7}{8}$ feet long.

Estimate how many whole pieces can be cut.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q5/5

587

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4 pieces

GRADING NOTE About $8 \div 2 = 4$; the exact quotient is 4.4, so 4 whole pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 15 • FORWARD 2 • Q5/5

Multiply multi-digit whole numbers using the standard algorithm

MAIN	Grade 5 main question	5.NBT.B.5
BACKWARD 1	Multiplying by a one-digit or two-digit number	4.NBT.B.5
BACKWARD 2	Multiplication facts	3.OA.C.7
BACKWARD 3	Adding partial products	4.NBT.B.4
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply integers	7.NS.A.2.c

SET 16 • 5.NBT.B.5

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$246 \times 37$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q1/5

590

QUESTION ID

S16-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

9,102

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method (area model, repeated addition) earns partial credit.

Partial products: $246 \times 7 = 1,722$ and $246 \times 30 = 7,380$; $1,722 + 7,380 = 9,102$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q1/5

Use the standard algorithm to multiply. Show your work.

$$1,385 \times 24$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q2/5

591

QUESTION ID

S16-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

33,240

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method (area model, repeated addition) earns partial credit.

Partial products: $1,385 \times 4 = 5,540$ and $1,385 \times 20 = 27,700$; $5,540 + 27,700 = 33,240$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q2/5

This is the standard algorithm for 508×63 with one partial product missing.

What is the missing partial product?

$$\begin{array}{r} 508 \\ \times 63 \\ \hline 1,524 \\ + ? \\ \hline 32,004 \end{array}$$

A. 1,524

C. 30,480

B. 31,904

D. 3,048

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q3/5

592

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 30,480

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q3/5

Use the standard algorithm to multiply. Show your work.

$$79 \times 86$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q4/5

593

QUESTION ID

S16-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

6,794

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method (area model, repeated addition) earns partial credit.

Partial products: $79 \times 6 = 474$ and $79 \times 80 = 6,320$; $474 + 6,320 = 6,794$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q4/5

A theater has 48 rows with 125 seats in each row.

Use the standard algorithm to find how many seats are in the theater. Show your work.

Seats: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q5/5

594

QUESTION ID

S16-M-Q5

TYPE

Answer with required work

CORRECT ANSWER

6,000

GRADING NOTE REQUIRED WORK: The standard algorithm: one partial product for each digit of the second factor (with regrouping), each lined up by place value, then the partial products added. Another method (area model, repeated addition) earns partial credit.

Partial products: $125 \times 8 = 1,000$ and $125 \times 40 = 5,000$; $1,000 + 5,000 = 6,000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 16 • MAIN • Q5/5

SET 16 • 5.NBT.B.5

BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$346 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q1/5

596

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2,422

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q1/5

596

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$23 \times 45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q2/5

597

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1,035

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q2/5

597

True or false?

$$1,203 \times 4 = 4,082$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q3/5

598

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q3/5

$$62 \times 18 = ?$$

- A.** 1,126
- B.** 496
- C.** 1,116
- D.** 1,006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q4/5

599

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1,116

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$58 \times 30$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q5/5

600

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1,740

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 16 • BACKWARD 1 • Q5/5

600

SET 16 • 5.NBT.B.5

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q1/5

602

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q2/5

603

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q2/5

603

True or false?

$$6 \times 9 = 56$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q3/5

604

QUESTION ID

S16-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q3/5

$$4 \times 8 = ?$$

- A.** 24
- B.** 36
- C.** 32
- D.** 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q4/5

605

QUESTION ID

S16-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q4/5

Multiply.

$$9 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q5/5

606

QUESTION ID

S16-B2-Q5

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 5.NBT.B.5

BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$7,380 + 1,722$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q1/5

608

QUESTION ID

S16-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9,102

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$27,700 + 5,540$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q2/5

609

QUESTION ID

S16-B3-Q2

TYPE

Short answer

CORRECT ANSWER

33,240

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q2/5

609

True or false?

$$30,480 + 1,524 = 31,904$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q3/5

610

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q3/5

$$6,400 + 394 = ?$$

- A.** 6,784
- B.** 6,794
- C.** 7,794
- D.** 6,694

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q4/5

611

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6,794

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$4,800 + 1,200$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q5/5

612

QUESTION ID

S16-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 3 • Q5/5

612

SET 16 • 5.NBT.B.5

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2.46 \times 3.7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q1/5

614

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9.102

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q1/5

614

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$13.85 \times 2.4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q2/5

615

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

33.24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q2/5

615

Multiply.

$$5.08 \times 6.3$$

- A.** 31.904
- B.** 320.04
- C.** 3.2004
- D.** 32.004

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q3/5

616

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 32.004

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.79 \times 8.6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q4/5

617

QUESTION ID

S16-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6.794

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q4/5

617

True or false?

$$1.25 \times 4.8 = 60$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q5/5

618

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 5.NBT.B.5

FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q1/5

620

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q1/5

Find the product.

$$(-12) \times (-35)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q2/5

621

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

420

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q2/5

Find the product.

$$18 \times (-21)$$

- A. 378
- B. -378
- C. -39
- D. -388

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q3/5

622

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -378

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q3/5

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q4/5

623

QUESTION ID

S16-F2-Q4

TYPE

Short answer

CORRECT ANSWER

−300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q4/5

True or false?

$$(-6)(-7) = -42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q5/5

624

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 16 • FORWARD 2 • Q5/5

Divide with up to four-digit dividends and two-digit divisors

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Multiplying to check a quotient	4.NBT.B.5
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Divide multi-digit numbers using the standard algorithm	6.NS.B.2
FORWARD 2	Divide integers	7.NS.A.2.b

SET 17 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$2,016 \div 24$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q1/5

627

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7,830 \div 45$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q2/5

628

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

174

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q2/5

628

Divide.

$$952 \div 17$$

- A.** 54
- B.** 46
- C.** 56
- D.** 57

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q3/5

629

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$3,724 \div 38$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q4/5

630

QUESTION ID

S17-M-Q4

TYPE

Short answer

CORRECT ANSWER

98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q4/5

630

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q5/5

631

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

31 R 8

GRADING NOTE Quotient 31 and remainder 8 (31 R 8, or "31 remainder 8"). The question asks for a remainder, so a mixed number or decimal (31 $\frac{1}{4}$, 31.25) is not accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 17 • MAIN • Q5/5

SET 17 • 5.NBT.B.6

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q1/5

633

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

142

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$1,935 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q2/5

634

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

387

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q2/5

634

True or false?

$$728 \div 8 = 901$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q3/5

635

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q3/5

$$504 \div 7 = ?$$

- A.** 82
- B.** 71
- C.** 62
- D.** 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q4/5

636

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q4/5

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q5/5

637

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

23 R 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 5.NBT.B.6

BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

True or false?

To check $2,016 \div 24 = 84$, you can multiply 24×84 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q1/5

639

QUESTION ID

S17-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q1/5

Multiply to check $2,016 \div 24 = 84$.

24×84

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q2/5

640

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2,016

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q2/5

Which product checks $952 \div 17 = 56$?

- A.** $17 + 56$
- B.** 17×56
- C.** $56 - 17$
- D.** 952×17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q3/5

641

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 17×56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$38 \times 98$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q4/5

642

QUESTION ID

S17-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3,724

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q4/5

True or false?

$$32 \times 31 + 8 = 1,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q5/5

643

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 5.NBT.B.6

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q1/5

645

QUESTION ID

S17-B3-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q1/5

Divide.

$$54 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q2/5

646

QUESTION ID

S17-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$$63 \div 7 = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q3/5

647

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q3/5

$$40 \div 5 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q4/5

648

QUESTION ID

S17-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$35 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q5/5

649

QUESTION ID

S17-B3-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • 5.NBT.B.6

FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$6,432 \div 48$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q1/5

651

QUESTION ID

S17-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

134

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q1/5

Use the standard algorithm to divide. Show your work.

$$25,875 \div 69$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q2/5

652

QUESTION ID

S17-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

375

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q2/5

Divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q3/5

653

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 270

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q3/5

Use the standard algorithm to divide. Show your work.

$$9,594 \div 41$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q4/5

654

QUESTION ID

S17-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

234

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q4/5

True or false?

$$11,700 \div 45 = 2,600$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q5/5

655

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 5.NBT.B.6

FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q1/5

657

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q1/5

Find the quotient.

$$(-952) \div (-17)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q2/5

658

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q2/5

Find the quotient.

$$780 \div (-15)$$

- A.** -52
- B.** -53
- C.** -765
- D.** 52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q3/5

659

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q3/5

Find the quotient.

$$-(144 \div 12)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q4/5

660

QUESTION ID

S17-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q4/5

True or false?

$$(-90) \div 18 = 90 \div (-18)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q5/5

661

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 17 • FORWARD 2 • Q5/5

Explain division using area models and equations

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Area as length times width	3.MD.C.7.b
BACKWARD 3	Splitting a rectangle to find its area	3.MD.C.7.c
FORWARD 1	Use the distributive property with the greatest common factor	6.NS.B.4
FORWARD 2	Factor and expand linear expressions	7.EE.A.1

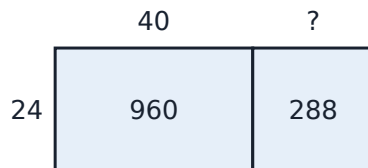
SET 18 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

The area model shows
 $1,248 \div 24$.

What is the quotient?



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q1/5

664

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

52

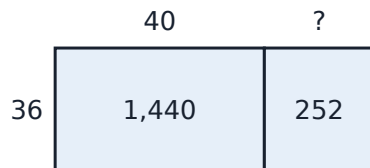
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q1/5

The area model shows
 $1,692 \div 36$.

What is the quotient?



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q2/5

665

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

47

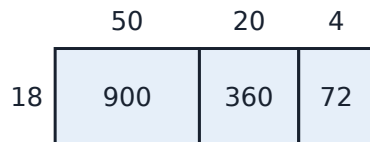
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q2/5

Which equation matches the area model?

- A. $1,332 \div 74 = 20$
- B. $1,332 \div 18 = 74$
- C. $18 \times 50 = 1,332$
- D. $900 \div 18 = 74$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q3/5

666

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $1,332 \div 18 = 74$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q3/5

Use an area model or equations to divide. Show your model or equations.

$$2,184 \div 28$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q4/5

667

QUESTION ID

S18-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

78

GRADING NOTE REQUIRED WORK: An area model or equations that split 2,184 into parts that 28 divides, for example $28 \times 70 = 1,960$ and $28 \times 8 = 224$, with $1,960 + 224 = 2,184$, so the quotient is $70 + 8 = 78$. The answer 78 without a valid model or equations earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q4/5

True or false?

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q5/5

668

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 18 • MAIN • Q5/5

SET 18 • 5.NBT.B.6

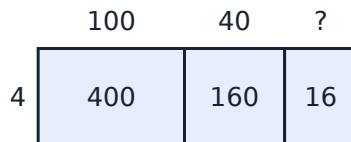
BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows

$$576 \div 4.$$

What is the quotient?



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q1/5

670

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q1/5

True or false?

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q2/5

671

QUESTION ID

S18-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q2/5

Which shows $245 \div 5$ split into easier parts?

- A. $(200 \div 5) + (45 \div 5)$
- B. $(245 \div 2) + (245 \div 3)$
- C. $200 \div (5 + 45)$
- D. $(200 \div 5) \times (45 \div 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q3/5

672

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

$(200 \div 5) + (45 \div 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q3/5

Divide.

$$735 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q4/5

673

QUESTION ID

S18-B1-Q4

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q4/5

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q5/5

674

QUESTION ID

S18-B1-Q5

TYPE

Short answer

CORRECT ANSWER

52 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 5.NBT.B.6

BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q1/5

676

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

63 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q1/5

True or false?

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q2/5

677

QUESTION ID

S18-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q2/5

Which equation gives the area of a 6-by-8 rectangle?

- A. $8 - 6 = 2$
- B. $6 + 8 = 14$
- C. $6 \times 8 = 48$
- D. $2 \times (6 + 8) = 28$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q3/5

678

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $6 \times 8 = 48$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q3/5

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q4/5

679

QUESTION ID

S18-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q5/5

680

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 5.NBT.B.6

BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

The area of a rectangle that is 4 by $(10 + 3)$ equals $4 \times 10 + 4 \times 3$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q1/5

682

QUESTION ID

S18-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q1/5

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q2/5

683

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

60 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q2/5

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

A. $7 \times 10 + 7 \times 3$

B. $7 \times 10 + 3$

C. $7 + 10 + 3$

D. $7 \times 10 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q3/5

684

QUESTION ID

S18-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $7 \times 10 + 7 \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q3/5

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

ID S18-B3-Q4

685

S18-B3-Q4

Short answer

4

ID S18-B3-Q4

685

True or false?

$$8 \times 12 = 8 \times 10 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q5/5

686

QUESTION ID

S18-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.c • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 5.NBT.B.6

FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 48$ as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q1/5

688

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12(3 + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q1/5

Find the greatest common factor of 32 and 56.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q2/5

689

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q2/5

Which expression is equal to $45 + 60$ and uses the greatest common factor?

- A. $15(3 + 6)$
- B. $15(3 + 4)$
- C. $3(15 + 20)$
- D. $5(9 + 12)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q3/5

690

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $15(3 + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q3/5

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q4/5

691

QUESTION ID

S18-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q4/5

True or false?

$$18 + 27 = 9(2 + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q5/5

692

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 5.NBT.B.6

FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q1/5

694

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$6(2x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q1/5

Expand.

$$7(4y - 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q2/5

695

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$28y - 21$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q2/5

Which expression is equivalent to $18a - 24$?

- A. $3(6a - 4)$
- B. $6(3a - 24)$
- C. $18(a - 6)$
- D. $6(3a - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q3/5

696

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $6(3a - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q3/5

Factor out -4 .

$$-8n - 20$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q4/5

697

QUESTION ID

S18-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-4(2n + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

$$4(2.5x + 3) = 10x + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q5/5

698

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 18 • FORWARD 2 • Q5/5

Draw division models and explain the strategy

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
FORWARD 1	Connect the standard division algorithm to partial quotients	6.NS.B.2
FORWARD 2	Solve real-world problems by dividing rational numbers	7.NS.A.3

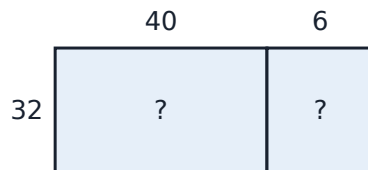
SET 60 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Complete the area model for $1,472 \div 32$ by writing the area of each part.

Then write the quotient.



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q1/5

701

QUESTION ID

S60-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: Areas
written in the parts:
1,280 (32×40) and
192 (32×6)**

Written: 46

GRADING NOTE Grade both: the two areas must be 1,280 and 192 (they add to 1,472), and the quotient $40 + 6 = 46$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q1/5

Draw an area model to divide $1,615 \div 19$.

Explain how your model shows the quotient.

Quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q2/5

702

QUESTION ID

S60-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

85

GRADING NOTE REQUIRED WORK: An area model with side 19 split into parts whose areas add to 1,615 (for example $19 \times 80 = 1,520$ and $19 \times 5 = 95$); the explanation says the part lengths ($80 + 5$) add up to the quotient because the part areas add up to the dividend.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q2/5

Lena finds $1,380 \div 23$ by thinking $23 \times 60 = 1,380$.

Which statement explains her strategy?

- A.** She multiplied 1,380 by 23 to get 60.
- B.** She divided 23 by 1,380 to get 60.
- C.** Division finds the missing factor:
 $23 \times ? = 1,380$, so the quotient is 60.
- D.** She subtracted 23 from 1,380 to get 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q3/5

703

QUESTION ID

S60-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**C. Division finds the missing factor:
 $23 \times ? = 1,380$, so the quotient is 60.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q3/5

Explain how these equations show $2,345 \div 35$.

$$35 \times 60 = 2,100$$

$$35 \times 7 = 245$$

$$2,100 + 245 = 2,345$$

Quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q4/5

704

QUESTION ID

S60-M-Q4

TYPE

Short answer

CORRECT ANSWER

67

GRADING NOTE Explanation: 2,345 is split into 2,100 and 245; 35 goes into them 60 times and 7 times, so it goes into 2,345 a total of $60 + 7 = 67$ times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q4/5

True or false?

An area model for $1,512 \div 36$ can use parts with areas 1,440 and 72, so the quotient is $40 + 2 = 42$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q5/5

705

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 60 • MAIN • Q5/5

SET 60 • 5.NBT.B.6

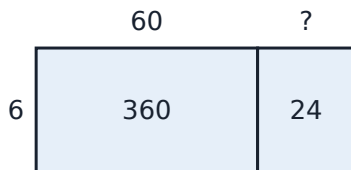
BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows

$$384 \div 6.$$

What is the missing length?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q1/5

707

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q1/5

True or false?

An area model for $528 \div 4$ can be split into $400 \div 4$, $120 \div 4$, and $8 \div 4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q2/5

708

QUESTION ID

S60-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q2/5

An area model for $945 \div 9$ has parts with areas 900 and 45.

What are the lengths of the parts?

- A.** 100 and 5 **B.** 90 and 5
C. 10 and 5 **D.** 100 and 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q3/5

709

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q3/5

Divide.

$$672 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q4/5

710

QUESTION ID

S60-B1-Q4

TYPE

Short answer

CORRECT ANSWER

84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q4/5

True or false?

In an area model for $945 \div 9$, the side length 9 is the quotient.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q5/5

711

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 5.NBT.B.6

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find the unknown factor.

$$6 \times ? = 42$$

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q1/5

713

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q1/5

True or false?

$32 \div 4 = 8$ because $4 \times 8 = 32$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q2/5

714

QUESTION ID

S60-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q2/5

Which multiplication fact helps you find $56 \div 7$?

- A. $8 + 7 = 15$
- B. $7 \times 7 = 49$
- C. $56 \times 7 = 392$
- D. $7 \times 8 = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q3/5

715

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $7 \times 8 = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q3/5

Find the unknown factor.

$$? \times 9 = 81$$

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q4/5

716

QUESTION ID

S60-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q4/5

True or false?

To find $27 \div 3$, you can ask "27 times what number is 3?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q5/5

717

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 5.NBT.B.6

FORWARD 1

Connect the standard division algorithm to partial quotients • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$8,640 \div 36$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q1/5

719

QUESTION ID

S60-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

240

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q1/5

In the standard algorithm for $5,712 \div 28$, the first step uses $28 \times 200 = 5,600$.

What does the 200 stand for?

- A.** Part of the quotient
- B.** The dividend
- C.** The remainder
- D.** Part of the divisor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q2/5

720

QUESTION ID

S60-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Part of the quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q2/5

True or false?

The steps of the standard algorithm for $1,836 \div 17$ match an area model with parts 17×100 and 17×8 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q3/5

721

QUESTION ID

S60-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q3/5

Use the standard algorithm to divide. Show your work.

$$13,248 \div 46$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q4/5

722

QUESTION ID

S60-F1-Q4

TYPE

Answer with required work

CORRECT ANSWER

288

GRADING NOTE REQUIRED WORK: The standard division algorithm with every step shown (divide, multiply, subtract, bring down). A correct quotient without the algorithm earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q4/5

True or false?

$$3,825 \div 45 = 95$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q5/5

723

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 60 • FORWARD 1 • Q5/5

SET 60 • 5.NBT.B.6

FORWARD 2

Solve real-world problems by dividing rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver goes down 126 feet in 18 equal stages.

Write the change in depth for each stage as a signed number.

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q1/5

725

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

–7 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q1/5

A business lost \$4,320 over 24 months, the same amount each month.

Which number shows the change each month?

- A.** $-\$108$ **B.** $\$180$
C. $-\$1,800$ **D.** $-\$180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q2/5

726

QUESTION ID

S60-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $-\$180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q2/5

The temperature fell 22.5°F in 9 hours at a steady rate.

Write the change per hour as a signed number.

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q3/5

727

QUESTION ID

S60-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-2.5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q3/5

True or false?

A change of $-\$1,512$ shared equally over 36 weeks is a change of $-\$42$ each week.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q4/5

728

QUESTION ID

S60-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q4/5

A tank loses $\frac{3}{4}$ gallon every hour.

After how many hours is the change in the amount of water -6 gallons?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q5/5

729

QUESTION ID

S60-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 60 • FORWARD 2 • Q5/5

Add decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1 Adding multi-digit whole numbers

4.NBT.B.4

BACKWARD 2 Adding tenths and hundredths

4.NF.C.5

BACKWARD 3 Writing decimals with the same number of places

4.NF.C.6

FORWARD 1 Add multi-digit decimals

6.NS.B.3

FORWARD 2 Add rational numbers

7.NS.A.1.d

SET 19 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$23.45 + 8.7$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q1/5

732

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

32.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q1/5

732

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$15.3 + 6.48$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q2/5

733

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

21.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q2/5

733

Add.

$$4.06 + 12.9$$

- A.** 16.15
- B.** 16.069
- C.** 5.35
- D.** 16.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q3/5

734

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$9.65 + 0.8$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q4/5

735

QUESTION ID

S19-M-Q4

TYPE

Short answer

CORRECT ANSWER

10.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q4/5

735

Maya had \$18.50. She earned \$6.75 more.

How much money does she have now?

Money: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q5/5

736

QUESTION ID

S19-M-Q5

TYPE

Short answer

CORRECT ANSWER

\$25.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • MAIN • Q5/5

SET 19 • 5.NBT.B.7

BACKWARD 1

Adding multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$2,345 + 870$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q1/5

738

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3,215

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q1/5

738

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$1,530 + 648$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q2/5

739

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2,178

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q2/5

739

True or false?

$$406 + 1,290 = 1,596$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q3/5

740

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q3/5

$$1,265 + 735 = ?$$

- A.** 2,100
- B.** 2,000
- C.** 1,900
- D.** 1,990

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q4/5

741

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$1,850 + 675$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q5/5

742

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2,525

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 19 • BACKWARD 1 • Q5/5

742

SET 19 • 5.NBT.B.7

BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q1/5

744

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

75/100**GRADING NOTE** 3/4 or 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q1/5

True or false?

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q2/5

745

QUESTION ID

S19-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q2/5

$$\frac{7}{10} + \frac{15}{100} = ?$$

A. $\frac{85}{100}$

B. $\frac{22}{100}$

C. $\frac{22}{110}$

D. $\frac{85}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q3/5

746

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 85/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q3/5

Write $\frac{4}{10}$ as hundredths.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q4/5

747

QUESTION ID

S19-B2-Q4

TYPE

Short answer

CORRECT ANSWER

40/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q4/5

True or false?

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q5/5

748

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 5.NBT.B.7

BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

True or false?

$$8.7 = 8.70$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q1/5

750

QUESTION ID

S19-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q1/5

Write 15.3 with two decimal places.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q2/5

751

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q2/5

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q3/5

752

QUESTION ID

S19-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q3/5

True or false?

$$6.4 = 6.04$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q4/5

753

QUESTION ID

S19-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q4/5

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q5/5

754

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

12.90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • 5.NBT.B.7

FORWARD 1

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.47 + 9.605$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q1/5

756

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

138.075

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$54.2 + 18.376$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q2/5

757

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

72.576

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q2/5

757

Add.

$$0.952 + 3.48 + 11.6$$

- A. 16.132
- B. 4.53
- C. 15.032
- D. 16.032

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q3/5

758

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 16.032

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q3/5

Add.

$$99.6 + 0.485$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q4/5

759

QUESTION ID

S19-F1-Q4

TYPE

Short answer

CORRECT ANSWER

100.085

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q4/5

True or false?

$$7.3 + 2.145 = 2.875$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q5/5

760

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 5.NBT.B.7

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q1/5

762

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-7.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q1/5

Find the value.

$$3.2 + (-8.45)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q2/5

763

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-5.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q2/5

Find the value.

$$-6.3 + (-2.85)$$

- A. 9.15
- B. -9.15
- C. -3.45
- D. 3.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q3/5

764

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -9.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q3/5

Find the value.

$$-4.6 + 10.25$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q4/5

765

QUESTION ID

S19-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q4/5

True or false?

$$-1.5 + 1.5 = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q5/5

766

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 19 • FORWARD 2 • Q5/5

Subtract decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers

4.NBT.B.4

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths

4.NF.B.3.c

FORWARD 1

Subtract multi-digit decimals

6.NS.B.3

FORWARD 2

Subtract rational numbers

7.NS.A.1.d

SET 61 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Subtract.

$$15.3 - 6.48$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q1/5

769

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

8.82

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$20 - 7.35$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q2/5

770

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

12.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q2/5

770

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q3/5

771

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

\$11.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q3/5

Subtract.

$$9.04 - 3.7$$

- A.** 6.34
- B.** 5.34
- C.** 8.67
- D.** 5.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q4/5

772

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5.34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q4/5

True or false?

$$6.5 - 2.75 = 3.75$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q5/5

773

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 61 • MAIN • Q5/5

SET 61 • 5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$1,530 - 648$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q1/5

775

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

882

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q1/5

775

$$2,000 - 735 = ?$$

- A.** 2,735
- B.** 1,275
- C.** 1,265
- D.** 1,365

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q2/5

776

QUESTION ID

S61-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 1,265

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q2/5

Subtract.

$$1,850 - 675$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q3/5

777

QUESTION ID

S61-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1,175

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q3/5

True or false?

$$5,000 - 1,234 = 3,766$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q4/5

778

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$4,206 - 1,519$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q5/5

779

QUESTION ID

S61-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2,687

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 61 • BACKWARD 1 • Q5/5

779

SET 61 • 5.NBT.B.7

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$5\frac{3}{10} - 2\frac{6}{10}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q1/5

781

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2 7/10

GRADING NOTE 27/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q1/5

True or false?

$$4 - \frac{35}{100} = 3\frac{65}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q2/5

782

QUESTION ID

S61-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q2/5

$$6\frac{2}{10} - 1\frac{9}{10} = ?$$

A. $5\frac{7}{10}$

B. $4\frac{7}{10}$

C. $4\frac{3}{10}$

D. $5\frac{3}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q3/5

783

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4 3/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q3/5

Subtract.

$$3\frac{40}{100} - 1\frac{25}{100}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q4/5

784

QUESTION ID

S61-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2 15/100

GRADING NOTE 2 3/20 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q4/5

True or false?

$$2\frac{1}{10} - \frac{5}{10} = 2\frac{4}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q5/5

785

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 5.NBT.B.7

FORWARD 1

Subtract multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$$54.2 - 18.376$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q1/5

787

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

35.824

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$100 - 0.485$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q2/5

788

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

99.515

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q2/5

788

True or false?

$$7.3 - 2.145 = 5.245$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q3/5

789

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$12.06 - 4.875$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q4/5

790

QUESTION ID

S61-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7.185

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q4/5

790

Subtract.

$$30 - 0.009$$

- A.** 29.991
- B.** 30.009
- C.** 29.001
- D.** 29.91

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q5/5

791

QUESTION ID

S61-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 29.991

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 61 • FORWARD 1 • Q5/5

SET 61 • 5.NBT.B.7

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$3.2 - 8.45$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q1/5

793

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-5.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q1/5

Find the value.

$$-4.6 - (-10.25)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q2/5

794

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q2/5

Find the value.

$$-2.75 - 4.5$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q3/5

795

QUESTION ID

S61-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-7.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q3/5

Find the value.

$$1.2 - (-3.65)$$

- A. 2.45
- B. -4.85
- C. -2.45
- D. 4.85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q4/5

796

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4.85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q4/5

True or false?

$$-8.1 - (-8.1) = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q5/5

797

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 61 • FORWARD 2 • Q5/5

Use place-value models to add and subtract decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Add tenths and hundredths	4.NF.C.5
BACKWARD 2	Rename a unit as ten of the next smaller unit	4.NBT.A.1
FORWARD 1	Add and subtract multi-digit decimals with the standard algorithm	6.NS.B.3
FORWARD 2	Add and subtract rational numbers	7.NS.A.1.d

SET 62 • 5.NBT.B.7

MAIN

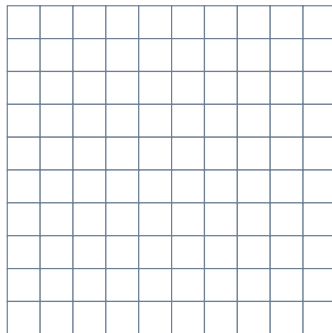
Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show

$$0.4 + 0.25.$$

Then write the sum.

Sum: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q1/5

800

QUESTION ID

S62-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: 65 of the 100 squares shaded: 40 squares (4 columns) for 0.4 and 25 more for 0.25

Written: 0.65

GRADING NOTE Grade both: 65 squares shaded, and the sum 0.65.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q1/5

Explain why you line up the decimal points to add
 $3.6 + 2.45$.

Sum: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q2/5

801

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

6.05

GRADING NOTE Explanation: lining up the decimal points lines up digits with the same place value (ones with ones, tenths with tenths, hundredths with hundredths), so like units are added; 3.6 can be written 3.60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q2/5

Which describes a place-value model for $1.3 - 0.45$?

- A. Add 4 tenths and 5 hundredths to 1 one and 3 tenths.
- B. Start with 1 one and 3 tenths, trade 1 tenth for 10 hundredths, then take away 4 tenths and 5 hundredths.
- C. Take away 1 one and 3 tenths from 4 tenths and 5 hundredths.
- D. Take away 4 ones and 5 tenths from 1 one and 3 tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q3/5

802

QUESTION ID

S62-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Start with 1 one and 3 tenths, trade 1 tenth for 10 hundredths, then take away 4 tenths and 5 hundredths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q3/5

True or false?

To find $0.7 - 0.25$, you can rename 0.7 as 70 hundredths and subtract 25 hundredths to get 45 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q4/5

803

QUESTION ID

S62-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q4/5

Raj added $4.2 + 1.35$ and got 1.77.

Find the correct sum and explain his mistake.

Correct sum: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q5/5

804

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

5.55

GRADING NOTE Explanation: he lined up the last digits instead of the decimal points, so he added as if 4.2 were 0.42. Lining up the decimal points gives $4.20 + 1.35 = 5.55$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 62 • MAIN • Q5/5

SET 62 • 5.NBT.B.7

BACKWARD 1

Add tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{4}{10} + \frac{25}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q1/5

806

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

65/100**GRADING NOTE** 0.65 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q1/5

True or false?

$$\frac{3}{10} = \frac{30}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q2/5

807

QUESTION ID

S62-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q2/5

$$\frac{8}{10} + \frac{7}{100} = ?$$

A. $\frac{87}{100}$

B. $\frac{15}{100}$

C. $\frac{15}{110}$

D. $\frac{87}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q3/5

808

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 87/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q3/5

Write $\frac{6}{10}$ as hundredths.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q4/5

809

QUESTION ID

S62-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q4/5

True or false?

$$\frac{2}{10} + \frac{9}{100} = \frac{11}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q5/5

810

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 5.NBT.B.7

BACKWARD 2

Rename a unit as ten of the next smaller unit • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

How many tens are equal to 1 hundred?

Tens: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q1/5

812

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q1/5

True or false?

3 hundreds 4 tens can be renamed as 2 hundreds 14 tens.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q2/5

813

QUESTION ID

S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q2/5

Which shows 52 renamed so that you can take away 8 ones?

- A. 6 tens 2 ones
- B. 4 tens 12 ones
- C. 4 tens 2 ones
- D. 5 tens 12 ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q3/5

814

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4 tens 12 ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q3/5

Rename 4 thousands as hundreds.

Hundreds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q4/5

815

QUESTION ID

S62-B2-Q4

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q4/5

True or false?

7 tens 3 ones = 6 tens 3 ones

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q5/5

816

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 5.NBT.B.7

FORWARD 1

Add and subtract multi-digit decimals with the standard algorithm • Grade 6 •
The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$36.075 + 8.9$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q1/5

818

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

44.975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q1/5

818

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$41.2 - 7.865$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q2/5

819

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

33.335

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q2/5

819

Add.

$$0.68 + 4.795$$

- A. 4.863
- B. 11.475
- C. 5.375
- D. 5.475

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q3/5

820

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5.475

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q3/5

Subtract.

$$9 - 2.468$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q4/5

821

QUESTION ID

S62-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6.532

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q4/5

True or false?

$$12.5 + 0.375 = 12.875$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q5/5

822

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • FORWARD 1 • Q5/5

SET 62 • 5.NBT.B.7

FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-3.25 + 7.6$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q1/5

824

QUESTION ID

S62-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q1/5

Find the value.

$$0.8 - 2.35$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q2/5

825

QUESTION ID

S62-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q2/5

Find the value.

$$-1.75 + (-0.6)$$

- A. 2.35
- B. 1.15
- C. -2.35
- D. -1.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q3/5

826

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. -2.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q3/5

True or false?

$$5.5 - 7.25 = -1.75$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q4/5

827

QUESTION ID

S62-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$-9.4 - (-2.65)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q5/5

828

QUESTION ID

S62-F2-Q5

TYPE

Short answer

CORRECT ANSWER

-6.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 62 • FORWARD 2 • Q5/5

828

Multiply decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiplying whole numbers	4.NBT.B.5
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
BACKWARD 4	Multiply a whole number by tenths or hundredths	4.NF.B.4.b
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 20 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.4 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q1/5

831

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

20.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q1/5

831

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.7 \times 0.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q2/5

832

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q2/5

832

Multiply.

$$2.5 \times 1.2$$

- A.** 3
- B.** 30
- C.** 3.7
- D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q3/5

833

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.06 \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q4/5

834

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q4/5

834

A pen costs \$1.25.

How much do 7 pens cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q5/5

835

QUESTION ID

S20-M-Q5

TYPE

Short answer

CORRECT ANSWER

\$8.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • MAIN • Q5/5

SET 20 • 5.NBT.B.7

BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$34 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q1/5

837

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

204

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q2/5

838

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q2/5

True or false?

$$125 \times 7 = 785$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q3/5

839

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q3/5

$$18 \times 4 = ?$$

- A.** 72
- B.** 82
- C.** 48
- D.** 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q4/5

840

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$106 \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q5/5

841

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

848

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q1/5

843

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q1/5

True or false?

0.4 is 4 tenths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q2/5

844

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q2/5

Which decimal is 204 tenths?

- A. 204
- B. 20.4
- C. 2.04
- D. 0.204

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q3/5

845

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q3/5

How many hundredths are in 0.48?

Hundredths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q4/5

846

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q4/5

True or false?

0.06 is 6 tenths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q5/5

847

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 5.NBT.B.7

BACKWARD 4

Multiply a whole number by tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$4 \times \frac{3}{10}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q1/5

849

QUESTION ID

S20-B4-Q1

TYPE

Short answer

CORRECT ANSWER

12/10

GRADING NOTE 1 2/10, 6/5, and 1.2 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q1/5

True or false?

$$6 \times \frac{5}{100} = \frac{30}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q2/5

850

QUESTION ID

S20-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q2/5

$$3 \times \frac{7}{10} = ?$$

A. $\frac{7}{30}$

B. $\frac{21}{10}$

C. $\frac{10}{10}$

D. $\frac{21}{30}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q3/5

851

QUESTION ID

S20-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 21/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q3/5

A cup holds $\frac{3}{10}$ liter.

How many liters do 5 cups hold? Write a fraction.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q4/5

852

QUESTION ID

S20-B4-Q4

TYPE

Short answer

CORRECT ANSWER

15/10

GRADING NOTE 1 $\frac{5}{10}$, $\frac{3}{2}$, 1 $\frac{1}{2}$, and 1.5 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q4/5

True or false?

$$8 \times \frac{2}{100} = \frac{16}{800}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q5/5

853

QUESTION ID

S20-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 20 • BACKWARD 4 • Q5/5

SET 20 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.45 \times 0.6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q1/5

855

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q1/5

855

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.08 \times 2.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q2/5

856

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

30.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q2/5

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q3/5

857

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$7.2 \times 1.15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q4/5

858

QUESTION ID

S20-F1-Q4

TYPE

Short answer

CORRECT ANSWER

8.28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q4/5

858

True or false?

$$0.03 \times 0.03 = 0.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q5/5

859

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-3.4 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q1/5

861

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-20.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q1/5

Find the product.

$$(-0.7) \times (-0.5)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q2/5

862

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q2/5

Find the product.

$$2.5 \times (-1.2)$$

- A. -0.3
- B. -30
- C. 3
- D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q3/5

863

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q3/5

Find the product.

$$(-0.06) \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q4/5

864

QUESTION ID

S20-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-0.48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q4/5

True or false?

$$(-1.25) \times (-4) = -5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q5/5

865

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 20 • FORWARD 2 • Q5/5

Use models and place value to multiply decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiply a fraction in tenths or hundredths by a whole number	4.NF.B.4.b
BACKWARD 2	Multiplication as equal groups	3.OA.A.1
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 63 • 5.NBT.B.7

MAIN

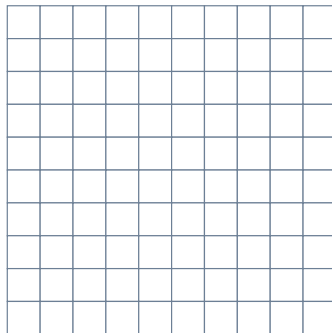
Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show

3×0.15 .

Then write the product.

Product: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q1/5

868

QUESTION ID

S63-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: 45 of the
100 squares shaded,
as three groups of
15 squares**

Written: 0.45

GRADING NOTE Grade both: three groups of 15 hundredths (45 squares) shaded, and the product 0.45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q1/5

On a hundredths grid, 0.3×0.4 is where 3 tenths (3 columns) and 4 tenths (4 rows) overlap.

Explain why the product is 0.12.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q2/5

869

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

The overlap is 3 columns by 4 rows of small squares, which is 12 squares; each small square is one hundredth, so $0.3 \times 0.4 = 12$ hundredths = 0.12.

GRADING NOTE Must say that tenths times tenths gives hundredths (each small square is $1/100$) and that $3 \times 4 = 12$ squares overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q2/5

Which explains why $4 \times 0.25 = 1$?

- A. 4 groups of 25 hundredths is 100 hundredths, which is 1 whole.
- B. $4 + 0.25$ is about 4, so the product is about 1.
- C. 0.25 has two decimal places, so the product is 0.01.
- D. $4 \times 25 = 100$, so the product is 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q3/5

870

QUESTION ID

S63-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 groups of 25 hundredths is 100 hundredths, which is 1 whole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q3/5

True or false?

$2 \times 0.35 = 0.70$ because 2 groups of 35 hundredths is 70 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q4/5

871

QUESTION ID

S63-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q4/5

Lia says $0.5 \times 0.6 = 3.0$.

Give the correct product and use place value to explain her mistake.

Correct product: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q5/5

872

QUESTION ID

S63-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 5 tenths $\times$ 6 tenths = 30 hundredths = 0.30, because tenths times tenths are hundredths. She counted the product as tenths. 0.30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 63 • MAIN • Q5/5

SET 63 • 5.NBT.B.7

BACKWARD 1

Multiply a fraction in tenths or hundredths by a whole number • Grade 4 •
Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$3 \times \frac{15}{100}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q1/5

874

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

45/100

GRADING NOTE 0.45 and 9/20 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q1/5

True or false?

$$2 \times \frac{35}{100} = \frac{70}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q2/5

875

QUESTION ID

S63-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q2/5

$$4 \times \frac{6}{10} = ?$$

A. $\frac{6}{40}$

B. $\frac{24}{40}$

C. $\frac{24}{10}$

D. $\frac{10}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q3/5

876

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 24/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q3/5

Write $7 \times \frac{2}{10}$ as one fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q4/5

877

QUESTION ID

S63-B1-Q4

TYPE

Short answer

CORRECT ANSWER

14/10

GRADING NOTE 1 $\frac{4}{10}$, $\frac{7}{5}$, and 1.4 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q4/5

True or false?

$$6 \times \frac{4}{100} = \frac{24}{600}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q5/5

878

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 5.NBT.B.7

BACKWARD 2

Multiplication as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

There are 3 groups with 15 counters in each group.

How many counters are there in all?

Counters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q1/5

880

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q1/5

True or false?

4×25 can mean 4 groups of 25.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q2/5

881

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q2/5

Which can 2×35 mean?

- A. 2 more than 35
- B. 2 groups of 3 and 5
- C. 35 take away 2
- D. 2 groups of 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q3/5

882

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 groups of 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q3/5

Write a multiplication equation for 6 groups of 4.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q4/5

883

QUESTION ID

S63-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6 × 4 = 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q4/5

True or false?

3×8 means 3 more than 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q5/5

884

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.24 \times 0.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q1/5

886

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.6 \times 0.15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q2/5

887

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q2/5

887

Multiply.

$$1.6 \times 0.45$$

- A.** 0.072
- B.** 0.72
- C.** 2.05
- D.** 7.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q3/5

888

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.08 \times 0.7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q4/5

889

QUESTION ID

S63-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.056

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q4/5

True or false?

$$2.5 \times 0.04 = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q5/5

890

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • FORWARD 1 • Q5/5

SET 63 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-0.3) \times 0.4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q1/5

892

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q1/5

Find the product.

$$(-2.5) \times (-0.6)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q2/5

893

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q2/5

Find the product.

$$3 \times (-0.15)$$

- A. -0.45
- B. 0.45
- C. -0.045
- D. -4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q3/5

894

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q3/5

True or false?

$$(-0.2) \times (-0.2) \times (-0.2) = -0.008$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q4/5

895

QUESTION ID

S63-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q4/5

Find the product.

$$-1.2 \times 0.05$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q5/5

896

QUESTION ID

S63-F2-Q5

TYPE

Short answer

CORRECT ANSWER

−0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 63 • FORWARD 2 • Q5/5

Divide decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Dividing whole numbers	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 4	Rename decimals as tenths or hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 21 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$8.4 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q1/5

899

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q1/5

899

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$3.6 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q2/5

900

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q2/5

900

Divide.

$$0.45 \div 0.05$$

- A.** 0.09
- B.** 0.9
- C.** 90
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q3/5

901

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.5 \div 0.3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q4/5

902

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q4/5

902

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q5/5

903

QUESTION ID

S21-M-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 21 • MAIN • Q5/5

SET 21 • 5.NBT.B.7

BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$84 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q1/5

905

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q1/5

905

Divide.

$$360 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q2/5

906

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$$450 \div 5 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q3/5

907

QUESTION ID

S21-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q3/5

$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q4/5

908

QUESTION ID

S21-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 250

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q4/5

Divide.

$$48 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q5/5

909

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 5.NBT.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 6$ by thinking $6 \times \square = 36$.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q1/5

911

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q1/5

True or false?

$7 \times 9 = 63$, so $63 \div 7 = 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q2/5

912

QUESTION ID

S21-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q2/5

Which fact helps you find $56 \div 8$?

- A. $56 \times 8 = 448$
- B. $8 \times 7 = 56$
- C. $8 + 48 = 56$
- D. $8 - 7 = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q3/5

913

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $8 \times 7 = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q3/5

What number goes in the box?

$$\square \times 5 = 45$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q4/5

914

QUESTION ID

S21-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q5/5

915

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 5.NBT.B.7

BACKWARD 4

Rename decimals as tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 3.6?

Tenths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q1/5

917

QUESTION ID

S21-B4-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q1/5

True or false?

0.45 is 45 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q2/5

918

QUESTION ID

S21-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q2/5

How many hundredths are in 0.05?

- A.** 500
- B.** 0.5
- C.** 5
- D.** 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q3/5

919

QUESTION ID

S21-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q3/5

How many tenths are in 0.6?

Tenths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q4/5

920

QUESTION ID

S21-B4-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q4/5

True or false?

7.5 is 75 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q5/5

921

QUESTION ID

S21-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 21 • BACKWARD 4 • Q5/5

SET 21 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q1/5

923

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$9.072 \div 2.4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q2/5

924

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q2/5

924

Divide.

$$0.081 \div 0.09$$

- A.** 90
- B.** 0.09
- C.** 0.9
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q3/5

925

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$52.5 \div 1.25$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q4/5

926

QUESTION ID

S21-F1-Q4

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q4/5

926

True or false?

$$1.8 \div 0.006 = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q5/5

927

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q1/5

929

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-2.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q1/5

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q2/5

930

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q2/5

Find the quotient.

$$0.45 \div (-0.05)$$

- A. -9
- B. -0.9
- C. 0.9
- D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q3/5

931

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q3/5

Find the quotient.

$$-7.5 \div 0.3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q4/5

932

QUESTION ID

S21-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q4/5

True or false?

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q5/5

933

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • FORWARD 2 • Q5/5

Use models and place value to divide decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Division as the number of equal groups	3.OA.A.2
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 64 • 5.NBT.B.7

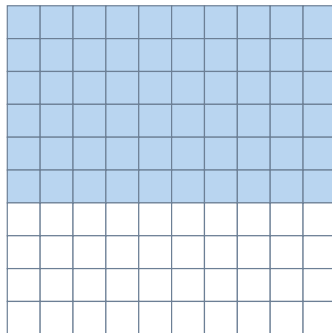
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.6 shaded.

Circle groups of 0.15 to find $0.6 \div 0.15$. Then write the quotient.

Quotient: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q1/5

936

QUESTION ID

S64-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The 60 shaded squares circled as 4 groups of 15 squares

Written: 4

GRADING NOTE Grade both: four groups of 15 hundredths circled, and the quotient 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q1/5

Which explains why $1.8 \div 0.3 = 6$?

- A. 0.3 has one decimal place, so the quotient is 0.6.
- B. There are 6 groups of 1.8 in 0.3.
- C. $1.8 - 0.3 = 6$
- D. There are 6 groups of 3 tenths in 18 tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q2/5

937

QUESTION ID

S64-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. There are 6 groups of 3 tenths in 18 tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q2/5

Explain how to find $2.4 \div 4$ by thinking of 2.4 as 24 tenths.

Quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q3/5

938

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

0.6

GRADING NOTE $24 \text{ tenths} \div 4 = 6 \text{ tenths} = 0.6$. The explanation must use the unit (tenths), not only "move the decimal point".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q3/5

True or false?

$4.5 \div 0.9 = 5$ because there are 5 groups of 9 tenths in 45 tenths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q4/5

939

QUESTION ID

S64-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q4/5

Sam says $0.8 \div 0.2 = 0.4$.

Give the correct quotient and use a model or place value to explain why he is wrong.

Correct quotient: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q5/5

940

QUESTION ID

S64-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE 0.8 is 8 tenths and 0.2 is 2 tenths; there are 4 groups of 2 tenths in 8 tenths, so the quotient is 4. Accept a described drawing or a units explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 64 • MAIN • Q5/5

SET 64 • 5.NBT.B.7

BACKWARD 1

Division as the number of equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

18 counters are put into groups of 3.

How many groups are there?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q1/5

942

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q1/5

True or false?

$24 \div 4$ can mean "how many groups of 4 are in 24?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q2/5

943

QUESTION ID

S64-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q2/5

Which question does $35 \div 5$ answer?

- A. What is 35 take away 5?
- B. What is 35 more than 5?
- C. How many groups of 5 are in 35?
- D. How many are in 5 groups of 35?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q3/5

944

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**C. How many groups
of 5 are in 35?**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q3/5

There are 28 apples. Each bag holds 7 apples.

How many bags are needed?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q4/5

945

QUESTION ID

S64-B1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q4/5

True or false?

$12 \div 3$ means 12 groups of 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q5/5

946

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 1.8?

Tenths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q1/5

948

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q1/5

True or false?

0.15 is 15 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q2/5

949

QUESTION ID

S64-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q2/5

How many hundredths are in 0.6?

- A.** 60
- B.** 0.6
- C.** 600
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q3/5

950

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q3/5

Write 45 tenths as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q4/5

951

QUESTION ID

S64-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q4/5

True or false?

2.4 is 24 hundredths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q5/5

952

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$3.75 \div 0.25$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q1/5

954

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q1/5

Divide.

$$0.096 \div 0.12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q2/5

955

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q2/5

Divide.

$$8.1 \div 0.09$$

- A.** 9
- B.** 90
- C.** 0.9
- D.** 900

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q3/5

956

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q3/5

Divide.

$$14.4 \div 1.2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q4/5

957

QUESTION ID

S64-F1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q4/5

True or false?

$$0.5 \div 0.02 = 2.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q5/5

958

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 64 • FORWARD 1 • Q5/5

SET 64 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-1.8 \div 0.3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q1/5

960

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q1/5

Find the quotient.

$$(-0.6) \div (-0.15)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q2/5

961

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q2/5

Find the quotient.

$$2.4 \div (-4)$$

- A. 0.6
- B. -6
- C. -0.06
- D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q3/5

962

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q3/5

True or false?

$$(-4.5) \div (-0.9) = -5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q4/5

963

QUESTION ID

S64-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$-0.08 \div 0.2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q5/5

964

QUESTION ID

S64-F2-Q5

TYPE

Short answer

CORRECT ANSWER

-0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 64 • FORWARD 2 • Q5/5

964

Solve real-world problems with decimal operations

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Word problems with money and measurement	4.MD.A.2
BACKWARD 2	Hundredths on a grid	4.NF.C.6
FORWARD 1	Real-world problems with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 22 • 5.NBT.B.7

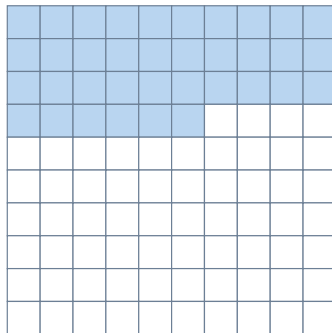
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q1/5

967

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q1/5

Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q2/5

968

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$10.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q2/5

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound **B.** 0.5 pound
C. 2 pounds **D.** 12.5 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q3/5

969

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.5 pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q3/5

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q4/5

970

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.85 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q5/5

971

QUESTION ID

S22-M-Q5

TYPE

Short answer

CORRECT ANSWER

\$5.40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 22 • MAIN • Q5/5

971

SET 22 • 5.NBT.B.7

BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q1/5

973

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

\$3.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q1/5

A board is 120 cm long. It is cut into 4 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q2/5

974

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q2/5

True or false?

Two quarters and three dimes are worth 65 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q3/5

975

QUESTION ID

S22-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q3/5

Ann has \$5. She spends \$2.25.

How much money is left?

- A. \$7.25
- B. \$3.25
- C. \$2.75
- D. \$2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q4/5

976

QUESTION ID

S22-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. \$2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q4/5

A jug holds 2 liters.

How many milliliters is that?

Milliliters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q5/5

977

QUESTION ID

S22-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2,000 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 22 • BACKWARD 1 • Q5/5

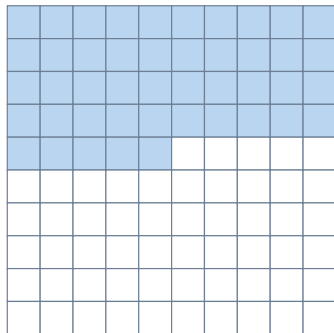
SET 22 • 5.NBT.B.7

BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal does the shaded part of the grid show?

Decimal: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q1/5

979

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

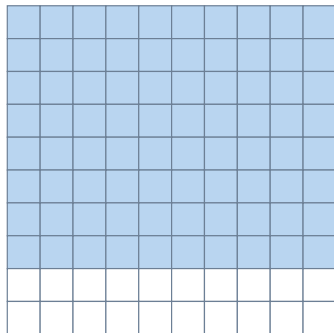
ID S22-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q1/5

True or false?

The shaded part of the grid shows 0.8.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q2/5

980

QUESTION ID

S22-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

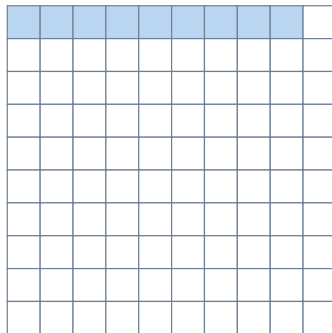
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q2/5

Which decimal does the shaded part of the grid show?

- A. 0.9
- B. 9
- C. 0.009
- D. 0.09



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q3/5

981

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.09

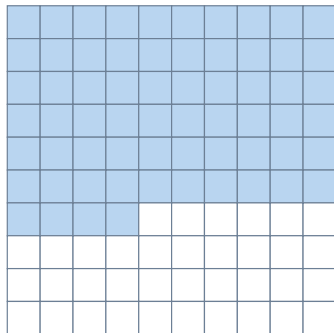
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q3/5

How many more squares must be shaded to fill the whole grid?

Squares: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q4/5

982

QUESTION ID

S22-B2-Q4

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

0.5 of a hundredths grid is 5 squares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q5/5

983

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 5.NBT.B.7

FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q1/5

985

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$46.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q1/5

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q2/5

986

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q2/5

Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A. \$3.00
- B. \$30.00
- C. \$2.70
- D. \$3.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q3/5

987

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. \$3.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q3/5

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q4/5

988

QUESTION ID

S22-F1-Q4

TYPE

Short answer

CORRECT ANSWER

\$23.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q4/5

True or false?

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q5/5

989

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 5.NBT.B.7

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A submarine at -45.5 m rises 12.75 m and then dives 20 m.

What is its final position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q1/5

991

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-52.75 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q1/5

A store's profit was $-\$125.40$ in May and $\$310.15$ in June.

What was the total profit for the two months?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q2/5

992

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\$184.75$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q2/5

The temperature was 4°F . It dropped 1.5°F each hour for 6 hours.

What was the final temperature?

- A.** -5°F **B.** 13°F
C. 5°F **D.** -9°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q3/5

993

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q3/5

An account changes by $-\$2.40$ each week for 5 weeks.

What is the total change?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q4/5

994

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-\$12.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

$$-3.6 \div 0.4 = -0.9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q5/5

995

QUESTION ID

S22-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 22 • FORWARD 2 • Q5/5

Add fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Adding fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Find the least common multiple	6.NS.B.4
FORWARD 2	Add rational numbers	7.NS.A.1.d

SET 23 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q1/5

998

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

11/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q1/5

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q2/5

999

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q2/5

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A. $1\frac{5}{24}$

B. $1\frac{1}{24}$

C. $\frac{8}{24}$

D. $\frac{8}{14}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q3/5

1000

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1 5/24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q3/5

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q4/5

1001

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

11/14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q5/5

1002

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

1 1/3**GRADING NOTE** 16/12 or 4/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q1/5

1004

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q1/5

True or false?

$$\frac{1}{4} = \frac{4}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q2/5

1005

QUESTION ID

S23-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{3}{5}$?

A. $\frac{6}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{10}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q3/5

1006

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q3/5

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q4/5

1007

QUESTION ID

S23-B1-Q4

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$$\frac{2}{7} = \frac{4}{21}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q5/5

1008

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 5.NF.A.1

BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q1/5

1010

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q1/5

True or false?

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q2/5

1011

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q2/5

$$\frac{7}{10} + \frac{6}{10} = ?$$

A. $1\frac{6}{10}$

B. $1\frac{3}{10}$

C. $\frac{13}{20}$

D. $\frac{1}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q3/5

1012

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1 3/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q3/5

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q4/5

1013

QUESTION ID

S23-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q4/5

True or false?

$$\frac{2}{5} + \frac{3}{5} = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q5/5

1014

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five positive multiples of 4.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q1/5

1016

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4, 8, 12, 16, 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q1/5

True or false?

10 is a multiple of both 3 and 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q2/5

1017

QUESTION ID

S23-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q2/5

Which number is a multiple of both 6 and 8?

- A.** 16
- B.** 30
- C.** 24
- D.** 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q3/5

1018

QUESTION ID

S23-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q3/5

Name a positive number that is a multiple of both 5 and 10.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q4/5

1019

QUESTION ID

S23-B3-Q4

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any positive multiple of 10 (10, 20, 30, ...) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q4/5

True or false?

18 is a multiple of both 4 and 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q5/5

1020

QUESTION ID

S23-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • 5.NF.A.1

FORWARD 1

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 10.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q1/5

1022

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q1/5

Find the least common multiple of 8 and 12.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q2/5

1023

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q2/5

What is the least common multiple of 4 and 9?

- A.** 72
- B.** 13
- C.** 18
- D.** 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q3/5

1024

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q3/5

Find the least common multiple of 5 and 15.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q4/5

1025

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q4/5

True or false?

The least common multiple of 6 and 9 is 54.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q5/5

1026

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 5.NF.A.1

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{2}{3} + \frac{1}{4}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q1/5

1028

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−5/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q1/5

Find the value.

$$\frac{3}{5} + \left(-\frac{7}{10}\right)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q2/5

1029

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q2/5

Find the value.

$$-\frac{1}{2} + (-\frac{3}{8})$$

A. $-\frac{7}{8}$

B. $-\frac{4}{10}$

C. $\frac{1}{8}$

D. $-\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q3/5

1030

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q3/5

Find the value.

$$-1\frac{1}{3} + \frac{5}{6}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q4/5

1031

QUESTION ID

S23-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q4/5

True or false?

$$-\frac{3}{4} + \frac{3}{4} = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q5/5

1032

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 23 • FORWARD 2 • Q5/5

Subtract fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Subtracting fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 24 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q1/5

1035

QUESTION ID

S24-M-Q1

TYPE

Short answer

CORRECT ANSWER

7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q1/5

Subtract.

$$\frac{7}{8} - \frac{1}{2}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q2/5

1036

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q2/5

Subtract.

$$\frac{4}{5} - \frac{2}{3}$$

A. $\frac{2}{2}$

B. $\frac{1}{5}$

C. $\frac{6}{15}$

D. $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q3/5

1037

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$\frac{9}{10} - \frac{2}{5}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q4/5

1038

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/2**GRADING NOTE** 5/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q4/5

1038

Subtract.

$$1 - \frac{3}{7}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q5/5

1039

QUESTION ID

S24-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{4} = \frac{?}{12}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q1/5

1041

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q1/5

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q2/5

1042

QUESTION ID

S24-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{2}{3}$?

A. $\frac{10}{15}$

B. $\frac{4}{9}$

C. $\frac{3}{4}$

D. $\frac{2}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q3/5

1043

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q3/5

Find the missing number.

$$\frac{4}{5} = \frac{?}{15}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q4/5

1044

QUESTION ID

S24-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q4/5

True or false?

$$\frac{3}{4} = \frac{9}{16}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q5/5

1045

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 5.NF.A.1

BACKWARD 2

Subtracting fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Subtract.

$$\frac{7}{9} - \frac{2}{9}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q1/5

1047

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q1/5

True or false?

$$\frac{5}{8} - \frac{3}{8} = \frac{2}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q2/5

1048

QUESTION ID

S24-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q2/5

$$\frac{11}{12} - \frac{4}{12} = ?$$

A. $\frac{7}{24}$

B. $\frac{7}{12}$

C. $\frac{15}{12}$

D. $\frac{4}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q3/5

1049

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q3/5

Subtract.

$$1 - \frac{3}{5}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q4/5

1050

QUESTION ID

S24-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q4/5

True or false?

$$\frac{9}{10} - \frac{4}{10} = \frac{5}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q5/5

1051

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four positive multiples of 6.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q1/5

1053

QUESTION ID

S24-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6, 12, 18, 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q1/5

True or false?

15 is a multiple of both 3 and 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q2/5

1054

QUESTION ID

S24-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q2/5

Which number is a multiple of both 4 and 5?

- A.** 15
- B.** 20
- C.** 9
- D.** 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q3/5

1055

QUESTION ID

S24-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q3/5

Name a positive number that is a multiple of both 3 and 8.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q4/5

1056

QUESTION ID

S24-B3-Q4

TYPE

Short answer

CORRECT ANSWER

24

GRADING NOTE Any positive multiple of 24 (24, 48, 72, ...) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q4/5

True or false?

16 is a multiple of both 4 and 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q5/5

1057

QUESTION ID

S24-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{3} = \frac{5}{6}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q1/5

1059

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q1/5

Solve for y.

$$y - \frac{2}{5} = \frac{3}{10}$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q2/5

1060

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q2/5

Solve for n.

$$\frac{3}{4} + n = 1\frac{1}{2}$$

A. $1\frac{1}{4}$

B. $\frac{3}{4}$

C. $2\frac{1}{4}$

D. $\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q3/5

1061

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q3/5

Solve for m.

$$m + \frac{5}{8} = 2$$

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q4/5

1062

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1 3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q4/5

True or false?

The solution of $x - \frac{1}{6} = \frac{2}{3}$ is $x = \frac{5}{6}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q5/5

1063

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$\frac{1}{4} - \frac{5}{6}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q1/5

1065

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q1/5

Find the value.

$$-\frac{2}{3} - \frac{1}{6}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q2/5

1066

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$-\frac{5}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q2/5

Find the value.

$$\frac{1}{2} - (-\frac{3}{4})$$

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. $1\frac{1}{4}$

D. $-1\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q3/5

1067

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q3/5

Find the value.

$$-\frac{3}{5} - (-\frac{1}{2})$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q4/5

1068

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q4/5

True or false?

$$\frac{2}{9} - \frac{5}{9} = \frac{1}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q5/5

1069

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 24 • FORWARD 2 • Q5/5

Add mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Adding mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Mixed numbers and fractions greater than 1	4.NF.B.3.b
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Solve equations of the form $px + q = r$ with fractions	7.EE.B.4.a

SET 25 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$2\frac{1}{3} + 1\frac{1}{2}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q1/5

1072

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

3 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q1/5

Add.

$$4\frac{3}{4} + 2\frac{1}{6}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q2/5

1073

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 11/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q2/5

Add.

$$1\frac{2}{5} + 3\frac{1}{2}$$

A. $4\frac{3}{7}$

B. $4\frac{3}{10}$

C. $5\frac{1}{10}$

D. $4\frac{9}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q3/5

1074

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4 9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q3/5

Add.

$$3\frac{5}{8} + 2\frac{3}{4}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q4/5

1075

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

6 3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q4/5

Add.

$$5\frac{2}{3} + 1\frac{5}{9}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q5/5

1076

QUESTION ID

S25-M-Q5

TYPE

Short answer

CORRECT ANSWER

7 2/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 25 • MAIN • Q5/5

SET 25 • 5.NF.A.1

BACKWARD 1

Adding mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Add.

$$1\frac{2}{5} + 2\frac{1}{5}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q1/5

1078

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q1/5

Add.

$$4\frac{1}{6} + 1\frac{4}{6}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q2/5

1079

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q2/5

True or false?

$$2\frac{3}{4} + 1\frac{3}{4} = 3\frac{6}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q3/5

1080

QUESTION ID

S25-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q3/5

$$2\frac{2}{3} + 1\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $4\frac{1}{3}$

D. $4\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q4/5

1081

QUESTION ID

S25-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 4 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$3\frac{7}{8} + \frac{3}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q5/5

1082

QUESTION ID

S25-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4 2/8**GRADING NOTE** 4 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q5/5

1082

SET 25 • 5.NF.A.1

BACKWARD 2

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{3}{4}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q1/5

1084

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

11/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q1/5

Write $\frac{17}{5}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q2/5

1085

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q2/5

True or false?

$$1\frac{1}{3} = \frac{4}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q3/5

1086

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q3/5

Which fraction is equal to $3\frac{1}{2}$?

A. $\frac{31}{2}$

B. $\frac{7}{2}$

C. $\frac{5}{2}$

D. $\frac{6}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q4/5

1087

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q4/5

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q5/5

1088

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 5.NF.A.1

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{8}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q1/5

1090

QUESTION ID

S25-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q1/5

True or false?

$$\frac{1}{6} = \frac{2}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q2/5

1091

QUESTION ID

S25-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q2/5

Which fraction is equivalent to $\frac{1}{2}$?

A. $\frac{5}{10}$

B. $\frac{2}{3}$

C. $\frac{2}{2}$

D. $\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q3/5

1092

QUESTION ID

S25-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q3/5

Find the missing number.

$$\frac{2}{5} = \frac{?}{10}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q4/5

1093

QUESTION ID

S25-B3-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q4/5

True or false?

$$\frac{5}{6} = \frac{10}{18}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q5/5

1094

QUESTION ID

S25-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 2\frac{1}{2} = 6\frac{1}{4}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q1/5

1096

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q1/5

Solve for y.

$$y - 1\frac{2}{3} = 3\frac{1}{6}$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q2/5

1097

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q2/5

Solve for n.

$$n + 4\frac{3}{8} = 7$$

A. $3\frac{5}{8}$

B. $2\frac{3}{8}$

C. $11\frac{3}{8}$

D. $2\frac{5}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q3/5

1098

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q3/5

A board was $8\frac{1}{2}$ feet long. After a piece was cut off, it was $5\frac{3}{4}$ feet long.

Write and solve an equation to find the length p of the piece.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q4/5

1099

QUESTION ID

S25-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$$8\frac{1}{2} - p = 5\frac{3}{4};$$

$$p = 2\frac{3}{4} \text{ feet}$$

GRADING NOTE Any equivalent equation is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q4/5

True or false?

The solution of $x - 1\frac{1}{4} = 2\frac{1}{2}$ is $x = 1\frac{1}{4}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q5/5

1100

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 5.NF.A.1

FORWARD 2

Solve equations of the form $px + q = r$ with fractions • Grade 7 • Expressions & Equations • 7.EE.B.4.a

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$2x + \frac{1}{2} = 3\frac{1}{2}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q1/5

1102

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{1}{3}x - 1\frac{1}{4} = 2\frac{3}{4}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q2/5

1103

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q2/5

Solve for x.

$$4x - 2\frac{1}{3} = 1$$

A. $1\frac{1}{12}$

B. $\frac{1}{6}$

C. $\frac{5}{6}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q3/5

1104

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q3/5

Jo bought 3 bags of flour that weigh the same and one $1\frac{1}{2}$ -pound bag of sugar. The total weight was $9\frac{3}{4}$ pounds.

How much does each bag of flour weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q4/5

1105

QUESTION ID

S25-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2 $\frac{3}{4}$ pounds

GRADING NOTE $3f + 1\frac{1}{2} = 9\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q4/5

True or false?

The solution of $\frac{1}{2}(x + 3) = 4$ is $x = 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q5/5

1106

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 25 • FORWARD 2 • Q5/5

Subtract mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Subtracting mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Rename a mixed number by decomposing a whole	4.NF.B.3.b
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 65 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$5\frac{3}{4} - 2\frac{1}{6}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q1/5

1109

QUESTION ID

S65-M-Q1

TYPE

Short answer

CORRECT ANSWER

3 7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q1/5

Subtract.

$$4\frac{1}{5} - 1\frac{1}{2}$$

A. $3\frac{3}{10}$

B. $2\frac{7}{10}$

C. $3\frac{7}{10}$

D. $2\frac{3}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q2/5

1110

QUESTION ID

S65-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 2 7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q2/5

Subtract.

$$6 - 2\frac{2}{3}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q3/5

1111

QUESTION ID

S65-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q3/5

Subtract.

$$7\frac{1}{4} - 3\frac{5}{6}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q4/5

1112

QUESTION ID

S65-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 5/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q4/5

True or false?

$$3\frac{1}{2} - 1\frac{2}{3} = 1\frac{5}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q5/5

1113

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 65 • MAIN • Q5/5

SET 65 • 5.NF.A.1

BACKWARD 1

Subtracting mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$4\frac{5}{6} - 1\frac{4}{6}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q1/5

1115

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q1/5

$$5\frac{1}{3} - 2\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $2\frac{2}{3}$

D. $2\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q2/5

1116

QUESTION ID

S65-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 2 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q2/5

Subtract.

$$6\frac{1}{4} - 3\frac{3}{4}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q3/5

1117

QUESTION ID

S65-B1-Q3

TYPE

Short answer

CORRECT ANSWER

2 2/4

GRADING NOTE 2 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q3/5

True or false?

$$7\frac{2}{5} - 4\frac{4}{5} = 3\frac{2}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q4/5

1118

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q4/5

Subtract.

$$5 - 1\frac{3}{8}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q5/5

1119

QUESTION ID

S65-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 5.NF.A.1

BACKWARD 2

Rename a mixed number by decomposing a whole • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Fill in the blank.

$$3\frac{1}{4} = 2\frac{?}{4}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q1/5

1121

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q1/5

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q2/5

1122

QUESTION ID

S65-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q2/5

Which is equal to $4\frac{1}{8}$?

A. $4\frac{9}{8}$

B. $5\frac{1}{8}$

C. $3\frac{1}{8}$

D. $3\frac{9}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q3/5

1123

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 9/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q3/5

Fill in the blank.

$$6 = 5\frac{?}{5}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q4/5

1124

QUESTION ID

S65-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q4/5

True or false?

$$2\frac{3}{10} = 1\frac{3}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q5/5

1125

QUESTION ID

S65-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 3\frac{2}{3} = 7\frac{1}{4}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q1/5

1127

QUESTION ID

S65-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3 7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q1/5

Solve for y.

$$9\frac{1}{2} - y = 4\frac{5}{6}$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q2/5

1128

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q2/5

Solve for n.

$$n + 1\frac{5}{8} = 4\frac{1}{4}$$

A. $2\frac{5}{8}$

B. $2\frac{3}{8}$

C. $3\frac{5}{8}$

D. $5\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q3/5

1129

QUESTION ID

S65-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q3/5

A tank had $10\frac{1}{4}$ gallons of water. After some water was used, $6\frac{2}{3}$ gallons were left.

Write and solve an equation to find the amount w used.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q4/5

1130

QUESTION ID

S65-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$10\frac{1}{4} - w = 6\frac{2}{3};$
 $w = 3\frac{7}{12}$ gallons**

GRADING NOTE Any equivalent equation is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q4/5

True or false?

The solution of $x + 2\frac{3}{4} = 5$ is $x = 2\frac{1}{4}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q5/5

1131

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 65 • FORWARD 1 • Q5/5

SET 65 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$2\frac{1}{4} - 5\frac{1}{2}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q1/5

1133

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q1/5

Find the value.

$$-1\frac{2}{3} - 2\frac{1}{6}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q2/5

1134

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$-3\frac{5}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q2/5

Find the value.

$$3\frac{1}{2} - (-1\frac{3}{4})$$

A. $1\frac{3}{4}$

B. $-1\frac{3}{4}$

C. $5\frac{1}{4}$

D. $-5\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q3/5

1135

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q3/5

Find the value.

$$-4\frac{1}{3} - (-1\frac{5}{6})$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q4/5

1136

QUESTION ID

S65-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q4/5

True or false?

$$1\frac{1}{4} - 3 = -1\frac{3}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q5/5

1137

QUESTION ID

S65-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 65 • FORWARD 2 • Q5/5

Solve word problems by adding and subtracting fractions

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Word problems with like denominators	4.NF.B.3.d
BACKWARD 2	Choosing addition or subtraction	2.OA.A.1
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Write and solve equations of the form $x + p = q$	6.EE.B.7
FORWARD 2	Write and solve equations of the form $px + q = r$	7.EE.B.4.a

SET 26 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Mia walked $\frac{3}{4}$ mile to school and then $\frac{2}{3}$ mile to the park.

How far did she walk in all?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q1/5

1140

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

1 $\frac{5}{12}$ miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q1/5

A jug had $\frac{7}{8}$ gallon of milk. Ben drank $\frac{1}{3}$ gallon.

How much milk is left?

Milk left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q2/5

1141

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{13}{24}$ gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q2/5

Ana used $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar.

How many cups did she use in all?

- A. $4\frac{1}{4}$ cups B. $3\frac{3}{4}$ cups
C. $4\frac{3}{4}$ cups D. $3\frac{1}{4}$ cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q3/5

1142

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 1/4 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q3/5

A ribbon is $5\frac{1}{6}$ feet long. Lea cuts off $2\frac{1}{2}$ feet.

How long is the ribbon now?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q4/5

1143

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

2 $\frac{2}{3}$ feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q4/5

Tom read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of the book on Tuesday.

What fraction of the book does he still have to read?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q5/5

1144

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 26 • MAIN • Q5/5

SET 26 • 5.NF.A.2

BACKWARD 1

Word problems with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

Kim ate $\frac{3}{8}$ of a pizza and Joe ate $\frac{2}{8}$ of the pizza.

What fraction of the pizza did they eat in all?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q1/5

1146

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q1/5

A rope is $\frac{9}{10}$ meter long. $\frac{4}{10}$ meter is cut off.

How much rope is left?

Rope left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q2/5

1147

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5/10 meter

GRADING NOTE 1/2 meter is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q2/5

True or false?

$\frac{2}{6}$ of a garden is corn and $\frac{3}{6}$ is beans. Corn and beans cover $\frac{5}{6}$ of the garden.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q3/5

1148

QUESTION ID

S26-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q3/5

Lu had $\frac{7}{12}$ of a cake. She gave away $\frac{5}{12}$ of the cake.

How much of the cake is left?

A. $\frac{12}{12}$

B. $\frac{2}{12}$

C. $\frac{35}{12}$

D. $\frac{2}{24}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q4/5

1149

QUESTION ID

S26-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q4/5

A tank is $\frac{3}{5}$ full. Water is added to fill another $\frac{1}{5}$ of the tank.

How full is the tank now?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q5/5

1150

QUESTION ID

S26-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 5.NF.A.2

BACKWARD 2

Choosing addition or subtraction • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

A class has 14 boys and 13 girls.

Which operation finds the number of students in all?

- A.** Subtraction
- B.** Division
- C.** Addition
- D.** Multiplication

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q1/5

1152

QUESTION ID

S26-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Addition

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q1/5

True or false?

To find how much is left after some is used, you subtract.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q2/5

1153

QUESTION ID

S26-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q2/5

Sam had 42 marbles. He lost 15 marbles.

How many marbles are left?

Marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q3/5

1154

QUESTION ID

S26-B2-Q3

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q3/5

A shelf holds 35 books. 18 books are on it now.
Which equation finds how many more books fit?

- A. $35 - 18 = \square$
- B. $18 - 35 = \square$
- C. $35 \times 18 = \square$
- D. $35 + 18 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q4/5

1155

QUESTION ID

S26-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $35 - 18 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q4/5

Liz has 26 stickers. She gets 17 more.
How many stickers does she have now?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q5/5

1156

QUESTION ID

S26-B2-Q5

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 5.NF.A.2

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{12}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q1/5

1158

QUESTION ID

S26-B4-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q1/5

True or false?

$$\frac{2}{3} = \frac{8}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q2/5

1159

QUESTION ID

S26-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q2/5

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{15}{16}$

B. $\frac{10}{12}$

C. $\frac{5}{12}$

D. $\frac{6}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q3/5

1160

QUESTION ID

S26-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q3/5

Rewrite $\frac{1}{2}$ and $\frac{1}{3}$ as fractions with the denominator 6.

$$\frac{1}{2} = \underline{\hspace{2cm}}$$

$$\frac{1}{3} = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q4/5

1161

QUESTION ID

S26-B4-Q4

TYPE

Short answer

CORRECT ANSWER

3/6 and 2/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q4/5

True or false?

$$\frac{3}{8} = \frac{6}{24}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q5/5

1162

QUESTION ID

S26-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 26 • BACKWARD 4 • Q5/5

SET 26 • 5.NF.A.2

FORWARD 1

Write and solve equations of the form $x + p = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

After using $\frac{2}{3}$ cup of oil, Ray has $\frac{3}{4}$ cup left.

Write and solve an equation to find how much oil c he started with.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q1/5

1164

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$c - \frac{2}{3} = \frac{3}{4}; c = 1$
 $\frac{5}{12}$ cups**

GRADING NOTE Any equivalent equation is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q1/5

A plant grew $\frac{5}{8}$ inch and is now $4\frac{1}{4}$ inches tall.

Write and solve an equation to find its starting height h .

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q2/5

1165

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$$h + 5/8 = 4 \frac{1}{4};$$

$$h = 3 \frac{5}{8} \text{ inches}$$

GRADING NOTE Any equivalent equation is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q2/5

$\frac{3}{10}$ inch of rain fell on day 1. In all, $\frac{9}{10}$ inch fell on days 1 and 2.

Which equation finds the rain r on day 2?

A. $\frac{3}{10}r = \frac{9}{10}$

B. $r + \frac{9}{10} = \frac{3}{10}$

C. $\frac{3}{10} + r = \frac{9}{10}$

D. $r - \frac{3}{10} = \frac{9}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q3/5

1166

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{3}{10} + r = \frac{9}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q3/5

Solve for x.

$$x + \frac{2}{9} = \frac{2}{3}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q4/5

1167

QUESTION ID

S26-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q4/5

True or false?

If $n - \frac{1}{4} = \frac{5}{8}$, then $n = \frac{7}{8}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q5/5

1168

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 26 • FORWARD 1 • Q5/5

SET 26 • 5.NF.A.2

FORWARD 2

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A recipe uses $\frac{3}{4}$ cup of sugar per batch, plus $\frac{1}{2}$ cup for frosting. Ali used $2\frac{3}{4}$ cups of sugar in all.

How many batches did he make?

Batches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q1/5

1170

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE $(\frac{3}{4})b + \frac{1}{2} = 2\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{2}{3}x + \frac{1}{6} = \frac{5}{6}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q2/5

1171

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q2/5

A 6-foot board is cut into 4 equal pieces, with $\frac{1}{2}$ foot left over.

How long is each piece?

- A. $1\frac{1}{2}$ feet B. $1\frac{5}{8}$ feet
C. $1\frac{3}{8}$ feet D. $5\frac{1}{2}$ feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q3/5

1172

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $1\frac{3}{8}$ feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q3/5

Solve for x.

$$5x - \frac{3}{4} = 4\frac{1}{4}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q4/5

1173

QUESTION ID

S26-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The solution of $2x + \frac{1}{3} = 1$ is $x = \frac{2}{3}$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q5/5

1174

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 26 • FORWARD 2 • Q5/5

Use benchmark fractions to estimate and to check answers

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Comparing fractions to $\frac{1}{2}$ and 1	4.NF.A.2
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Reason about the size of a quotient of fractions	6.NS.A.1
FORWARD 2	Use estimation to judge whether an answer is reasonable	7.EE.B.3

SET 27 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

Ella says $\frac{3}{4} + \frac{1}{6} = \frac{4}{10}$. Her answer must be wrong, because $\frac{4}{10}$ is less than $\frac{3}{4}$, one of the fractions she added.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q1/5

1177

QUESTION ID

S27-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q1/5

Which is the best estimate of $\frac{7}{8} + \frac{5}{12}$?

- A. About $\frac{1}{2}$
- B. About $2\frac{1}{2}$
- C. About $1\frac{1}{2}$
- D. About 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q2/5

1178

QUESTION ID

S27-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. About 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q2/5

Is $\frac{3}{5} + \frac{4}{9}$ more or less than 1?

Use benchmark fractions to explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q3/5

1179

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

More than 1

GRADING NOTE $\frac{3}{5}$ is more than $\frac{1}{2}$ by $\frac{1}{10}$, and $\frac{4}{9}$ is less than $\frac{1}{2}$ by only $\frac{1}{18}$, so the sum is a little more than $\frac{1}{2} + \frac{1}{2} = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q3/5

Jen ran $\frac{9}{10}$ mile and walked $\frac{1}{8}$ mile.

Estimate the total distance to the nearest whole mile.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q4/5

1180

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

1 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q4/5

Leo says $\frac{5}{6} - \frac{1}{3} = \frac{4}{3}$.

Which statement explains why his answer is not reasonable?

- A.** The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.
- B.** The denominators are different.
- C.** $\frac{4}{3}$ is less than 1.
- D.** The answer should be 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q5/5

1181

QUESTION ID

S27-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.2 • Set 27 • MAIN • Q5/5

SET 27 • 5.NF.A.2

BACKWARD 1

Comparing fractions to $\frac{1}{2}$ and 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{5}{8}$ is less than $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q1/5

1183

QUESTION ID

S27-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q1/5

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{3}{8}$

C. $\frac{2}{5}$

D. $\frac{4}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q2/5

1184

QUESTION ID

S27-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 4/7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Is $\frac{5}{9}$ closest to 0, $\frac{1}{2}$, or 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q3/5

1185

QUESTION ID

S27-B1-Q3

TYPE

Short answer

CORRECT ANSWER

 $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q3/5

1185

True or false?

$\frac{6}{10}$ is less than $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q4/5

1186

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q4/5

Is $\frac{11}{12}$ closer to $\frac{1}{2}$ or to 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q5/5

1187

QUESTION ID

S27-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

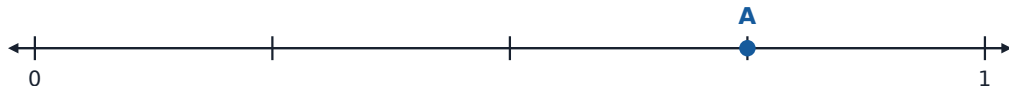
Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 5.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q1/5

1189

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q1/5

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q2/5

1190

QUESTION ID

S27-B2-Q2

TYPE

True or false

CORRECT ANSWER

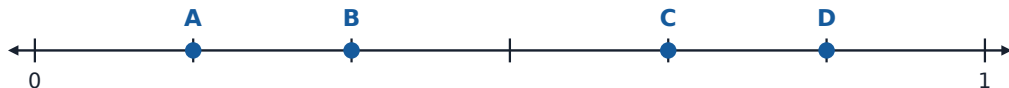
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q2/5

Which point is at $\frac{1}{3}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q3/5

1191

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

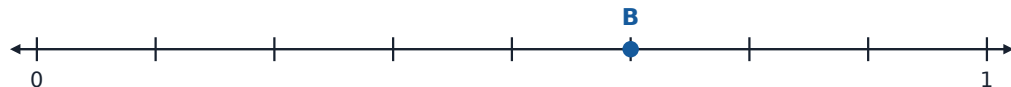
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q3/5

What fraction is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q4/5

1192

QUESTION ID

S27-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q4/5

True or false?

$\frac{7}{8}$ is closer to 1 than to $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q5/5

1193

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 5.NF.A.2

FORWARD 1

Reason about the size of a quotient of fractions • Grade 6 • The Number System • 6.NS.A.1

True or false?

$\frac{3}{4} \div \frac{1}{8}$ is greater than 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q1/5

1195

QUESTION ID

S27-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q1/5

Without dividing, which quotient is less than 1?

A. $\frac{3}{4} \div \frac{1}{2}$

B. $\frac{1}{3} \div \frac{1}{2}$

C. $2 \div \frac{1}{2}$

D. $\frac{5}{6} \div \frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q2/5

1196

QUESTION ID

S27-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $\frac{1}{3} \div \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q2/5

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q3/5

1197

QUESTION ID

S27-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q3/5

Estimate $2\frac{7}{8} \div \frac{1}{2}$ by first rounding $2\frac{7}{8}$ to the nearest whole number.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q4/5

1198

QUESTION ID

S27-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE $3 \div 1/2 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q4/5

True or false?

$\frac{1}{2} \div 4$ is greater than $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q5/5

1199

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 5.NF.A.2

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A board $9\frac{7}{8}$ feet long is cut into 3 equal pieces.

Round the length of the board to the nearest whole foot.
Then use it to estimate the length of each piece.

Rounded length: _____

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q1/5

1201

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 feet; about 3 $\frac{1}{3}$ feet

GRADING NOTE $10 \div 3 = 3\frac{1}{3}$ feet. Both parts are required. The exact length, $3\frac{7}{24}$ feet, is not the estimate asked for.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q1/5

Ann earns \$11.85 per hour and works $7\frac{3}{4}$ hours.

Which is the best estimate of her pay?

- A. \$96
- B. \$84
- C. \$120
- D. \$77

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q2/5

1202

QUESTION ID

S27-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. \$96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q2/5

True or false?

$\frac{3}{4}$ of \$79.95 is about \$60.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q3/5

1203

QUESTION ID

S27-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q3/5

Kai says $4\frac{5}{6} + 3\frac{1}{8}$ is about 9.

Round each mixed number to the nearest whole number and estimate the sum. Is the exact sum more or less than 9?

Your estimate: _____

More or less than 9? _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q4/5

1204

QUESTION ID

S27-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8; less

GRADING NOTE $5 + 3 = 8$. The exact sum, $7\frac{23}{24}$, is less than 9. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q4/5

A recipe needs $2\frac{1}{3}$ cups of flour per batch.

Round $2\frac{1}{3}$ to the nearest whole number. Then estimate the flour needed for 4 batches.

Rounded amount: _____

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q5/5

1205

QUESTION ID

S27-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2 cups; 8 cups

GRADING NOTE $2 \times 4 = 8$ cups. Both parts are required. The exact amount, $9\frac{1}{3}$ cups, is not the estimate asked for.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.3 • Set 27 • FORWARD 2 • Q5/5

Interpret a fraction as division of the numerator by the denominator

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	Division with remainders	4.NBT.B.6
BACKWARD 3	Mixed numbers and fractions greater than 1	4.NF.B.3.b
FORWARD 1	Unit rates written as fractions	6.RP.A.2
FORWARD 2	Write fractions as decimals using division	7.NS.A.2.d

SET 28 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 5$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q1/5

1208

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q1/5

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Each friend gets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q2/5

1209

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/4 pizza

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q2/5

Which division expression is equal to $\frac{7}{4}$?

- A. $7 \div 4$
- B. $7 - 4$
- C. $4 \div 7$
- D. 7×4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q3/5

1210

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $7 \div 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q3/5

Write $\frac{13}{6}$ as a division expression. Then write the quotient as a mixed number.

Division: _____

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q4/5

1211

QUESTION ID

S28-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 ÷ 6; 2 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q4/5

True or false?

When 2 pounds of rice are shared equally among 3 bags, each bag gets $\frac{2}{3}$ pound.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q5/5

1212

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 28 • MAIN • Q5/5

SET 28 • 5.NF.B.3

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ is one of 4 equal parts of a whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q1/5

1214

QUESTION ID

S28-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q1/5

A sandwich is cut into 3 equal parts.

What fraction of the sandwich is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q2/5

1215

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q2/5

What fraction of the bar is shaded?



A. $\frac{8}{3}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q3/5

1216

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q3/5

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q4/5

1217

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q5/5

1218

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 5.NF.B.3

BACKWARD 2

Division with remainders • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide. Write the quotient and the remainder.

$$17 \div 5$$

Quotient and remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q1/5

1220

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 R 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q1/5

True or false?

$$23 \div 4 = 6 \text{ R } 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q2/5

1221

QUESTION ID

S28-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q2/5

$$29 \div 6 = ?$$

- A.** 5 R 5
- B.** 5 R 1
- C.** 4 R 3
- D.** 4 R 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q3/5

1222

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4 R 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q3/5

Divide. Write the quotient and the remainder.

$$50 \div 8$$

Quotient and remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q4/5

1223

QUESTION ID

S28-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6 R 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q4/5

True or false?

$$15 \div 2 = 7 \text{ R } 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q5/5

1224

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 5.NF.B.3

BACKWARD 3

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{9}{4}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q1/5

1226

QUESTION ID

S28-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2 $\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q1/5

Write $1\frac{2}{3}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q2/5

1227

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q2/5

True or false?

$$\frac{11}{5} = 1\frac{1}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q3/5

1228

QUESTION ID

S28-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q3/5

Which mixed number is equal to $\frac{7}{3}$?

A. $2\frac{2}{3}$

B. $3\frac{1}{3}$

C. $2\frac{1}{3}$

D. $1\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q4/5

1229

QUESTION ID

S28-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q4/5

Write $\frac{15}{4}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q5/5

1230

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 28 • BACKWARD 3 • Q5/5

SET 28 • 5.NF.B.3

FORWARD 1

Unit rates written as fractions • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A recipe uses 3 cups of flour for every 4 cups of water.

How many cups of flour are used per cup of water?

Cups of flour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q1/5

1232

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3/4 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q1/5

8 apples cost \$6.

What is the cost per apple?

Cost per apple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q2/5

1233

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$0.75

GRADING NOTE 3/4 dollar is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q2/5

A car goes 150 miles on 4 gallons of gas.

What is the rate in miles per gallon?

- A. $\frac{4}{150}$
- B. 600
- C. 154
- D. 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q3/5

1234

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q3/5

There are 5 cups of juice for every 2 pitchers.
How many cups of juice are there per pitcher?

Cups per pitcher: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q4/5

1235

QUESTION ID

S28-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2 1/2 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q4/5

True or false?

The ratio 3 : 8 has the unit rate $\frac{3}{8}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q5/5

1236

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • 5.NF.B.3

FORWARD 2

Write fractions as decimals using division • Grade 7 • The Number System •
7.NS.A.2.d

Write $5 \div 8$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q1/5

1238

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.625

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q1/5

Use long division to write $\frac{7}{12}$ as a decimal. Show your work.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q2/5

1239

QUESTION ID

S28-F2-Q2

TYPE

Answer with required work

CORRECT ANSWER

0.58333... (the 3 repeats)

GRADING NOTE REQUIRED WORK: Long division of the numerator by the denominator, shown step by step, until the remainder is 0 or the digits start to repeat. A correct decimal without the division earns partial credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q2/5

Which fraction has a decimal form that repeats?

A. $\frac{7}{20}$

B. $\frac{5}{6}$

C. $\frac{9}{25}$

D. $\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q3/5

1240

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q3/5

Write $11 \div 4$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q4/5

1241

QUESTION ID

S28-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q4/5

True or false?

$$\frac{1}{3} = 0.3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q5/5

1242

QUESTION ID

S28-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 28 • FORWARD 2 • Q5/5

Solve sharing problems whose answers are fractions or mixed numbers

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	Whole numbers as fractions	3.NF.A.3.c
FORWARD 1	Solve unit rate problems	6.RP.A.3.b
FORWARD 2	Unit rates with fractional quantities	7.RP.A.1

SET 29 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

6 friends share 10 granola bars equally.

How many bars does each friend get? Write a mixed number.

Bars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q1/5

1245

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

1 $\frac{2}{3}$ bars

GRADING NOTE 1 $\frac{4}{6}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q1/5

A 50-pound bag of rice is shared equally among 8 families.

How many pounds does each family get?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q2/5

1246

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 1/4 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q2/5

9 yards of ribbon are cut into 4 equal pieces.

How long is each piece?

A. $4\frac{1}{2}$ yards

B. $\frac{4}{9}$ yard

C. $2\frac{1}{9}$ yards

D. $2\frac{1}{4}$ yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q3/5

1247

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 $\frac{1}{4}$ yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q3/5

7 liters of juice are poured equally into 3 jugs.

How many liters are in each jug? Between which two whole numbers is your answer?

Liters: _____

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q4/5

1248

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

**2 1/3 liters; between
2 and 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q4/5

True or false?

When 5 cookies are shared equally by 2 children, each child gets $2\frac{1}{2}$ cookies.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q5/5

1249

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 29 • MAIN • Q5/5

SET 29 • 5.NF.B.3

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

24 stickers are shared equally by 6 friends.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q1/5

1251

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q1/5

35 chairs are set out in 5 equal rows.

How many chairs are in each row?

- A. 30
- B. 6
- C. 40
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q2/5

1252

QUESTION ID

S29-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q2/5

True or false?

If 4 boxes hold 28 toys, with the same number in each box, each box holds 7 toys.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q3/5

1253

QUESTION ID

S29-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q3/5

A baker puts 45 muffins into bags of 9.

How many bags does she fill?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q4/5

1254

QUESTION ID

S29-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q4/5

18 pencils are shared equally by 3 students.

How many pencils does each student get?

Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q5/5

1255

QUESTION ID

S29-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 5.NF.B.3

BACKWARD 2

Whole numbers as fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

True or false?

$$\frac{6}{3} = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q1/5

1257

QUESTION ID

S29-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q1/5

Write 4 as a fraction with a denominator of 1.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q2/5

1258

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4/1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q2/5

Which fraction is equal to 3?

A. $\frac{3}{3}$

B. $\frac{2}{6}$

C. $\frac{3}{2}$

D. $\frac{6}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q3/5

1259

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q3/5

How many halves are in 5 wholes?

Halves: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q4/5

1260

QUESTION ID

S29-B2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q4/5

True or false?

$$\frac{8}{8} = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q5/5

1261

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 5.NF.B.3

FORWARD 1

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

ANSWER KEY

Reduced copy of the question page, with its answer

3 pounds of grapes cost \$7.

What is the cost per pound?

Cost per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q1/5

1263

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$2 1/3**GRADING NOTE** About \$2.33 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q1/5

1263

A car travels 100 miles in 3 hours at a constant speed.

At this rate, how far does it travel in 6 hours?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q2/5

1264

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

200 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q2/5

8 cans of soup cost \$6.

At this rate, how much do 12 cans cost?

- A. \$16
- B. \$8
- C. \$9
- D. \$4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q3/5

1265

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. \$9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q3/5

A runner runs 5 miles in 40 minutes at a constant pace.
At this pace, how long does she take to run 8 miles?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q4/5

1266

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

64 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q4/5

True or false?

If 4 pens cost \$3, then 10 pens cost \$7.00.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q5/5

1267

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 29 • FORWARD 1 • Q5/5

SET 29 • 5.NF.B.3

FORWARD 2

Unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail crawls $\frac{2}{3}$ foot in $\frac{1}{4}$ hour.

What is its speed in feet per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q1/5

1269

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2 $\frac{2}{3}$ feet per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q1/5

A recipe uses $\frac{3}{4}$ cup of sugar for every $\frac{1}{2}$ cup of milk.

How many cups of sugar are used per cup of milk?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q2/5

1270

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1 1/2 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q2/5

Jo walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour.

What is her speed?

- A.** 2 miles per hour **B.** 3 miles per hour
C. $\frac{1}{3}$ mile per hour **D.** $\frac{1}{12}$ mile per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q3/5

1271

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q3/5

A pump fills $\frac{5}{8}$ of a tank in $\frac{1}{2}$ hour.

How much of the tank does it fill per hour?

Tanks per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q4/5

1272

QUESTION ID

S29-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1 $\frac{1}{4}$ tanks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q4/5

True or false?

$\frac{1}{2}$ cup of juice for every $\frac{1}{4}$ cup of water is a rate of 2 cups of juice per cup of water.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q5/5

1273

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 29 • FORWARD 2 • Q5/5

Multiply a fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A fraction times a whole number as repeated unit fractions	4.NF.B.4.b
BACKWARD 2	Parts of a whole	3.NF.A.1
BACKWARD 3	Splitting a number into equal parts	3.OA.A.2
FORWARD 1	Find a percent of a quantity	6.RP.A.3.c
FORWARD 2	Multistep percent problems	7.RP.A.3

SET 30 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q1/5

1276

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times \frac{2}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q2/5

1277

QUESTION ID

S30-M-Q2

TYPE

Short answer

CORRECT ANSWER

3 1/3**GRADING NOTE** 10/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q2/5

1277

What is $\frac{2}{5}$ of 20?

- A.** 10
- B.** 8
- C.** 4
- D.** 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q3/5

1278

QUESTION ID

S30-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\frac{5}{8} \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q4/5

1279

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 3/4**GRADING NOTE** 30/8 or 15/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q4/5

1279

True or false?

$\frac{1}{3} \times 15$ means 15 split into 3 equal parts, with 1 of the parts taken.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q5/5

1280

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 30 • MAIN • Q5/5

SET 30 • 5.NF.B.4.a

BACKWARD 1

A fraction times a whole number as repeated unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Write $\frac{4}{5}$ as a whole number times $\frac{1}{5}$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q1/5

1282

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 × 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q1/5

True or false?

$$3 \times \frac{2}{7} = \frac{6}{21}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q2/5

1283

QUESTION ID

S30-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q2/5

$$3 \times \frac{3}{8} = ?$$

A. $3\frac{3}{8}$

B. $\frac{6}{8}$

C. $\frac{9}{8}$

D. $\frac{9}{24}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q3/5

1284

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q3/5

Multiply.

$$4 \times \frac{2}{9}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q4/5

1285

QUESTION ID

S30-B1-Q4

TYPE

Short answer

CORRECT ANSWER

8/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q4/5

True or false?

$$2 \times \frac{3}{5} = \frac{6}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q5/5

1286

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 5.NF.B.4.a

BACKWARD 2

Parts of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ of a bar means 1 of 4 equal parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q1/5

1288

QUESTION ID

S30-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q1/5

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q2/5

1289

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q2/5

What fraction is 2 parts of size $\frac{1}{3}$?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{3}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q3/5

1290

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q3/5

How many $\frac{1}{8}$ s are in $\frac{5}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q4/5

1291

QUESTION ID

S30-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

$\frac{3}{4}$ means 3 parts of size $\frac{1}{4}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q5/5

1292

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 5.NF.B.4.a

BACKWARD 3

Splitting a number into equal parts • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.2

12 is split into 4 equal parts.

How much is in each part?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q1/5

1294

QUESTION ID

S30-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q1/5

True or false?

20 split into 5 equal parts gives 4 in each part.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q2/5

1295

QUESTION ID

S30-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q2/5

Which expression shows 18 split into 3 equal parts?

- A.** $3 \div 18$
- B.** $18 - 3$
- C.** $18 \div 3$
- D.** 18×3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q3/5

1296

QUESTION ID

S30-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $18 \div 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$24 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q4/5

1297

QUESTION ID

S30-B3-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q4/5

1297

ANSWER KEY

Reduced copy of the question page, with its answer

30 is split into 6 equal parts.

How much is in each part?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q5/5

1298

QUESTION ID

S30-B3-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 30 • BACKWARD 3 • Q5/5

1298

SET 30 • 5.NF.B.4.a

FORWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 48?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q1/5

1300

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is 60% of 35?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q2/5

1301

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q2/5

1301

A \$40 game is 15% off.

How many dollars are taken off the price?

- A.** \$6
- B.** \$15
- C.** \$4
- D.** \$34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q3/5

1302

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q3/5

A class has 30 students. 70% of them ride the bus.

How many students ride the bus?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q4/5

1303

QUESTION ID

S30-F1-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q4/5

True or false?

50% of 18 is 36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q5/5

1304

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 30 • FORWARD 1 • Q5/5

SET 30 • 5.NF.B.4.a

FORWARD 2

Multistep percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

ANSWER KEY

Reduced copy of the question page, with its answer

A \$60 coat is 30% off.

What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q1/5

1306

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

\$42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q1/5

1306

A meal costs \$25. A 20% tip is added.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q2/5

1307

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

\$30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q2/5

A \$80 bike has a 6% sales tax added.

What is the total cost?

- A. \$86.00
- B. \$4.80
- C. \$85.60
- D. \$84.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q3/5

1308

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. \$84.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q3/5

A town of 500 people grew by 12%.

What is the new population?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q4/5

1309

QUESTION ID

S30-F2-Q4

TYPE

Short answer

CORRECT ANSWER

560

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q4/5

True or false?

A 25% markup on a \$40 item makes the price \$50.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q5/5

1310

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 30 • FORWARD 2 • Q5/5

Multiply a fraction by a fraction

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A whole number times a fraction	4.NF.B.4.b
BACKWARD 2	Equivalent fractions and simplest form	4.NF.A.1
FORWARD 1	Divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 31 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{7}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q1/5

1313

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

3/14

GRADING NOTE 6/28 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q1/5

Multiply.

$$\frac{3}{4} \times \frac{1}{6}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q2/5

1314

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/8

GRADING NOTE 3/24 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q2/5

Multiply.

$$\frac{5}{6} \times \frac{3}{10}$$

A. $\frac{15}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q3/5

1315

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is $\frac{1}{2}$ of $\frac{3}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q4/5

1316

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

3/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q4/5

1316

True or false?

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q5/5

1317

QUESTION ID

S31-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 31 • MAIN • Q5/5

SET 31 • 5.NF.B.4.a

BACKWARD 1

A whole number times a fraction • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$2 \times \frac{3}{10}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q1/5

1319

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6/10

GRADING NOTE 3/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q1/5

True or false?

$$5 \times \frac{1}{4} = \frac{5}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q2/5

1320

QUESTION ID

S31-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q2/5

$$4 \times \frac{2}{3} = ?$$

A. $\frac{8}{12}$

B. $\frac{8}{3}$

C. $\frac{6}{3}$

D. $\frac{2}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q3/5

1321

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q3/5

Multiply.

$$3 \times \frac{5}{12}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q4/5

1322

QUESTION ID

S31-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15/12

GRADING NOTE 1 1/4 or 5/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q4/5

True or false?

$$6 \times \frac{1}{6} = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q5/5

1323

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 31 • BACKWARD 1 • Q5/5

SET 31 • 5.NF.B.4.a

BACKWARD 2

Equivalent fractions and simplest form • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{6}{8}$ as an equivalent fraction with a denominator of 4.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q1/5

1325

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q1/5

True or false?

$$\frac{10}{15} = \frac{3}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q2/5

1326

QUESTION ID

S31-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q2/5

Which fraction is equivalent to $\frac{4}{12}$?

A. $\frac{4}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q3/5

1327

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q3/5

Find the missing number.

$$\frac{9}{12} = \frac{?}{4}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q4/5

1328

QUESTION ID

S31-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q4/5

True or false?

$$\frac{5}{20} = \frac{1}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q5/5

1329

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 5.NF.B.4.a

FORWARD 1

Divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{4}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q1/5

1331

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q1/5

Divide.

$$\frac{3}{4} \div \frac{1}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q2/5

1332

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4 1/2

GRADING NOTE 9/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q2/5

Divide.

$$\frac{5}{8} \div \frac{1}{4}$$

A. $\frac{5}{32}$

B. $\frac{1}{2}$

C. $2\frac{1}{2}$

D. $1\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q3/5

1333

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q3/5

How many $\frac{3}{8}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q4/5

1334

QUESTION ID

S31-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q5/5

1335

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-\frac{2}{3} \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q1/5

1337

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q1/5

Find the product.

$$\left(-\frac{4}{5}\right) \times \left(-\frac{5}{6}\right)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q2/5

1338

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q2/5

Find the product.

$$\frac{3}{7} \times \left(-\frac{7}{9}\right)$$

A. $-\frac{3}{7}$

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. $\frac{10}{16}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q3/5

1339

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $-\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q3/5

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q4/5

1340

QUESTION ID

S31-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q4/5

True or false?

$$\left(-\frac{2}{5}\right) \times \frac{5}{2} = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q5/5

1341

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 31 • FORWARD 2 • Q5/5

Interpret fraction products with tape diagrams and area models

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	Fractions of a whole in a model	3.NF.A.1
BACKWARD 2	Split a whole number into equal parts	3.OA.A.2
FORWARD 1	Use visual models to divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

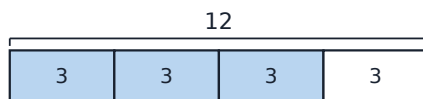
SET 66 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

The tape diagram shows 12 split into 4 equal parts, with 3 parts shaded.

Which product does the shaded part show?



- A. $\frac{3}{4} \times 12$ B. $\frac{4}{3} \times 12$ C. $\frac{1}{4} \times 3$ D. 3×12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q1/5

1344

QUESTION ID

S66-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $\frac{3}{4} \times 12$

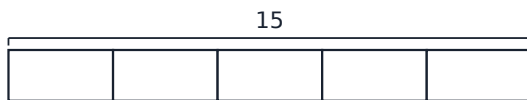
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q1/5

Shade the tape diagram to show $\frac{2}{5} \times 15$.

Then write the product.



Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q2/5

1345

QUESTION ID

S66-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: 2 of the 5
equal parts shaded
(each part is 3)**

Written: 6

GRADING NOTE Grade both: exactly 2 of the 5 parts shaded, and the product 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

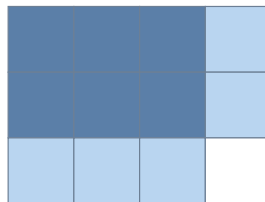
Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q2/5

In the model, the darker part shows a product of two fractions.

Which equation does the model show?

A. $\frac{2}{3} + \frac{3}{4} = \frac{5}{7}$ B. $\frac{2}{3} \times \frac{3}{4} = \frac{5}{7}$

C. $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$ D. $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q3/5

1346

QUESTION ID

S66-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$

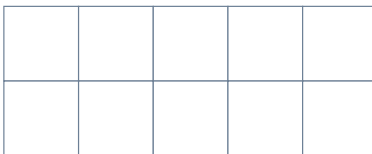
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q3/5

Shade 3 of the 5 columns.
Then shade 1 of the 2 rows.

What product of fractions
does the part shaded twice
show, and what is it?



Product: _____

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q4/5

1347

QUESTION ID

S66-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: 3 of the 5
columns shaded and 1
of the 2 rows shaded; 3
parts are shaded twice**

Written: $\frac{1}{2} \times \frac{3}{5}$; $\frac{3}{10}$

GRADING NOTE Grade both: the shading (3 columns, 1 row, 3 parts overlap) and the written product $\frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$ ($\frac{3}{5} \times \frac{1}{2}$ is also correct).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q4/5

True or false?

In a model of $\frac{3}{4} \times 8$, 8 is split into 4 equal parts and 3 of the parts are taken, so the product is 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q5/5

1348

QUESTION ID

S66-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

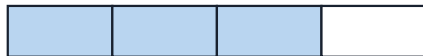
Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 66 • MAIN • Q5/5

SET 66 • 5.NF.B.4.a

BACKWARD 1

Fractions of a whole in a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q1/5

1350

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q1/5

True or false?

$\frac{2}{3}$ of a bar means 2 of 3 equal parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q2/5

1351

QUESTION ID

S66-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q2/5

A bar is split into 6 equal parts and 4 parts are shaded.

What fraction is shaded?

A. $\frac{6}{4}$

B. $\frac{2}{6}$

C. $\frac{4}{6}$

D. $\frac{4}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q3/5

1352

QUESTION ID

S66-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q3/5

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q4/5

1353

QUESTION ID

S66-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q4/5

True or false?

A bar split into 8 equal parts with 3 parts shaded shows $\frac{3}{5}$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q5/5

1354

QUESTION ID

S66-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • 5.NF.B.4.a

BACKWARD 2

Split a whole number into equal parts • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is split into 5 equal parts.

How much is in each part?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q1/5

1356

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q1/5

True or false?

21 split into 3 equal parts gives 7 in each part.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q2/5

1357

QUESTION ID

S66-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q2/5

Which expression shows 32 split into 4 equal parts?

- A. $32 \div 4$
- B. $32 - 4$
- C. $4 \div 32$
- D. 32×4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q3/5

1358

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $32 \div 4$

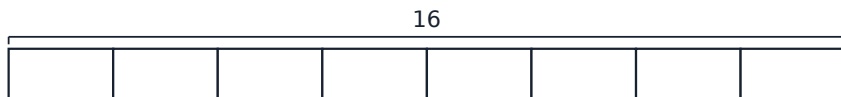
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q3/5

The tape diagram shows 16 split into 8 equal parts.

What number goes in each part?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q4/5

1359

QUESTION ID

S66-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q4/5

True or false?

36 split into 6 equal parts gives 5 in each part.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q5/5

1360

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • 5.NF.B.4.a

FORWARD 1

Use visual models to divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{6}$ s are in $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q1/5

1362

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q1/5

True or false?

$\frac{3}{4} \div \frac{1}{8} = 6$ because there are 6 eighths in $\frac{3}{4}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q2/5

1363

QUESTION ID

S66-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q2/5

A model shows 2 wholes, each cut into thirds.

How many $\frac{1}{3}$ s are in 2?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q3/5

1364

QUESTION ID

S66-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q3/5

Divide.

$$\frac{5}{6} \div \frac{1}{12}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q4/5

1365

QUESTION ID

S66-F1-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q4/5

True or false?

$$\frac{2}{5} \div \frac{1}{10} = \frac{2}{50}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q5/5

1366

QUESTION ID

S66-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 66 • FORWARD 1 • Q5/5

SET 66 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$\frac{3}{4} \times (-8)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q1/5

1368

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q1/5

True or false?

$$-\frac{2}{3} \times \frac{3}{4} = -\frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q2/5

1369

QUESTION ID

S66-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q2/5

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{3}{5}\right)$$

A. $-\frac{3}{10}$

B. $\frac{3}{10}$

C. $-\frac{4}{7}$

D. $\frac{4}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q3/5

1370

QUESTION ID

S66-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q3/5

Find the product.

$$-\frac{2}{5} \times 15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q4/5

1371

QUESTION ID

S66-F2-Q4

TYPE

Short answer

CORRECT ANSWER

−6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q4/5

True or false?

$$(-1) \times \left(-\frac{5}{6}\right) = -\frac{5}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q5/5

1372

QUESTION ID

S66-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 66 • FORWARD 2 • Q5/5

Find the area of a rectangle with fractional side lengths

MAIN

Grade 5 main question

5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares

3.MD.C.7.a

BACKWARD 2

The area formula for rectangles

4.MD.A.3

BACKWARD 4

Multiply a whole number by a fraction

4.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures

6.G.A.1

FORWARD 2

Area of composite figures

7.G.B.6

SET 32 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q1/5

1375

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ square foot

GRADING NOTE 15/30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q1/5

A rectangle is $2\frac{1}{2}$ inches long and $\frac{4}{5}$ inch wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q2/5

1376

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q2/5

A rectangle that is $\frac{2}{3}$ unit by $\frac{3}{4}$ unit is tiled with small rectangles that are each $\frac{1}{3}$ unit by $\frac{1}{4}$ unit.

What is the area of the large rectangle?

- A.** $\frac{6}{12}$ square unit **B.** $\frac{6}{7}$ square unit
C. $\frac{5}{7}$ square unit **D.** $1\frac{5}{12}$ square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q3/5

1377

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6/12 square unit

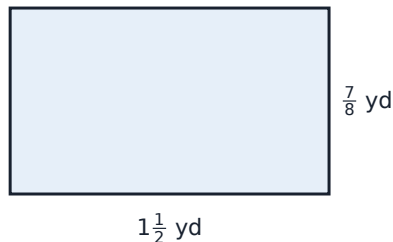
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q3/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q4/5

1378

QUESTION ID

S32-M-Q4

TYPE

Short answer

CORRECT ANSWER

1 5/16 square yards

GRADING NOTE 21/16 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q4/5

True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{2}$ foot has an area of 1 square foot.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q5/5

1379

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 32 • MAIN • Q5/5

SET 32 • 5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares, with no gaps.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q1/5

1381

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q1/5

True or false?

Tiling a 3-by-5 rectangle takes 8 unit squares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q2/5

1382

QUESTION ID

S32-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q2/5

How many unit squares cover a 7-by-2 rectangle?

- A. 72
- B. 9
- C. 14
- D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q3/5

1383

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q3/5

A rectangle has 5 rows of unit squares with 8 squares in each row.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q4/5

1384

QUESTION ID

S32-B1-Q4

TYPE

Short answer

CORRECT ANSWER

40 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q4/5

True or false?

The number of unit squares that cover a rectangle equals its length times its width.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q5/5

1385

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 5.NF.B.4.b

BACKWARD 2

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 5 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q1/5

1387

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q1/5

A rectangle has an area of 54 square feet and a length of 9 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q2/5

1388

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q2/5

True or false?

The formula $A = l \times w$ gives the area of a rectangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q3/5

1389

QUESTION ID

S32-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q3/5

A rectangle is 15 cm long and 4 cm wide.

What is its area?

- A. 60 square cm
- B. 45 square cm
- C. 38 square cm
- D. 19 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q4/5

1390

QUESTION ID

S32-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 60 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q5/5

1391

QUESTION ID

S32-B2-Q5

TYPE

Short answer

CORRECT ANSWER

40 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 5.NF.B.4.b

BACKWARD 4

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

A rectangle is 3 units long and $\frac{2}{5}$ unit wide.

Write its area as a fraction.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q1/5

1393

QUESTION ID

S32-B4-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{6}{5}$ square units

GRADING NOTE 1 $\frac{1}{5}$ square units is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q1/5

True or false?

$$4 \times \frac{3}{8} = \frac{12}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q2/5

1394

QUESTION ID

S32-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q2/5

$$5 \times \frac{2}{3} = ?$$

A. $\frac{10}{3}$

B. $\frac{10}{15}$

C. $\frac{2}{15}$

D. $\frac{7}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q3/5

1395

QUESTION ID

S32-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q3/5

Multiply.

$$6 \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q4/5

1396

QUESTION ID

S32-B4-Q4

TYPE

Short answer

CORRECT ANSWER

18/4

GRADING NOTE 4 1/2 and 9/2 are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q4/5

True or false?

$$2 \times \frac{5}{6} = \frac{10}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q5/5

1397

QUESTION ID

S32-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 32 • BACKWARD 4 • Q5/5

SET 32 • 5.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures • Grade 6 •
Geometry • 6.G.A.1

A triangle has a base of $4\frac{1}{2}$ inches and a height of 2 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q1/5

1399

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4 1/2 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q1/5

A parallelogram has a base of $3\frac{1}{4}$ feet and a height of 4 feet.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q2/5

1400

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q2/5

A triangle has a base of $\frac{3}{4}$ m and a height of $\frac{2}{3}$ m.

What is its area?

- A. $\frac{1}{2}$ square meter B. $\frac{5}{7}$ square meter
C. $\frac{1}{4}$ square meter D. $\frac{6}{7}$ square meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q3/5

1401

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4 square meter

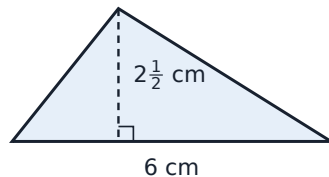
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q3/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q4/5

1402

QUESTION ID

S32-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7 1/2 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q4/5

True or false?

A trapezoid with bases of $2\frac{1}{2}$ m and $3\frac{1}{2}$ m and a height of 2 m has an area of 6 square meters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q5/5

1403

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 5.NF.B.4.b

FORWARD 2

Area of composite figures • Grade 7 • Geometry • 7.G.B.6

A figure is made of a rectangle that is $6\frac{1}{2}$ ft by 4 ft and a triangle with a base of 4 ft and a height of 3 ft.

What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q1/5

1405

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

32 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q1/5

A rectangular garden is 10 m by $7\frac{1}{2}$ m. A square patio with $2\frac{1}{2}$ -m sides is inside the garden.

What is the area of the garden that is not patio?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q2/5

1406

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**68 $\frac{3}{4}$ square
meters**

GRADING NOTE 68.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q2/5

A $\frac{1}{2}$ -by-2 rectangle is cut out of a 5-by-4 rectangle.

What is the area of the part that is left?

- A. 21 square units
- B. 18 square units
- C. 19 square units
- D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q3/5

1407

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q3/5

A kite is made of two triangles. Each triangle has a base of 8 in. and a height of $3\frac{1}{2}$ in.

What is the area of the kite?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q4/5

1408

QUESTION ID

S32-F2-Q4

TYPE

Short answer

CORRECT ANSWER

28 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

A figure made of two $\frac{1}{2}$ -by-3 rectangles that do not overlap has an area of 6 square units.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q5/5

1409

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 32 • FORWARD 2 • Q5/5

Tile a rectangle with unit-fraction squares and connect the tiling to the area

MAIN	Grade 5 main question	5.NF.B.4.b
BACKWARD 1	Measure area by counting unit squares	3.MD.C.6
BACKWARD 2	Unit fractions of a length	3.NF.A.1
FORWARD 1	Pack a prism with unit-fraction cubes to find its volume	6.G.A.2
FORWARD 2	Area of composite figures with fractional measures	7.G.B.6

SET 67 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

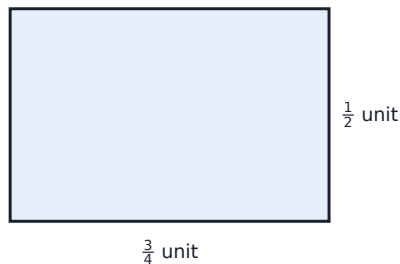
The rectangle is $\frac{3}{4}$ unit long and $\frac{1}{2}$ unit wide.

Draw lines to tile it with squares that are $\frac{1}{4}$ unit on each side.

How many squares fit, and what is the area?

Squares: _____

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q1/5

1412

QUESTION ID

S67-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Lines that split the rectangle into 3 columns and 2 rows of $\frac{1}{4}$ -unit squares (6 squares)

**Written: 6 squares;
 $\frac{6}{16} = \frac{3}{8}$ square unit**

GRADING NOTE Grade both: the tiling (3 by 2 squares, no gaps or overlaps) and the area $\frac{6}{16}$ or $\frac{3}{8}$ square unit. Each square is $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$ square unit, and $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$.

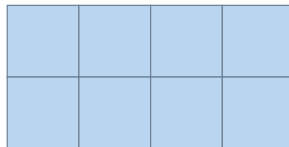
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q1/5

The rectangle is tiled with squares that are $\frac{1}{3}$ unit on each side.

What is its area? Explain how the tiling shows it.



Area: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q2/5

1413

QUESTION ID

S67-M-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{8}{9}$ square unit

GRADING NOTE Each square is $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$ square unit, and there are 8 squares, so the area is $\frac{8}{9}$. This matches $\frac{4}{3} \times \frac{2}{3} = \frac{8}{9}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q2/5

How many squares with $\frac{1}{6}$ -unit sides tile a rectangle that is $\frac{5}{6}$ unit by $\frac{1}{2}$ unit?

- A. 5
- B. 30
- C. 8
- D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q3/5

1414

QUESTION ID

S67-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q3/5

True or false?

Tiling a square with $\frac{2}{3}$ -unit sides takes 4 squares with $\frac{1}{3}$ -unit sides, so its area is $\frac{4}{9}$ square unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q4/5

1415

QUESTION ID

S67-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q4/5

A rectangle is tiled with 3 rows of 4 squares. Each square has $\frac{1}{4}$ -unit sides.

What are the rectangle's side lengths and its area? Show that multiplying the side lengths gives the same area.

Side lengths: _____

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q5/5

1416

QUESTION ID

S67-M-Q5

TYPE

Short answer

CORRECT ANSWER

**1 unit by $\frac{3}{4}$ unit;
 $\frac{3}{4}$ square unit**

GRADING NOTE $4 \times \frac{1}{4} = 1$ and $3 \times \frac{1}{4} = \frac{3}{4}$. The tiling gives $12 \times \frac{1}{16} = \frac{12}{16} = \frac{3}{4}$, and $1 \times \frac{3}{4} = \frac{3}{4}$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 67 • MAIN • Q5/5

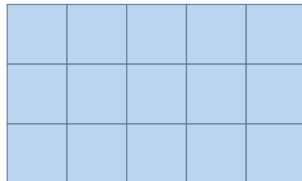
SET 67 • 5.NF.B.4.b

BACKWARD 1

Measure area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

Each small square is 1 square unit.

What is the area of the rectangle?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q1/5

1418

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q1/5

True or false?

A shape covered by 9 unit squares with no gaps or overlaps has an area of 9 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q2/5

1419

QUESTION ID

S67-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q2/5

A rectangle is covered by 2 rows of 6 unit squares.

What is its area?

- A. 8 square units
- B. 6 square units
- C. 12 square units
- D. 16 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q3/5

1420

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 square units

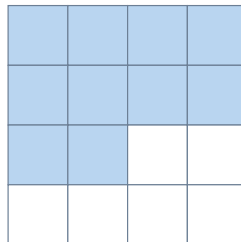
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q3/5

Each small square is 1 square inch.

What is the area of the shaded part?



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q4/5

1421

QUESTION ID

S67-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q4/5

True or false?

If the unit squares overlap, counting them still gives the exact area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q5/5

1422

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • 5.NF.B.4.b

BACKWARD 2

Unit fractions of a length • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

A 1-unit side is split into 3 equal parts.

How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q1/5

1424

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/3 unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q1/5

True or false?

Four lengths of $\frac{1}{4}$ unit make 1 unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q2/5

1425

QUESTION ID

S67-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q2/5

How many lengths of $\frac{1}{6}$ unit make $\frac{5}{6}$ unit?

- A. 11
- B. 5
- C. 6
- D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q3/5

1426

QUESTION ID

S67-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q3/5

A side is made of 3 lengths of $\frac{1}{4}$ unit.

How long is the side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q4/5

1427

QUESTION ID

S67-B2-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q4/5

True or false?

Two lengths of $\frac{1}{8}$ unit make $\frac{2}{16}$ unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q5/5

1428

QUESTION ID

S67-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • 5.NF.B.4.b

FORWARD 1

Pack a prism with unit-fraction cubes to find its volume • Grade 6 • Geometry •
6.G.A.2

A prism is 1 unit by $\frac{1}{2}$ unit by $\frac{1}{2}$ unit.

How many cubes with edges $\frac{1}{2}$ unit long pack it?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q1/5

1430

QUESTION ID

S67-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q1/5

True or false?

A cube with edges $\frac{1}{2}$ unit long has a volume of $\frac{1}{8}$ cubic unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q2/5

1431

QUESTION ID

S67-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q2/5

A box is packed with 24 cubes that each have edges $\frac{1}{2}$ unit long.

What is its volume?

- A.** 12 cubic units **B.** 24 cubic units
C. 3 cubic units **D.** 6 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q3/5

1432

QUESTION ID

S67-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q3/5

Find the volume of a prism that is $\frac{3}{2}$ units by 1 unit by $\frac{1}{2}$ unit.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q4/5

1433

QUESTION ID

S67-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ cubic unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q4/5

True or false?

Packing a 1-by-1-by-1 cube with cubes that have edges $\frac{1}{3}$ unit long takes 9 cubes.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q5/5

1434

QUESTION ID

S67-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 67 • FORWARD 1 • Q5/5

SET 67 • 5.NF.B.4.b

FORWARD 2

Area of composite figures with fractional measures • Grade 7 • Geometry •
7.G.B.6

A $\frac{1}{4}$ -m by $\frac{1}{3}$ -m rectangle is cut out of a $\frac{3}{4}$ -m by $\frac{2}{3}$ -m rectangle.

What area is left?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q1/5

1436

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{5}{12}$ square meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q1/5

True or false?

Two rectangles that do not overlap, $\frac{1}{2}$ ft by 3 ft and $\frac{1}{2}$ ft by 1 ft, have a total area of 2 square feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q2/5

1437

QUESTION ID

S67-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q2/5

A square with sides of $\frac{5}{2}$ units has a square with sides of $\frac{1}{2}$ unit removed.

What is the area of the figure?

- A.** 5 square units **B.** 6 square units
C. 2 square units **D.** $6\frac{1}{4}$ square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q3/5

1438

QUESTION ID

S67-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q3/5

A rug is $2\frac{1}{2}$ ft by 4 ft. A square corner with sides of $1\frac{1}{2}$ ft is cut off.

What is the area of the rug that is left?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q4/5

1439

QUESTION ID

S67-F2-Q4

TYPE

Short answer

CORRECT ANSWER

7 $\frac{3}{4}$ square feet

GRADING NOTE $10 - 2\frac{1}{4} = 7\frac{3}{4}$; 7.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

A triangle with a base of $\frac{3}{4}$ unit and a height of $\frac{2}{3}$ unit has an area of $\frac{1}{2}$ square unit.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q5/5

1440

QUESTION ID

S67-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 67 • FORWARD 2 • Q5/5

Compare the size of a product to the size of one factor without multiplying

MAIN	Grade 5 main question	5.NF.B.5.a
BACKWARD 1	Multiplicative comparison	4.OA.A.2
BACKWARD 2	Comparing fractions to 1	4.NF.A.2
FORWARD 1	Scale quantities in ratio tables	6.RP.A.3.a
FORWARD 2	Use a scale to find lengths in scale drawings	7.G.A.1

SET 33 • 5.NF.B.5.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.a

Without multiplying, how does $\frac{3}{4} \times 20$ compare to 20?

- A. It is less than 20.
- B. It is equal to 20.
- C. It is 4 times 20.
- D. It is greater than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q1/5

1443

QUESTION ID

S33-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. It is less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q1/5

True or false?

$5 \times 3\frac{1}{2}$ is greater than 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q2/5

1444

QUESTION ID

S33-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q2/5

Without multiplying, is $\frac{2}{3} \times 15$ greater than, less than, or equal to 15?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q3/5

1445

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

Less than 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q3/5

Which product is greater than 8?

A. $\frac{9}{8} \times 8$

B. $\frac{1}{2} \times 8$

C. $\frac{7}{8} \times 8$

D. $\frac{8}{8} \times 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q4/5

1446

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $9/8 \times 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q4/5

A photo is 6 inches wide. It is resized to $\frac{5}{4}$ of its width.

Is the new width greater than or less than 6 inches?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q5/5

1447

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Greater than 6
inches**

GRADING NOTE $\frac{5}{4}$ is greater than 1. The new width is $7\frac{1}{2}$ inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.a • Set 33 • MAIN • Q5/5

SET 33 • 5.NF.B.5.a

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.2

True or false?

3×12 is 3 more than 12.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q1/5

1449

QUESTION ID

S33-B1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q1/5

A dog weighs 4 times as much as a 6-pound cat.

How much does the dog weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q2/5

1450

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

24 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q2/5

Which number is 5 times as much as 7?

- A.** 57
- B.** 12
- C.** 2
- D.** 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q3/5

1451

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q3/5

True or false?

2×9 is less than 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q4/5

1452

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q4/5

Fill in the blank.

42 is ____ times as much as 7.

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q5/5

1453

QUESTION ID

S33-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 5.NF.B.5.a

BACKWARD 2

Comparing fractions to 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{4}{5}$ is greater than 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q1/5

1455

QUESTION ID

S33-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q1/5

Which fraction is less than 1?

A. $\frac{6}{7}$

B. $\frac{9}{9}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q2/5

1456

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 6/7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q2/5

Write $>$, $<$, or $=$ to compare.

$$\frac{8}{8} \text{ ____ } 1$$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q3/5

1457

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

$$\frac{3}{5} > 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q4/5

1458

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q4/5

Is $\frac{11}{10}$ greater than, less than, or equal to 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q5/5

1459

QUESTION ID

S33-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Greater than 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 5.NF.B.5.a

FORWARD 1

Scale quantities in ratio tables • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

A recipe uses 2 cups of rice for every 3 cups of water.
How many cups of water are needed for 8 cups of rice?

Cups of water: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q1/5

1461

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q1/5

A ratio table starts with 4 : 10.

Which ratio could also be in the table?

- A. 4 : 5
- B. 2 : 5
- C. 8 : 15
- D. 10 : 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q2/5

1462

QUESTION ID

S33-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 2 : 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q2/5

The table shows equivalent ratios.

Find the missing value.

Red	3	6	?
Blue	5	10	25

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q3/5

1463

QUESTION ID

S33-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q3/5

True or false?

Multiplying both numbers in $6 : 9$ by $\frac{1}{3}$ gives $2 : 3$,
an equivalent ratio.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q4/5

1464

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q4/5

A map shows 5 km for every 2 cm.
How many kilometers do 7 cm show?

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q5/5

1465

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

17.5 km

GRADING NOTE 17 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 33 • FORWARD 1 • Q5/5

SET 33 • 5.NF.B.5.a

FORWARD 2

Use a scale to find lengths in scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 inch = 4 feet. A wall is $3\frac{1}{2}$ inches long in the drawing.

How long is the real wall?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q1/5

1467

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

14 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q1/5

A room is 15 feet long.

How long is it on a drawing with a scale of 1 inch = 3 feet?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q2/5

1468

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q2/5

A drawing is reduced to $\frac{1}{2}$ of its size.

A 10-cm side becomes how long?

- A. 2 cm
- B. 20 cm
- C. 10 cm
- D. 5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q3/5

1469

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q3/5

A drawing that is 4 cm by 6 cm is enlarged by a scale factor of $\frac{3}{2}$.

What are the new dimensions?

Dimensions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q4/5

1470

QUESTION ID

S33-F2-Q4

TYPE

Short answer

CORRECT ANSWER

6 cm by 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q4/5

True or false?

Using a scale factor less than 1 makes a drawing smaller.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q5/5

1471

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 33 • FORWARD 2 • Q5/5

Explain the effect of multiplying by a fraction greater than, less than, or equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 2	Fractions equal to 1	3.NF.A.3.c
BACKWARD 3	Multiply a whole number by a fraction	4.NF.B.4.b
FORWARD 1	Percents greater than and less than 100%	6.RP.A.3.c
FORWARD 2	Use a multiplier for percent increase and decrease	7.RP.A.3

SET 34 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

True or false?

Multiplying 12 by $\frac{7}{6}$ gives a product greater than 12.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q1/5

1474

QUESTION ID

S34-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q1/5

Which statement explains why $\frac{4}{5} \times 30$ is less than 30?

- A. $\frac{4}{5}$ is less than 1.
- B. $\frac{4}{5}$ is greater than 1.
- C. 30 is greater than 1.
- D. The product is a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q2/5

1475

QUESTION ID

S34-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 4/5 is less than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q2/5

Explain why multiplying 24 by $\frac{5}{8}$ gives a product less than 24.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q3/5

1476

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

$\frac{5}{8}$ is less than 1, so $\frac{5}{8} \times 24$ is only part of 24 (5 of 8 equal parts of 24, which is 15).

GRADING NOTE Must say the factor $\frac{5}{8}$ is less than 1, so the product is a fraction of 24 and smaller than 24. Computing 15 alone is not an explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q3/5

Which expression has a value equal to 9?

A. $\frac{6}{5} \times 9$

B. $\frac{5}{6} \times 9$

C. $\frac{1}{5} \times 9$

D. $\frac{5}{5} \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q4/5

1477

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\frac{5}{5} \times 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q4/5

Without multiplying, order the products from least to greatest.

$$\frac{3}{4} \times 10, \frac{4}{3} \times 10, \frac{4}{4} \times 10$$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q5/5

1478

QUESTION ID

S34-M-Q5

TYPE

Short answer

CORRECT ANSWER

**$\frac{3}{4} \times 10, \frac{4}{4} \times 10,$
 $\frac{4}{3} \times 10$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 34 • MAIN • Q5/5

SET 34 • 5.NF.B.5.b

BACKWARD 2

Fractions equal to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

True or false?

$$\frac{4}{4} = 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q1/5

1480

QUESTION ID

S34-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q1/5

Write 1 as a fraction with a denominator of 6.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q2/5

1481

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q2/5

Which fraction is equal to 1?

A. $\frac{0}{8}$

B. $\frac{8}{1}$

C. $\frac{1}{8}$

D. $\frac{8}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q3/5

1482

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q3/5

True or false?

$\frac{6}{6}$ is greater than 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q4/5

1483

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q4/5

How many eighths make 1 whole?

Eighths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q5/5

1484

QUESTION ID

S34-B2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.c • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 5.NF.B.5.b

BACKWARD 3

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

True or false?

$$3 \times \frac{2}{3} = 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q1/5

1486

QUESTION ID

S34-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q1/5

Multiply.

$$4 \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q2/5

1487

QUESTION ID

S34-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 12/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q2/5

$$6 \times \frac{3}{2} = ?$$

A. 4

B. $\frac{9}{2}$

C. $\frac{18}{12}$

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q3/5

1488

QUESTION ID

S34-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q3/5

Multiply.

$$5 \times \frac{2}{5}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q4/5

1489

QUESTION ID

S34-B3-Q4

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE 10/5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q4/5

True or false?

$$2 \times \frac{7}{8} = \frac{14}{16}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q5/5

1490

QUESTION ID

S34-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.b • Set 34 • BACKWARD 3 • Q5/5

SET 34 • 5.NF.B.5.b

FORWARD 1

Percents greater than and less than 100% • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

True or false?

150% of 40 is less than 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q1/5

1492

QUESTION ID

S34-F1-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is 120% of 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q2/5

1493

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q2/5

1493

Which value is less than 80?

- A.** 200% of 80
- B.** 125% of 80
- C.** 75% of 80
- D.** 100% of 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q3/5

1494

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 75% of 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

What is 100% of 37?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q4/5

1495

QUESTION ID

S34-F1-Q4

TYPE

Short answer

CORRECT ANSWER

37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q4/5

1495

ANSWER KEY

Reduced copy of the question page, with its answer

What is 250% of 12?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q5/5

1496

QUESTION ID

S34-F1-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 34 • FORWARD 1 • Q5/5

1496

SET 34 • 5.NF.B.5.b

FORWARD 2

Use a multiplier for percent increase and decrease • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A \$60 price increases by 15%.

Which expression gives the new price?

- A. 1.15×60
- B. $60 + 15$
- C. 0.85×60
- D. 0.15×60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q1/5

1498

QUESTION ID

S34-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 1.15×60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

A \$90 jacket is 20% off.

Multiply by a decimal to find the sale price.

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q2/5

1499

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

\$72**GRADING NOTE** $0.8 \times 90 = 72$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q2/5

True or false?

Multiplying a price by 0.7 decreases it by 30%.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q3/5

1500

QUESTION ID

S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q3/5

A population of 2,000 grows by 5%.

What is the new population?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q4/5

1501

QUESTION ID

S34-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2,100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q4/5

Which multiplier shows a 40% decrease?

- A.** 0.6
- B.** 0.4
- C.** 1.4
- D.** 1.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q5/5

1502

QUESTION ID

S34-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 34 • FORWARD 2 • Q5/5

Explain equivalent fractions as multiplying by a fraction equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Equivalent fractions with a model	3.NF.A.3.b
FORWARD 1	Find equivalent ratios	6.RP.A.3.a
FORWARD 2	Test for a proportional relationship with equivalent ratios	7.RP.A.2.a

SET 68 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

Use the fact that $\frac{4}{4} = 1$ to explain why $\frac{2}{3} = \frac{8}{12}$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q1/5

1505

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

**$\frac{2}{3} \times \frac{4}{4} = \frac{8}{12}$, and
 $\frac{4}{4} = 1$, so
multiplying by $\frac{4}{4}$
does not change the
value.**

GRADING NOTE The response must connect the equivalent fraction to multiplying by a fraction equal to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q1/5

Which fraction equal to 1 turns $\frac{3}{5}$ into $\frac{9}{15}$?

A. $\frac{3}{3}$

B. $\frac{15}{15}$

C. $\frac{9}{9}$

D. $\frac{5}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q2/5

1506

QUESTION ID

S68-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q2/5

True or false?

$\frac{5}{6} \times \frac{2}{2} = \frac{10}{12}$, and $\frac{10}{12} = \frac{5}{6}$ because $\frac{2}{2} = 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q3/5

1507

QUESTION ID

S68-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q3/5

What fraction equal to 1 can you multiply $\frac{7}{10}$ by to get $\frac{28}{40}$?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q4/5

1508

QUESTION ID

S68-M-Q4

TYPE

Short answer

CORRECT ANSWER

4/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q4/5

True or false?

$\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$ shows that $\frac{3}{4} = \frac{6}{12}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q5/5

1509

QUESTION ID

S68-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.5.b • Set 68 • MAIN • Q5/5

SET 68 • 5.NF.B.5.b

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

To write a fraction equal to $\frac{3}{5}$, you can multiply its numerator and denominator by the same nonzero whole number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q1/5

1511

QUESTION ID

S68-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{3}{5} = \frac{?}{20}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q2/5

1512

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{7}{8}$?

A. $\frac{7}{16}$

B. $\frac{14}{24}$

C. $\frac{21}{24}$

D. $\frac{8}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q3/5

1513

QUESTION ID

S68-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 21/24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q3/5

Find the missing number.

$$\frac{1}{3} = \frac{4}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q4/5

1514

QUESTION ID

S68-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q4/5

True or false?

$$\frac{2}{9} = \frac{4}{27}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q5/5

1515

QUESTION ID

S68-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • 5.NF.B.5.b

BACKWARD 2

Equivalent fractions with a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

True or false?

$$\frac{1}{2} = \frac{2}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q1/5

1517

QUESTION ID

S68-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q1/5

Find the missing number.

$$\frac{2}{3} = \frac{?}{6}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q2/5

1518

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q2/5

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{4}{3}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q3/5

1519

QUESTION ID

S68-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q3/5

The bar shows $\frac{1}{3}$ shaded. Each part is cut in half.

How many sixths are shaded?



Sixths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q4/5

1520

QUESTION ID

S68-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q4/5

True or false?

$$\frac{2}{4} = \frac{2}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q5/5

1521

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.3.b • Set 68 • BACKWARD 2 • Q5/5

SET 68 • 5.NF.B.5.b

FORWARD 1

Find equivalent ratios • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Write a ratio equivalent to 3 : 5 by multiplying both numbers by 4.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q1/5

1523

QUESTION ID

S68-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q1/5

True or false?

6 : 8 and 9 : 12 are equivalent ratios.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q2/5

1524

QUESTION ID

S68-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q2/5

Which ratio is equivalent to 2 : 7?

- A.** 4 : 9
- B.** 7 : 2
- C.** 2 : 14
- D.** 6 : 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q3/5

1525

QUESTION ID

S68-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 : 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q3/5

Find the missing value.

$$5 : 6 = 20 : ?$$

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q4/5

1526

QUESTION ID

S68-F1-Q4

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q4/5

True or false?

4 : 10 and 6 : 12 are equivalent ratios.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q5/5

1527

QUESTION ID

S68-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 68 • FORWARD 1 • Q5/5

SET 68 • 5.NF.B.5.b

FORWARD 2

Test for a proportional relationship with equivalent ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	2	4	6
y	5	10	15

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q1/5

1529

QUESTION ID

S68-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q1/5

Which table shows a proportional relationship?

- A. x: 1, 2, 3 and y: 4, 6, 8
- B. x: 1, 2, 3 and y: 4, 8, 12
- C. x: 1, 2, 3 and y: 3, 5, 7
- D. x: 1, 2, 3 and y: 5, 6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q2/5

1530

QUESTION ID

S68-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 2, 3 and y: 4, 8, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q2/5

Are 3 : 4 and 12 : 16 equivalent ratios? Explain using a fraction equal to 1.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q3/5

1531

QUESTION ID

S68-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE $\frac{3}{4} \times \frac{4}{4} = \frac{12}{16}$; multiplying by $\frac{4}{4} = 1$ does not change the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q3/5

True or false?

3 : 5 and 9 : 20 are equivalent ratios.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q4/5

1532

QUESTION ID

S68-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q4/5

The ratios 2 : 9 and 8 : y are equivalent.

Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q5/5

1533

QUESTION ID

S68-F2-Q5

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 68 • FORWARD 2 • Q5/5

Solve real-world problems by multiplying fractions and mixed numbers

MAIN	Grade 5 main question	5.NF.B.6
BACKWARD 1	Word problems: a whole number times a fraction	4.NF.B.4.c
BACKWARD 2	Mixed numbers as fractions	4.NF.B.3.b
BACKWARD 4	Word problems that add fractions with like denominators	4.NF.B.3.d
FORWARD 1	Word problems with division of fractions	6.NS.A.1
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 35 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q1/5

1536

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

1 $\frac{7}{8}$ cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q1/5

A garden is 6 m long. $\frac{2}{3}$ of its length is planted with beans.

How many meters are planted with beans?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q2/5

1537

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q2/5

Sam walks $\frac{5}{8}$ mile each day.

How far does he walk in 4 days?

- A. $\frac{5}{32}$ mile B. $\frac{9}{8}$ miles
C. $2\frac{1}{2}$ miles D. $4\frac{5}{8}$ miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q3/5

1538

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 1/2 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q3/5

A board is $3\frac{1}{3}$ feet long. Jo uses $\frac{3}{5}$ of the board.

How many feet does she use?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q4/5

1539

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q4/5

A pitcher holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the pitcher?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q5/5

1540

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

1 liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 35 • MAIN • Q5/5

SET 35 • 5.NF.B.6

BACKWARD 1

Word problems: a whole number times a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each serving is $\frac{2}{3}$ cup.

How many cups are in 5 servings?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q1/5

1542

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3 $\frac{1}{3}$ cups

GRADING NOTE 10/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q1/5

True or false?

4 children each drink $\frac{3}{8}$ liter of milk. They drink $1\frac{1}{2}$ liters in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q2/5

1543

QUESTION ID

S35-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q2/5

Each piece of ribbon is $\frac{5}{6}$ yard long.

How long are 3 pieces in all?

- A. $\frac{5}{18}$ yard B. $2\frac{1}{2}$ yards
C. $3\frac{5}{6}$ yards D. $\frac{15}{18}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q3/5

1544

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 1/2 yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q3/5

A bug crawls $\frac{3}{10}$ meter each minute.

How far does it crawl in 7 minutes?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q4/5

1545

QUESTION ID

S35-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2 $\frac{1}{10}$ meters

GRADING NOTE 21/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q4/5

Each bag holds $\frac{1}{4}$ pound of nuts.

How many pounds are in 6 bags?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q5/5

1546

QUESTION ID

S35-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 1/2 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 5.NF.B.6

BACKWARD 2

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $2\frac{1}{2}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q1/5

1548

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q1/5

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q2/5

1549

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q2/5

True or false?

$$1\frac{1}{2} = \frac{11}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q3/5

1550

QUESTION ID

S35-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q3/5

Which fraction is equal to $4\frac{2}{5}$?

A. $\frac{22}{5}$

B. $\frac{42}{5}$

C. $\frac{11}{5}$

D. $\frac{24}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q4/5

1551

QUESTION ID

S35-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 22/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q4/5

Write $\frac{15}{8}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q5/5

1552

QUESTION ID

S35-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 5.NF.B.6

BACKWARD 4

Word problems that add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

A recipe uses $\frac{3}{4}$ cup of oats. Ann makes the recipe twice.

How many cups of oats does she use? Write a fraction.

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q1/5

1554

QUESTION ID

S35-B4-Q1

TYPE

Short answer

CORRECT ANSWER

6/4 cups

GRADING NOTE 1 $\frac{2}{4}$, 1 $\frac{1}{2}$, and $\frac{3}{2}$ cups are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q1/5

True or false?

$$\frac{2}{5} + \frac{2}{5} + \frac{2}{5} = \frac{6}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q2/5

1555

QUESTION ID

S35-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q2/5

Ty walks $\frac{3}{8}$ mile to school and $\frac{3}{8}$ mile home.

How far does he walk in all?

A. $\frac{9}{8}$ mile

B. $\frac{6}{8}$ mile

C. $\frac{3}{8}$ mile

D. $\frac{6}{16}$ mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q3/5

1556

QUESTION ID

S35-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6/8 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q3/5

A plant grew $\frac{1}{6}$ inch on Monday and $\frac{4}{6}$ inch on Tuesday.

How much did it grow in all?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q4/5

1557

QUESTION ID

S35-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{5}{6}$ inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q4/5

True or false?

$$\frac{5}{12} + \frac{5}{12} = \frac{10}{24}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q5/5

1558

QUESTION ID

S35-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.d • Set 35 • BACKWARD 4 • Q5/5

SET 35 • 5.NF.B.6

FORWARD 1

Word problems with division of fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{3}{4}$ -cup servings are in 6 cups of soup?

Servings: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q1/5

1560

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q1/5

A $\frac{7}{8}$ -mile trail is split into $\frac{1}{8}$ -mile sections.

How many sections are there?

Sections: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q2/5

1561

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q2/5

A rectangle has an area of $\frac{3}{4}$ square foot and a width of $\frac{1}{2}$ foot.

What is its length?

- A.** $\frac{3}{8}$ foot **B.** $1\frac{1}{4}$ feet
C. $1\frac{1}{2}$ feet **D.** $\frac{2}{3}$ foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q3/5

1562

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 1/2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q3/5

Ann has $2\frac{1}{2}$ pounds of clay. Each pot uses $\frac{5}{8}$ pound.

How many pots can she make?

Pots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q4/5

1563

QUESTION ID

S35-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q4/5

True or false?

$$\frac{2}{3} \div \frac{1}{6} = \frac{1}{9}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q5/5

1564

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 5.NF.B.6

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot per second for $6\frac{1}{2}$ seconds.

What is the diver's change in depth?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q1/5

1566

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−4 $\frac{7}{8}$ feet

GRADING NOTE "4 $\frac{7}{8}$ feet down" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q1/5

The temperature was 2°F . It dropped $\frac{1}{2}^{\circ}\text{F}$ each hour for 9 hours.

What was the final temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q2/5

1567

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$-2 \frac{1}{2}^{\circ}\text{F}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q2/5

A stock loses $\frac{3}{8}$ dollar each day for 4 days.

What is the total change in value?

- A.** $-4\frac{3}{8}$ dollars **B.** $-\frac{3}{32}$ dollar
C. $-1\frac{1}{2}$ dollars **D.** $1\frac{1}{2}$ dollars

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q3/5

1568

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $-1\frac{1}{2}$ dollars

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q3/5

A tank contains $12\frac{1}{2}$ gallons of water. $\frac{2}{5}$ of the water drains out.

How many gallons drain out?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q4/5

1569

QUESTION ID

S35-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q4/5

True or false?

$$-\frac{2}{3} \times 4\frac{1}{2} = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q5/5

1570

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 35 • FORWARD 2 • Q5/5

Solve real-world problems by multiplying mixed numbers

MAIN

Grade 5 main question

5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1

4.NF.B.3.b

BACKWARD 3

Breaking apart a factor to multiply

3.OA.B.5

BACKWARD 4

Measurement word problems with fractions

4.MD.A.2

FORWARD 1

Volume of prisms with fractional edge lengths

6.G.A.2

FORWARD 2

Surface area of prisms with fractional edge lengths

7.G.B.6

SET 36 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A bag of flour weighs $2\frac{1}{2}$ pounds.

How much do 3 bags weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q1/5

1573

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

7 1/2 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q1/5

A garden bed is $1\frac{1}{3}$ yards long and $1\frac{1}{2}$ yards wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q2/5

1574

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

2 square yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q2/5

A hiker walks $2\frac{1}{4}$ miles each hour.

How far does she walk in $1\frac{2}{3}$ hours?

- A.** $3\frac{1}{6}$ miles **B.** $4\frac{1}{2}$ miles
C. $2\frac{2}{12}$ miles **D.** $3\frac{3}{4}$ miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q3/5

1575

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 3/4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q3/5

A poster is $3\frac{1}{5}$ feet long. A copy is made $2\frac{1}{2}$ times as long.

How long is the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q4/5

1576

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

8 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q4/5

True or false?

A recipe uses $1\frac{1}{2}$ cups of milk. Making $1\frac{1}{2}$ times the recipe uses $1\frac{1}{4}$ cups of milk.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q5/5

1577

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 36 • MAIN • Q5/5

SET 36 • 5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $1\frac{3}{4}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q1/5

1579

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q1/5

Write $\frac{13}{3}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q2/5

1580

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4 $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q2/5

True or false?

$$2\frac{2}{3} = \frac{22}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q3/5

1581

QUESTION ID

S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q3/5

Which mixed number is equal to $\frac{9}{2}$?

A. $2\frac{1}{9}$

B. $4\frac{1}{9}$

C. $9\frac{1}{2}$

D. $4\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q4/5

1582

QUESTION ID

S36-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q4/5

Write $5\frac{1}{2}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q5/5

1583

QUESTION ID

S36-B1-Q5

TYPE

Short answer

CORRECT ANSWER

11/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 5.NF.B.6

BACKWARD 3

Breaking apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking
• 3.OA.B.5

True or false?

$$4 \times (5 + 2) = 4 \times 5 + 4 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q1/5

1585

QUESTION ID

S36-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q1/5

$$6 \times 12 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

ID S36-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q2/5

1586

S36-B3-Q2

Short answer

2

ID S36-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q2/5

Which expression is equal to 3×15 ?

- A. $3 \times 10 + 5$
- B. $3 + 10 \times 5$
- C. $3 \times 10 \times 5$
- D. $3 \times 10 + 3 \times 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q3/5

1587

QUESTION ID

S36-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $3 \times 10 + 3 \times 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q3/5

True or false?

$$7 \times 13 = 7 \times 10 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q4/5

1588

QUESTION ID

S36-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q4/5

Fill in the blank.

$$8 \times (4 + 1) = 8 \times 4 + 8 \times \underline{\hspace{1cm}}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q5/5

1589

QUESTION ID

S36-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • 5.NF.B.6

BACKWARD 4

Measurement word problems with fractions • Grade 4 • Measurement & Data •
4.MD.A.2

A ribbon is $\frac{3}{4}$ yard long.

How long are 4 ribbons in all?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q1/5

1591

QUESTION ID

S36-B4-Q1

TYPE

Short answer

CORRECT ANSWER

3 yards

GRADING NOTE 12/4 yards is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q1/5

True or false?

Each lap is $\frac{1}{2}$ mile. 6 laps is 3 miles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q2/5

1592

QUESTION ID

S36-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q2/5

A jug holds 2 liters.

How many milliliters are in 3 jugs?

- A. 6,000 mL
- B. 5,000 mL
- C. 2,003 mL
- D. 600 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q3/5

1593

QUESTION ID

S36-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6,000 mL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q3/5

Each box weighs $1\frac{1}{4}$ pounds.

How much do 2 boxes weigh?

Weight: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q4/5

1594

QUESTION ID

S36-B4-Q4

TYPE

Short answer

CORRECT ANSWER

2 1/2 pounds

GRADING NOTE 2 2/4 pounds is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q4/5

True or false?

3 pieces of wood, each $\frac{2}{5}$ meter long, make 1 meter in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q5/5

1595

QUESTION ID

S36-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B4-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 36 • BACKWARD 4 • Q5/5

SET 36 • 5.NF.B.6

FORWARD 1

Volume of prisms with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $2\frac{1}{2}$ ft by 2 ft by $1\frac{1}{2}$ ft.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q1/5

1597

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7 1/2 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q1/5

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q2/5

1598

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

15 5/8 cubic inches

GRADING NOTE 125/8 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q2/5

A prism has a base area of $3\frac{1}{3}$ square cm and a height of $1\frac{1}{2}$ cm.

What is its volume?

- A.** $3\frac{1}{2}$ cubic cm **B.** 10 cubic cm
C. $4\frac{5}{6}$ cubic cm **D.** 5 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q3/5

1599

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5 cubic cm

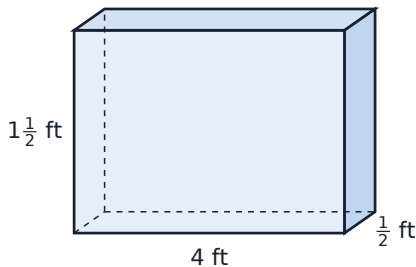
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q3/5

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q4/5

1600

QUESTION ID

S36-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q5/5

1601

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 5.NF.B.6

FORWARD 2

Surface area of prisms with fractional edge lengths • Grade 7 • Geometry •
7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q1/5

1603

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

59 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q1/5

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q2/5

1604

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

37 1/2 square inches

GRADING NOTE 37.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q2/5

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its surface area?

- A.** 30 square inches
- B.** 40 square inches
- C.** 67 square inches
- D.** $33\frac{1}{2}$ square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q3/5

1605

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 67 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q3/5

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q4/5

1606

QUESTION ID

S36-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**13 1/2 square
meters**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q4/5

True or false?

A prism that is 2 ft by $1\frac{1}{2}$ ft by 1 ft has a surface area of 13 square feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q5/5

1607

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 36 • FORWARD 2 • Q5/5

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{7} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q1/5

1610

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q1/5

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q2/5

1611

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q2/5

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q3/5

1612

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q3/5

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q4/5

1613

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q4/5

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q5/5

1614

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 37 • MAIN • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q1/5

1616

QUESTION ID

S37-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q1/5

Which unit fraction is the smallest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q2/5

1617

QUESTION ID

S37-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q2/5

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q3/5

1618

QUESTION ID

S37-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q3/5

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q4/5

1619

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q4/5

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q5/5

1620

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q1/5

1622

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q1/5

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q2/5

1623

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q2/5

Which expression shows 21 shared equally among 7 people?

A. $21 \div 7$

B. $21 - 7$

C. $7 \div 21$

D. 21×7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q3/5

1624

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $21 \div 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q3/5

Divide.

$$16 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q4/5

1625

QUESTION ID

S37-B2-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q4/5

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q5/5

1626

QUESTION ID

S37-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 5.NF.B.7.a

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{3} = \frac{?}{12}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q1/5

1628

QUESTION ID

S37-B4-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q1/5

True or false?

$$\frac{1}{2} = \frac{5}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q2/5

1629

QUESTION ID

S37-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q2/5

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{1}{8}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q3/5

1630

QUESTION ID

S37-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{1}{5} = \frac{3}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q4/5

1631

QUESTION ID

S37-B4-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q4/5

True or false?

$$\frac{1}{6} = \frac{3}{12}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q5/5

1632

QUESTION ID

S37-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 37 • BACKWARD 4 • Q5/5

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q1/5

1634

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q1/5

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q2/5

1635

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q2/5

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q3/5

1636

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q3/5

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q4/5

1637

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q4/5

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q5/5

1638

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q1/5

1640

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q1/5

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q2/5

1641

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q2/5

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q3/5

1642

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q3/5

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q4/5

1643

QUESTION ID

S37-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$-\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q4/5

True or false?

$$-(\frac{1}{2} \div 5) = (-\frac{1}{2}) \div 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q5/5

1644

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 37 • FORWARD 2 • Q5/5

Use models to divide a unit fraction by a whole number and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 69 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

The bar shows $\frac{1}{3}$ shaded.

Split every third into 2 equal parts to find $\frac{1}{3} \div 2$. Then write the quotient.



Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q1/5

1647

QUESTION ID

S69-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Each of the 3 parts split into 2 equal parts (6 small parts in all); the shaded third is now 2 small parts

Written: $\frac{1}{6}$

GRADING NOTE Grade both: all 3 parts split in half, and the quotient $\frac{1}{6}$ (one small part is $\frac{1}{6}$ of the whole bar).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q1/5

Explain how you can use multiplication to check that

$$\frac{1}{4} \div 3 = \frac{1}{12}.$$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q2/5

1648

QUESTION ID

S69-M-Q2

TYPE

Short answer

CORRECT ANSWER

**Multiply the quotient by the divisor:
 $\frac{1}{12} \times 3 = \frac{3}{12} = \frac{1}{4}$,
which is the number
that was divided, so
the quotient is correct.**

GRADING NOTE Must multiply $\frac{1}{12}$ by 3 (or $3 \times \frac{1}{12}$) and get back $\frac{1}{4}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q2/5

Which equation checks $\frac{1}{5} \div 4 = \frac{1}{20}$?

A. $\frac{1}{20} \div 4 = \frac{1}{80}$

B. $\frac{1}{5} + 4 = 4\frac{1}{5}$

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

D. $\frac{1}{5} \times 4 = \frac{4}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q3/5

1649

QUESTION ID

S69-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q3/5

True or false?

$\frac{1}{2} \div 3 = \frac{1}{6}$ because 3 groups of $\frac{1}{6}$ make $\frac{1}{2}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q4/5

1650

QUESTION ID

S69-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q4/5

4 friends share $\frac{1}{2}$ of a pizza equally.

How much of the whole pizza does each friend get?

Describe a model that shows it.

Share: _____

Model: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q5/5

1651

QUESTION ID

S69-M-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ of the pizza

GRADING NOTE Model: the half is split into 4 equal parts; the whole pizza then has 8 such parts, so each part is $\frac{1}{8}$. Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 69 • MAIN • Q5/5

SET 69 • 5.NF.B.7.a

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A bar is cut into 8 equal parts.

What fraction of the bar is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q1/5

1653

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q1/5

True or false?

2 parts of size $\frac{1}{6}$ make $\frac{2}{6}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q2/5

1654

QUESTION ID

S69-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q2/5

What fraction of the bar is shaded?



A. $\frac{5}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{6}$

D. $\frac{6}{1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q3/5

1655

QUESTION ID

S69-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q3/5

A whole is cut into 3 equal parts.

What fraction of the whole is one part?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q4/5

1656

QUESTION ID

S69-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{4}$ is one of 3 equal parts of a whole.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q5/5

1657

QUESTION ID

S69-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • 5.NF.B.7.a

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{3}{12}$ as a whole number times $\frac{1}{12}$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q1/5

1659

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 × 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q1/5

True or false?

$$4 \times \frac{1}{8} = \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q2/5

1660

QUESTION ID

S69-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q2/5

Which fraction equals $6 \times \frac{1}{10}$?

A. $\frac{1}{60}$

B. $\frac{6}{10}$

C. $\frac{10}{6}$

D. $\frac{6}{60}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q3/5

1661

QUESTION ID

S69-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q3/5

Multiply.

$$2 \times \frac{1}{6}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q4/5

1662

QUESTION ID

S69-B2-Q4

TYPE

Short answer

CORRECT ANSWER

2/6

GRADING NOTE 1/3 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q4/5

True or false?

$$3 \times \frac{1}{9} = \frac{3}{27}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q5/5

1663

QUESTION ID

S69-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 69 • BACKWARD 2 • Q5/5

SET 69 • 5.NF.B.7.a

FORWARD 1

Divide fractions and check with multiplication • Grade 6 • The Number System
• 6.NS.A.1

Divide.

$$\frac{2}{3} \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q1/5

1665

QUESTION ID

S69-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE 2/12 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q1/5

True or false?

$$\frac{3}{4} \div \frac{1}{4} = 3 \text{ because } 3 \times \frac{1}{4} = \frac{3}{4}.$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q2/5

1666

QUESTION ID

S69-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q2/5

Which equation checks $\frac{2}{3} \div \frac{3}{4} = \frac{8}{9}$?

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

B. $\frac{3}{4} \div \frac{8}{9} = \frac{27}{32}$

C. $\frac{8}{9} \times \frac{2}{3} = \frac{16}{27}$

D. $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q3/5

1667

QUESTION ID

S69-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q3/5

Divide.

$$\frac{4}{5} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q4/5

1668

QUESTION ID

S69-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2} \div \frac{1}{3} = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q5/5

1669

QUESTION ID

S69-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 69 • FORWARD 1 • Q5/5

SET 69 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{4} \div 2$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q1/5

1671

QUESTION ID

S69-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q1/5

True or false?

$$-\frac{1}{3} \div 5 = -\frac{1}{15}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q2/5

1672

QUESTION ID

S69-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q2/5

Find the quotient.

$$\frac{1}{6} \div (-3)$$

A. $\frac{1}{18}$

B. $-\frac{1}{2}$

C. -18

D. $-\frac{1}{18}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q3/5

1673

QUESTION ID

S69-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $-\frac{1}{18}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q3/5

Find the quotient.

$$\left(-\frac{1}{2}\right) \div (-4)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q4/5

1674

QUESTION ID

S69-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q4/5

True or false?

$$\left(-\frac{1}{5}\right) \div 2 = \frac{1}{10}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q5/5

1675

QUESTION ID

S69-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 69 • FORWARD 2 • Q5/5

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q1/5

1678

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q1/5

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q2/5

1679

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q2/5

How many $\frac{1}{3}$ s are in 7?

- A. 10
- B. 21
- C. $\frac{7}{3}$
- D. $\frac{1}{21}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q3/5

1680

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q3/5

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q4/5

1681

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q4/5

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q5/5

1682

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 38 • MAIN • Q5/5

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q1/5

1684

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q1/5

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q2/5

1685

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q2/5

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q3/5

1686

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $24 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q3/5

How many 4s are in 32?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q4/5

1687

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q4/5

A 30-inch ribbon is cut into 5-inch pieces.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q5/5

1688

QUESTION ID

S38-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q1/5

1690

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q1/5

True or false?

There are 3 thirds in 2 wholes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q2/5

1691

QUESTION ID

S38-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q2/5

How many $\frac{1}{6}$ s make 2 wholes?

- A. 2
- B. 8
- C. 12
- D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q3/5

1692

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q3/5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q4/5

1693

QUESTION ID

S38-B2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

There are 2 halves in 2 wholes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q5/5

1694

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q1/5

1696

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q1/5

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q2/5

1697

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q2/5

How many $\frac{2}{3}$ -cup servings are in 8 cups?

- A. 6
- B. $\frac{1}{12}$
- C. $\frac{16}{3}$
- D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q3/5

1698

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q3/5

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q4/5

1699

QUESTION ID

S38-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q4/5

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q5/5

1700

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q1/5

1702

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q1/5

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q2/5

1703

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q2/5

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A. $\frac{1}{12}$ mile per hour
- B. 3 miles per hour
- C. 8 miles per hour
- D. 12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q3/5

1704

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q3/5

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q4/5

1705

QUESTION ID

S38-F2-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q4/5

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q5/5

1706

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 38 • FORWARD 2 • Q5/5

Use models to divide a whole number by a unit fraction and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.b
BACKWARD 1	Write whole numbers as sums of unit fractions	4.NF.B.3.b
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide whole numbers by fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 70 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Each bar is 1 whole.

Split each whole into fourths to find $2 \div \frac{1}{4}$. Then write the quotient.

1 whole

1 whole

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q1/5

1709

QUESTION ID

S70-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

**Drawing: Each of the
2 bars split into 4
equal parts (8
fourths in all)**

Written: 8

GRADING NOTE Grade both: both wholes split into fourths, and the quotient 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q1/5

Explain how to use multiplication to check that

$$5 \div \frac{1}{3} = 15.$$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q2/5

1710

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

**$15 \times 1/3 = 15/3 = 5$,
which is the number
that was divided, so
15 is correct.**

GRADING NOTE Must multiply 15 by $1/3$ (or $1/3 \times 15$) and get back 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q2/5

Which equation checks $6 \div \frac{1}{2} = 12$?

A. $12 \div \frac{1}{2} = 24$

B. $6 \times \frac{1}{2} = 3$

C. $12 \times \frac{1}{2} = 6$

D. $6 \times 12 = 72$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q3/5

1711

QUESTION ID

S70-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $12 \times (1/2) = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q3/5

True or false?

$3 \div \frac{1}{5} = 15$ because each of the 3 wholes has 5 fifths.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q4/5

1712

QUESTION ID

S70-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q4/5

Write a story problem for $4 \div \frac{1}{2}$ and solve it. Describe a model that shows the answer.

Story and answer: _____

Model: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q5/5

1713

QUESTION ID

S70-M-Q5

TYPE

Short answer

CORRECT ANSWER

Answer 8. Example story: How many 1/2-cup scoops are in 4 cups?

GRADING NOTE The story must ask how many groups of $\frac{1}{2}$ are in 4 (a story about sharing 4 between 2 people gives 2 and is not correct). Model: 4 wholes, each split into 2 halves, make 8 halves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 70 • MAIN • Q5/5

SET 70 • 5.NF.B.7.b

BACKWARD 1

Write whole numbers as sums of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write 1 as a sum of thirds.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q1/5

1715

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3} + \frac{1}{3} + \frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q1/5

True or false?

$$2 = \frac{5}{5} + \frac{5}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q2/5

1716

QUESTION ID

S70-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q2/5

Which sum equals 1?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

C. $\frac{1}{4} + \frac{1}{4}$

D. $\frac{1}{2} + \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q3/5

1717

QUESTION ID

S70-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q3/5

Write 2 as a fraction with a denominator of 6.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q4/5

1718

QUESTION ID

S70-B1-Q4

TYPE

Short answer

CORRECT ANSWER

12/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q4/5

True or false?

$$3 = \frac{3}{3}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q5/5

1719

QUESTION ID

S70-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 70 • BACKWARD 1 • Q5/5

SET 70 • 5.NF.B.7.b

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{8}{4}$ as a whole number times $\frac{1}{4}$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q1/5

1721

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 × 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q1/5

True or false?

$$15 \times \frac{1}{3} = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q2/5

1722

QUESTION ID

S70-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q2/5

What is $10 \times \frac{1}{5}$?

A. 2

B. $\frac{1}{50}$

C. 50

D. $\frac{10}{50}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q3/5

1723

QUESTION ID

S70-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q3/5

Multiply.

$$6 \times \frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q4/5

1724

QUESTION ID

S70-B2-Q4

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE 6/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q4/5

True or false?

$$12 \times \frac{1}{4} = 48$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q5/5

1725

QUESTION ID

S70-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.a • Set 70 • BACKWARD 2 • Q5/5

SET 70 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions and check with multiplication • Grade 6 •
The Number System • 6.NS.A.1

Divide.

$$3 \div \frac{3}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q1/5

1727

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q1/5

True or false?

$$2 \div \frac{2}{3} = 3 \text{ because } 3 \times \frac{2}{3} = 2.$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q2/5

1728

QUESTION ID

S70-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q2/5

Which equation checks $4 \div \frac{2}{7} = 14$?

A. $4 \times 14 = 56$

B. $4 \times \frac{2}{7} = \frac{8}{7}$

C. $14 \div \frac{2}{7} = 49$

D. $14 \times \frac{2}{7} = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q3/5

1729

QUESTION ID

S70-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $14 \times (2/7) = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q3/5

How many $\frac{3}{4}$ -foot pieces can be cut from a 9-foot board?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q4/5

1730

QUESTION ID

S70-F1-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q4/5

True or false?

$$6 \div \frac{3}{4} = \frac{9}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q5/5

1731

QUESTION ID

S70-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 70 • FORWARD 1 • Q5/5

SET 70 • 5.NF.B.7.b

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-3 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q1/5

1733

QUESTION ID

S70-F2-Q1

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q1/5

True or false?

$$4 \div \left(-\frac{1}{2}\right) = -8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q2/5

1734

QUESTION ID

S70-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q2/5

Find the quotient.

$$(-5) \div (-\frac{1}{3})$$

A. 15

B. $\frac{5}{3}$

C. -15

D. $-\frac{5}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q3/5

1735

QUESTION ID

S70-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q3/5

Find the quotient.

$$-2 \div \frac{1}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q4/5

1736

QUESTION ID

S70-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q4/5

True or false?

$$(-6) \div \frac{1}{2} = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q5/5

1737

QUESTION ID

S70-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 70 • FORWARD 2 • Q5/5

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 3	Word problems: a whole number times a unit fraction	4.NF.B.4.c
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{4}$ -cup scoops of birdseed are in 3 cups?

Scoops: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q1/5

1740

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q1/5

4 students share $\frac{1}{3}$ liter of paint equally.

How much paint does each student get?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q2/5

1741

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{12}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q2/5

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q3/5

1742

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q3/5

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q4/5

1743

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q4/5

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q5/5

1744

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 39 • MAIN • Q5/5

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.

How many bracelets can she make?

Bracelets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q1/5

1746

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q1/5

True or false?

If 32 students sit 4 at each table, there are 6 tables.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q2/5

1747

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q2/5

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A.** 8
- B.** 200
- C.** 35
- D.** 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q3/5

1748

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q3/5

A 27-foot fence is split into 9 equal sections.

How long is each section?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q4/5

1749

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q4/5

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q5/5

1750

QUESTION ID

S39-B1-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 5.NF.B.7.c

BACKWARD 3

Word problems: a whole number times a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each scoop holds $\frac{1}{4}$ cup.

How many cups are in 8 scoops?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q1/5

1752

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2 cups

GRADING NOTE 8/4 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q1/5

True or false?

5 servings of $\frac{1}{3}$ pound each weigh $\frac{5}{3}$ pounds in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q2/5

1753

QUESTION ID

S39-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q2/5

Each piece of ribbon is $\frac{1}{2}$ meter long.

How long are 6 pieces in all?

- A.** $\frac{1}{12}$ meter **B.** $6\frac{1}{2}$ meters
C. 12 meters **D.** 3 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q3/5

1754

QUESTION ID

S39-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q3/5

Each lap is $\frac{1}{8}$ mile.

How far are 16 laps?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q4/5

1755

QUESTION ID

S39-B3-Q4

TYPE

Short answer

CORRECT ANSWER

2 miles

GRADING NOTE 16/8 miles is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q4/5

True or false?

4 cups that each hold $\frac{1}{3}$ liter hold $\frac{4}{12}$ liter in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q5/5

1756

QUESTION ID

S39-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.4.c • Set 39 • BACKWARD 3 • Q5/5

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q1/5

1758

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q1/5

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q2/5

1759

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q2/5

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q3/5

1760

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q3/5

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q4/5

1761

QUESTION ID

S39-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{4}{5}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q5/5

1762

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q1/5

1764

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

–36 feet per minute

GRADING NOTE "36 feet down per minute" is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q1/5

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q2/5

1765

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-20°F per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q2/5

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

- A.** \$135 **B.** $-\$45$
C. $-\$15$ **D.** $-\$135$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q3/5

1766

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $-\$135$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q3/5

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q4/5

1767

QUESTION ID

S39-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q4/5

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q5/5

1768

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 39 • FORWARD 2 • Q5/5

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 4	Multiply whole numbers by 10, 100, and 1,000	4.NBT.A.1
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q1/5

1771

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q1/5

How many ounces are in 3 pounds?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q2/5

1772

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q2/5

A track is 2.5 kilometers long.

How many meters is that?

- A.** 250 m
- B.** 25,000 m
- C.** 2,500 m
- D.** 25 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q3/5

1773

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2,500 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q3/5

Convert.

6 yards = ____ feet

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q4/5

1774

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q4/5

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q5/5

1775

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 40 • MAIN • Q5/5

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q1/5

1777

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q1/5

True or false?

1 pound = 16 ounces

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q2/5

1778

QUESTION ID

S40-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q2/5

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q3/5

1779

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q3/5

How many meters are in 1 kilometer?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q4/5

1780

QUESTION ID

S40-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

1 yard = 12 inches

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q5/5

1781

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$16 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q1/5

1783

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q1/5

1783

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$1,000 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q2/5

1784

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q2/5

1784

True or false?

$$3 \times 36 = 98$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q3/5

1785

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q3/5

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q4/5

1786

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$60 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q5/5

1787

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 40 • BACKWARD 2 • Q5/5

1787

SET 40 • 5.MD.A.1

BACKWARD 4

Multiply whole numbers by 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$25 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q1/5

1789

QUESTION ID

S40-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q1/5

1789

True or false?

$$7 \times 1,000 = 700$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q2/5

1790

QUESTION ID

S40-B4-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q2/5

$$42 \times 10 = ?$$

- A.** 4.2
- B.** 420
- C.** 4,200
- D.** 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q3/5

1791

QUESTION ID

S40-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 420

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$16 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q4/5

1792

QUESTION ID

S40-B4-Q4

TYPE

Short answer

CORRECT ANSWER

16,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q4/5

1792

True or false?

3×100 is 10 times as much as 3×10 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q5/5

1793

QUESTION ID

S40-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1 • Set 40 • BACKWARD 4 • Q5/5

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.
How many centimeters are in 5 inches?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q1/5

1795

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q1/5

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q2/5

1796

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q2/5

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q3/5

1797

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q3/5

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q4/5

1798

QUESTION ID

S40-F1-Q4

TYPE

Short answer

CORRECT ANSWER

33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q4/5

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q5/5

1799

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 40 • FORWARD 1 • Q5/5

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q1/5

1801

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/5 foot per minute

GRADING NOTE 0.2 foot per minute is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q1/5

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q2/5

1802

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q2/5

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q3/5

1803

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q3/5

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q4/5

1804

QUESTION ID

S40-F2-Q4

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q5/5

1805

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • FORWARD 2 • Q5/5

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 4	Write tenths and hundredths as decimals	4.NF.C.6
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Convert.

30 inches = ____ feet

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q1/5

1808

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/2**GRADING NOTE** 2.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q1/5

1808

Convert.

750 milliliters = ____ liters

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q2/5

1809

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

GRADING NOTE 3/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q2/5

A rope is 45 feet long.

How many yards is that?

- A.** 15
- B.** 5
- C.** 135
- D.** 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q3/5

1810

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Convert.

40 ounces = ____ pounds

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q4/5

1811

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

2 1/2**GRADING NOTE** 2.5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q4/5

1811

ANSWER KEY

Reduced copy of the question page, with its answer

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q5/5

1812

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

1.35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 41 • MAIN • Q5/5

1812

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q1/5

1814

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q1/5

True or false?

1 kilogram = 1,000 grams

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q2/5

1815

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q2/5

How many feet are in 1 yard?

- A.** 10
- B.** 3
- C.** 36
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q3/5

1816

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q3/5

How many cups are in 1 pint?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q4/5

1817

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q4/5

True or false?

1 foot = 10 inches

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q5/5

1818

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$36 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q1/5

1820

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q1/5

1820

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$64 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q2/5

1821

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q2/5

1821

True or false?

$$84 \div 7 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q3/5

1822

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q3/5

$$90 \div 3 = ?$$

- A.** 33
- B.** 300
- C.** 27
- D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q4/5

1823

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$5,000 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q5/5

1824

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 41 • BACKWARD 2 • Q5/5

1824

SET 41 • 5.MD.A.1

BACKWARD 4

Write tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{75}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q1/5

1826

QUESTION ID

S41-B4-Q1

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q1/5

True or false?

$$\frac{5}{10} = 0.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q2/5

1827

QUESTION ID

S41-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q2/5

Which decimal is equal to $\frac{45}{100}$?

- A. 45
- B. 0.045
- C. 0.45
- D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q3/5

1828

QUESTION ID

S41-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q3/5

Write $2\frac{6}{10}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q4/5

1829

QUESTION ID

S41-B4-Q4

TYPE

Short answer

CORRECT ANSWER

2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q4/5

True or false?

$$\frac{8}{10} = 0.08$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q5/5

1830

QUESTION ID

S41-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 41 • BACKWARD 4 • Q5/5

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q1/5

1832

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8.5

GRADING NOTE 8 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q1/5

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q2/5

1833

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q2/5

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q3/5

1834

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q3/5

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q4/5

1835

QUESTION ID

S41-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q4/5

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q5/5

1836

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 41 • FORWARD 1 • Q5/5

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q1/5

1838

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 2.54i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q1/5

Write an equation for the number of feet f in y yards.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q2/5

1839

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$f = 3y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q2/5

Which equation converts k kilograms to g grams?

- A. $k = 1,000g$
- B. $g = 1,000k$
- C. $g = k \div 1,000$
- D. $g = k + 1,000$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q3/5

1840

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $g = 1,000k$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q3/5

Write an equation for the number of pounds p in z ounces.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q4/5

1841

QUESTION ID

S41-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$p = z/16$

GRADING NOTE $p = z \div 16$ or $p = (1/16)z$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q4/5

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q5/5

1842

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 41 • FORWARD 2 • Q5/5

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.

How many feet of ribbon are left?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q1/5

1845

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q1/5

A recipe needs 2 pounds of flour. Jo has 20 ounces.

How many more ounces does she need?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q2/5

1846

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q2/5

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A. 1,050 m
- B. 2,050 m
- C. 9,700 m
- D. 862 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q3/5

1847

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2,050 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q3/5

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q4/5

1848

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q4/5

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q5/5

1849

QUESTION ID

S42-M-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 42 • MAIN • Q5/5

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.

What time does it end?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q1/5

1851

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3:55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q1/5

True or false?

3 quarters and 2 dimes make 85 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q2/5

1852

QUESTION ID

S42-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q2/5

Jake has 2 meters of string. He uses 75 cm.

How many centimeters are left?

- A. 1.25 cm
- B. 75 cm
- C. 125 cm
- D. 275 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q3/5

1853

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 125 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q3/5

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q4/5

1854

QUESTION ID

S42-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q4/5

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q5/5

1855

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1,600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q1/5

1857

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q1/5

True or false?

1 gallon = 4 quarts

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q2/5

1858

QUESTION ID

S42-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q2/5

How many centimeters are in 1 meter?

- A. 1,000
- B. 100
- C. 10
- D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q3/5

1859

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q3/5

How many feet are in 3 yards?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q4/5

1860

QUESTION ID

S42-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q4/5

True or false?

1 liter = 100 milliliters

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q5/5

1861

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q1/5

1863

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

8 miles per quart

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q1/5

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q2/5

1864

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q2/5

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

- A. \$72
- B. \$18
- C. \$24
- D. \$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q3/5

1865

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. \$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q3/5

A 24-ounce bag of rice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q4/5

1866

QUESTION ID

S42-F1-Q4

TYPE

Short answer

CORRECT ANSWER

\$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q5/5

1867

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 42 • FORWARD 1 • Q5/5

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q1/5

1869

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The 2-liter bottle

GRADING NOTE Bottle: \$0.90 per liter; can: \$1.20 per liter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q1/5

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q2/5

1870

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 minutes

GRADING NOTE 7.5 cm = 15 km; 15 km at 60 km per hour takes 1/4 hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q2/5

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

- A.** 1,050 **B.** 3
C. 4 **D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q3/5

1871

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q3/5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q4/5

1872

QUESTION ID

S42-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.5 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q4/5

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q5/5

1873

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 42 • FORWARD 2 • Q5/5

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare the difference in centers with the variability	7.SP.B.3

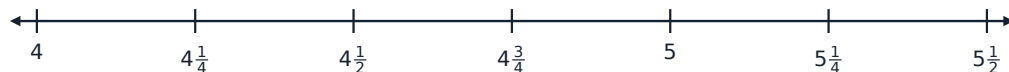
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q1/5

1876

QUESTION ID

S43-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 3 X's at $4\frac{1}{2}$, 1 X at $4\frac{3}{4}$, 2 X's at 5, 1 X at $5\frac{1}{4}$

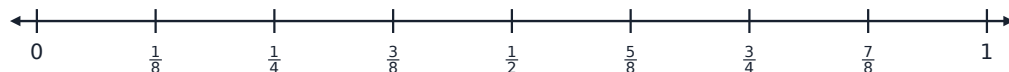
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q1/5

Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q2/5

1877

QUESTION ID

S43-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 1 X at $\frac{1}{8}$,
3 X's at $\frac{1}{4}$, 1 X at
 $\frac{3}{8}$, 2 X's at $\frac{1}{2}$**

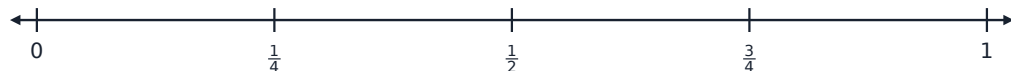
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q2/5

Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q3/5

1878

QUESTION ID

S43-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 1 X at $\frac{1}{4}$,
2 X's at $\frac{1}{2}$, 3 X's at
 $\frac{3}{4}$, 1 X at 1**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q3/5

Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q4/5

1879

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

3

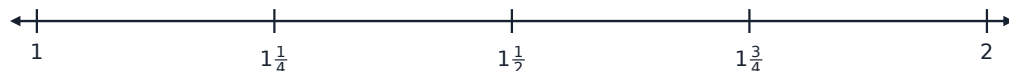
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q4/5

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q5/5

1880

QUESTION ID

S43-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 1 X at $1\frac{1}{4}$, 3 X's at $1\frac{1}{2}$, 1 X at $1\frac{3}{4}$, 2 X's at 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 43 • MAIN • Q5/5

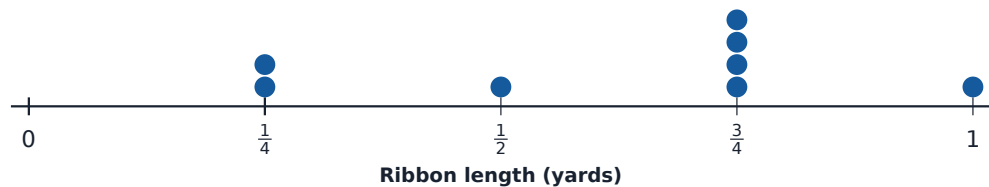
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q1/5

1882

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

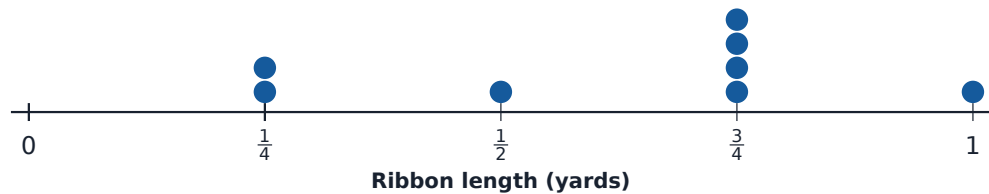
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q1/5

True or false?

Two ribbons are $\frac{1}{4}$ yard long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q2/5

1883

QUESTION ID

S43-B1-Q2

TYPE

True or false

CORRECT ANSWER

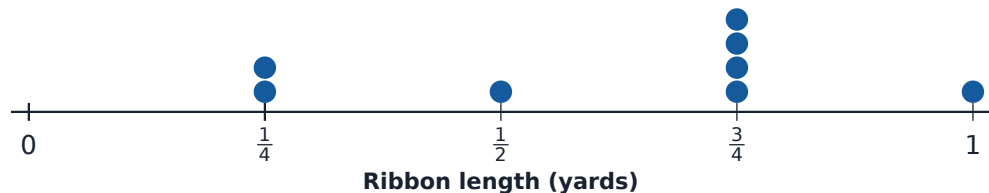
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q2/5

Which ribbon length is the most common?



- A. 1 yard B. $\frac{1}{2}$ yard C. $\frac{1}{4}$ yard D. $\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q3/5

1884

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

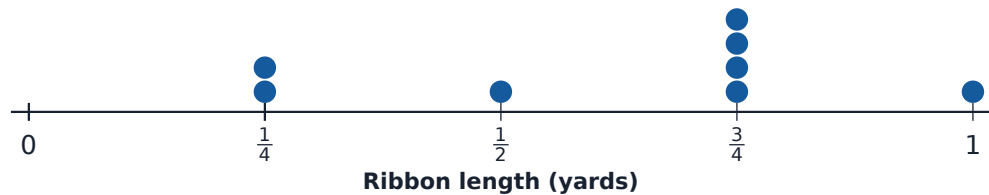
D. $\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q3/5

How many ribbons were measured?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q4/5

1885

QUESTION ID

S43-B1-Q4

TYPE

Short answer

CORRECT ANSWER

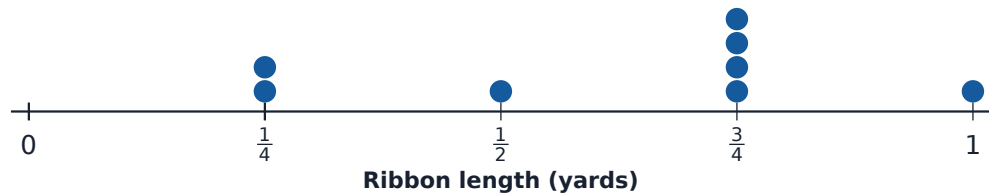
8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q4/5

What is the difference between the longest and the shortest ribbon?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q5/5

1886

QUESTION ID

S43-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q1/5

1888

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q1/5

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q2/5

1889

QUESTION ID

S43-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q2/5

Into how many equal parts is the space from 0 to 1 divided?



A. 5

B. 7

C. 6

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q3/5

1890

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

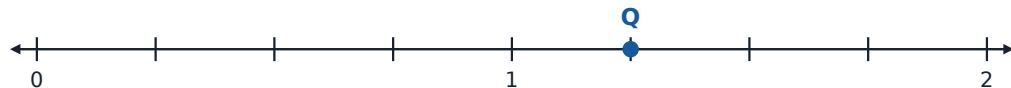
C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q3/5

What number is at point Q?



Q = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q4/5

1891

QUESTION ID

S43-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1 1/4

GRADING NOTE 5/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q4/5

True or false?

On a number line from 0 to 1 marked in eighths,
 $\frac{6}{8}$ is one mark to the left of 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q5/5

1892

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q1/5

1894

QUESTION ID

S43-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q1/5

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q2/5

1895

QUESTION ID

S43-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q2/5

Which number line would you use for a line plot of 3, 4, 4, 7?

- A. A number line from 0 to 8
- B. A number line from 0 to 3
- C. A number line from 5 to 6
- D. A number line from 10 to 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q3/5

1896

QUESTION ID

S43-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. A number line
from 0 to 8**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q3/5

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q4/5

1897

QUESTION ID

S43-B3-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 1 X at 2, 2 X's at 4, 1 X at 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q4/5

Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q5/5

1898

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

10 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q1/5

1900

QUESTION ID

S43-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 1 dot at 7,
2 dots at 8, 3 dots at
9, 1 dot at 10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q1/5

Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q2/5

1901

QUESTION ID

S43-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 2 dots at 0,
3 dots at 1, 2 dots at
2, 1 dot at 3**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q2/5

Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q3/5

1902

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**C. A number line
from 10 to 20**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q3/5

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q4/5

1903

QUESTION ID

S43-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q4/5

True or false?

A dot plot shows every value in a data set.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q5/5

1904

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 5.MD.B.2

FORWARD 2

Compare the difference in centers with the variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Plot A has a mean of 5 inches and Plot B has a mean of 8 inches. Each plot has a mean absolute deviation (MAD) of 1 inch.

The difference in the means is how many times the MAD?



Times the MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q1/5

1906

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q1/5

True or false?

The difference in the means (3 inches) is 3 times the MAD, so the two distributions overlap only a little.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q2/5

1907

QUESTION ID

S43-F2-Q2

TYPE

True or false

CORRECT ANSWER

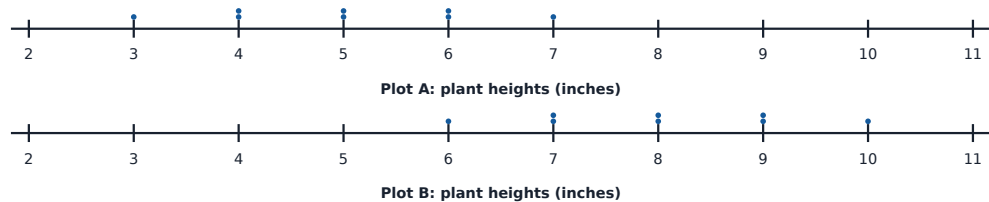
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q2/5

Which statement compares the two plots correctly?



- A. The centers differ by less than 1 MAD, so the plots are about the same.
- B. Plot A has much more variation than Plot B.
- C. The plots cannot be compared because they show different plants.
- D. The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q3/5

1908

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q3/5

Two teams have mean heights of 150 cm and 156 cm.
Each team has a MAD of 3 cm.

The difference in the means is how many times the
MAD?

Times the MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q4/5

1909

QUESTION ID

S43-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q4/5

True or false?

Two data sets have means of 20 and 21, and each has a MAD of 4. The difference in the means is large compared with the variability.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q5/5

1910

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 43 • FORWARD 2 • Q5/5

1910

Use operations on fractions to solve problems with line plot data

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 4	Rename halves and fourths as eighths	4.NF.A.1
BACKWARD 5	Add fractions with like denominators	4.NF.B.3.a
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.

What is the total amount of water?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q1/5

1913

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 $\frac{3}{8}$ liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q1/5

If all of the water were shared equally among the beakers, how much would each beaker hold?



Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q2/5

1914

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

19/64 liter

GRADING NOTE Total $2 \frac{3}{8}$ liters $\div$ 8 beakers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q2/5

How much more water is in the fullest beaker than in the least full beaker?



- A. $\frac{1}{2}$ liter B. $\frac{1}{8}$ liter C. $\frac{5}{8}$ liter D. $\frac{3}{8}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q3/5

1915

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\frac{3}{8}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q3/5

What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q4/5

1916

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{4}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q4/5

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q5/5

1917

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

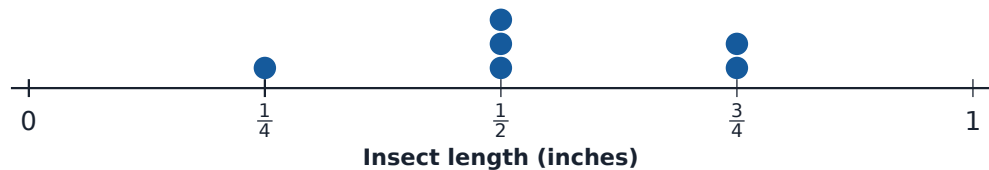
Grade 5 • Measurement & Data • 5.MD.B.2 • Set 44 • MAIN • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 1

Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q1/5

1919

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

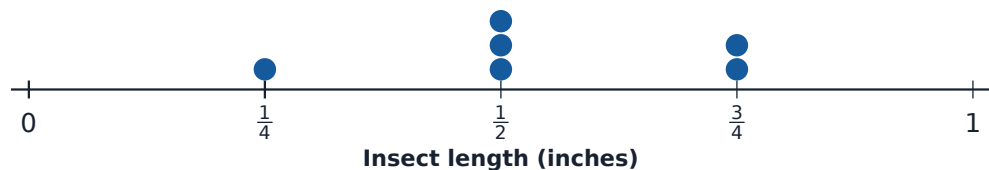
3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q1/5

What is the difference between the longest and the shortest insect?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q2/5

1920

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/2 inch

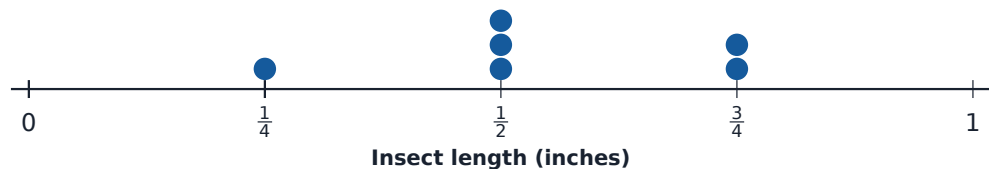
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q3/5

1921

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

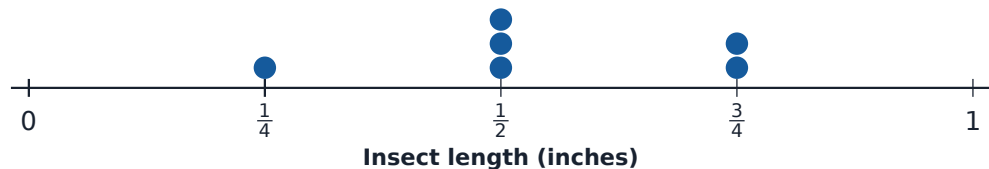
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q3/5

How many insects were measured?



A. 4

B. 3

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q4/5

1922

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

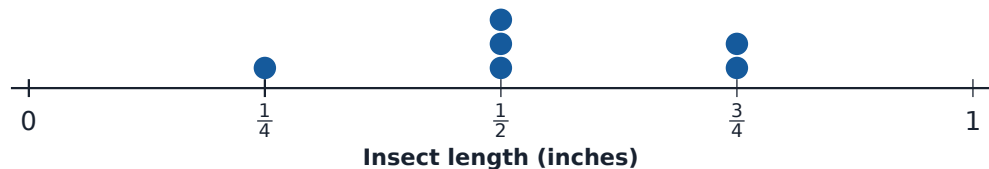
D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q4/5

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q5/5

1923

QUESTION ID

S44-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 1/2 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 4

Rename halves and fourths as eighths • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{1}{4}$ as eighths.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q1/5

1925

QUESTION ID

S44-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q1/5

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q2/5

1926

QUESTION ID

S44-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q2/5

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{7}{8}$

D. $\frac{4}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q3/5

1927

QUESTION ID

S44-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q4/5

1928

QUESTION ID

S44-B4-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q4/5

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q5/5

1929

QUESTION ID

S44-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 44 • BACKWARD 4 • Q5/5

SET 44 • 5.MD.B.2

BACKWARD 5

Add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{1}{8} + \frac{2}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q1/5

1931

QUESTION ID

S44-B5-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q1/5

True or false?

$$\frac{3}{8} + \frac{3}{8} = \frac{6}{16}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q2/5

1932

QUESTION ID

S44-B5-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q2/5

$$\frac{2}{8} + \frac{4}{8} + \frac{1}{8} = ?$$

A. $\frac{6}{8}$

B. $\frac{7}{8}$

C. $\frac{7}{24}$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q3/5

1933

QUESTION ID

S44-B5-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q3/5

Add.

$$\frac{5}{8} + \frac{3}{8}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q4/5

1934

QUESTION ID

S44-B5-Q4

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE 8/8 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q4/5

True or false?

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q5/5

1935

QUESTION ID

S44-B5-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.a • Set 44 • BACKWARD 5 • Q5/5

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q1/5

1937

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q1/5

Find the mean.

12, 15, 21

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q2/5

1938

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q2/5

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q3/5

1939

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q3/5

The dot plot shows the goals scored in 6 games.

What is the mean number of goals?



Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q4/5

1940

QUESTION ID

S44-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 2 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q4/5

True or false?

The mean of 4, 4, and 10 is 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q5/5

1941

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 44 • FORWARD 1 • Q5/5

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q1/5

1943

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q1/5

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A. The populations are the same.
- B. Plants in the second population tend to be taller.
- C. Plants in the first population tend to be taller.
- D. No conclusion is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q2/5

1944

QUESTION ID

S44-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Plants in the second population tend to be taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q2/5

True or false?

Comparing the means of random samples is one way to compare two populations.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q3/5

1945

QUESTION ID

S44-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q3/5

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q4/5

1946

QUESTION ID

S44-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Class Y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q4/5

A random sample of 30 students from School A spends a mean of 40 minutes on homework. A random sample of 30 students from School B spends a mean of 50 minutes. The samples have similar variability.

What can you infer about all the students at the two schools?

Inference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q5/5

1947

QUESTION ID

S44-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Students at School B probably tend to spend more time on homework than students at School A.

GRADING NOTE Must draw a conclusion about the populations (all students at each school), for example that School B students typically spend about 10 minutes more. Comparing only the two sample means, without a conclusion about the schools, is not enough.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 44 • FORWARD 2 • Q5/5

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q1/5

1950

QUESTION ID

S45-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q1/5

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q2/5

1951

QUESTION ID

S45-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 1 cubic unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q2/5

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q3/5

1952

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

1 cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q3/5

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q4/5

1953

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A sugar cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q4/5

True or false?

Volume measures the amount of space inside a solid figure.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q5/5

1954

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.a • Set 45 • MAIN • Q5/5

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q1/5

1956

QUESTION ID

S45-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q1/5

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q2/5

1957

QUESTION ID

S45-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 1 square unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q2/5

How many unit squares cover a 3-by-3 square?

Unit squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q3/5

1958

QUESTION ID

S45-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q3/5

True or false?

Area is measured in cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q4/5

1959

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q4/5

A square has sides that are 1 inch long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q5/5

1960

QUESTION ID

S45-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 square inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.5.a • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q1/5

1962

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q1/5

True or false?

All the faces of a cube are squares.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q2/5

1963

QUESTION ID

S45-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q2/5

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q3/5

1964

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q3/5

How many vertices (corners) does a cube have?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q4/5

1965

QUESTION ID

S45-B2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q4/5

True or false?

All the edges of a cube are the same length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q5/5

1966

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q1/5

1968

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{8}$ cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q1/5

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q2/5

1969

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q2/5

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

- A. $\frac{1}{3}$ cubic foot B. $\frac{1}{9}$ cubic foot
C. $\frac{1}{27}$ cubic foot D. 1 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q3/5

1970

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{1}{27}$ cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q3/5

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q4/5

1971

QUESTION ID

S45-F1-Q4

TYPE

Short answer

CORRECT ANSWER

2 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q4/5

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q5/5

1972

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q1/5

1974

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q1/5

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q2/5

1975

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

27 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q2/5

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

- A.** 60 cubic inches
- B.** 30 cubic inches
- C.** 16 cubic inches
- D.** 600 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q3/5

1976

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q3/5

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q4/5

1977

QUESTION ID

S45-F2-Q4

TYPE

Short answer

CORRECT ANSWER

60 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q4/5

True or false?

Volume is measured in cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q5/5

1978

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 45 • FORWARD 2 • Q5/5

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q1/5

1981

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

30 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q1/5

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q2/5

1982

QUESTION ID

S46-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q2/5

A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units **B.** 18 cubic units
C. 54 cubic units **D.** 6 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q3/5

1983

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18 cubic units

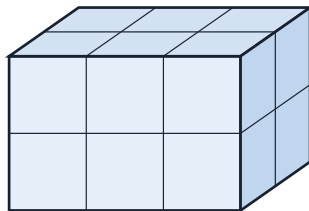
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q3/5

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q4/5

1984

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q4/5

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q5/5

1985

QUESTION ID

S46-M-Q5

TYPE

Short answer

CORRECT ANSWER

9 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 46 • MAIN • Q5/5

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q1/5

1987

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q1/5

True or false?

Overlapping squares give a wrong count for area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q2/5

1988

QUESTION ID

S46-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q2/5

A rectangle is covered by 20 unit squares.

What is its area?

- A. 10 square units
- B. 40 square units
- C. 20 cubic units
- D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q3/5

1989

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q3/5

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q4/5

1990

QUESTION ID

S46-B1-Q4

TYPE

Short answer

CORRECT ANSWER

15 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q4/5

True or false?

Area can be found by counting the unit squares that cover a shape.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q5/5

1991

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.6 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q1/5

1993

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q1/5

True or false?

3 rows of 5 make 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q2/5

1994

QUESTION ID

S46-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q2/5

Which equation matches an array with 2 rows of 5?

- A. $2 + 5 = 7$
- B. $5 + 5 = 10$
- C. $5 - 2 = 3$
- D. $2 + 2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q3/5

1995

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5 + 5 = 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q3/5

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q4/5

1996

QUESTION ID

S46-B2-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q4/5

Write an addition equation for 3 rows of 4.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q5/5

1997

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q1/5

1999

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q1/5

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q2/5

2000

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q2/5

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

- A.** 27 cubic feet **B.** 9 cubic feet
C. 1 cubic foot **D.** 3 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q3/5

2001

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q3/5

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q4/5

2002

QUESTION ID

S46-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q4/5

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q5/5

2003

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6

A fish tank is 30 cm by 20 cm by 25 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q1/5

2005

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

15,000 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q1/5

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q2/5

2006

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q2/5

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

- A.** 84 cubic meters **B.** 42 cubic meters
C. 21 cubic meters **D.** 14 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q3/5

2007

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 42 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q3/5

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q4/5

2008

QUESTION ID

S46-F2-Q4

TYPE

Short answer

CORRECT ANSWER

1.2 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q4/5

True or false?

Two prisms with the same base area and the same height have the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q5/5

2009

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 46 • FORWARD 2 • Q5/5

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

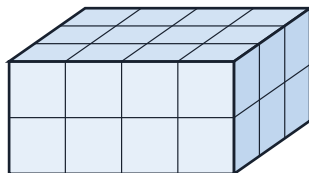
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q1/5

2012

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

24 cubic cm

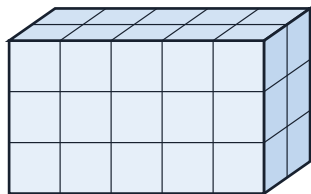
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q1/5

Each cube is 1 cubic inch.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q2/5

2013

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

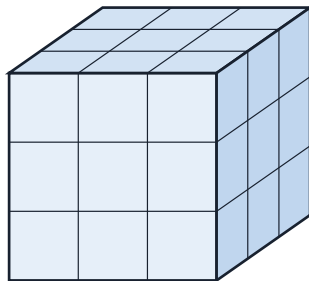
ID S47-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q2/5

Each cube is 1 cubic foot.

What is the volume of the prism?

- A.** 18 cubic feet **B.** 54 cubic feet
C. 27 cubic feet **D.** 9 cubic feet



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q3/5

2014

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 27 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q3/5

A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q4/5

2015

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

72 number cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

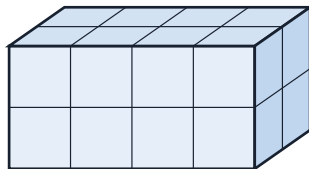
ID S47-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q4/5

True or false?

Each cube is 1 cubic cm.

The volume of the prism is
16 cubic cm.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q5/5

2016

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 47 • MAIN • Q5/5

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.

How many cubes are in the layer?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q1/5

2018

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q1/5

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q2/5

2019

QUESTION ID

S47-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q2/5

How many unit squares cover a 4-by-6 rectangle?

- A.** 10
- B.** 46
- C.** 24
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q3/5

2020

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q3/5

A rectangle is tiled with 7 rows of 3 unit squares.

How many squares are there?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q4/5

2021

QUESTION ID

S47-B1-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q4/5

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q5/5

2022

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.

How many cubes are there in all?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q1/5

2024

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q1/5

True or false?

4 layers of 10 cubes is 14 cubes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q2/5

2025

QUESTION ID

S47-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q2/5

Which expression shows 5 layers of 8 cubes?

- A. $8 - 5$
- B. 5×8
- C. $5 + 8$
- D. 58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q3/5

2026

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 9$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q4/5

2027

QUESTION ID

S47-B2-Q4

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q4/5

2027

There are 2 layers with 15 cubes in each layer.

How many cubes are there in all?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q5/5

2028

QUESTION ID

S47-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q1/5

2030

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q1/5

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q2/5

2031

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q2/5

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A. 4
- B. 8
- C. 16
- D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q3/5

2032

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

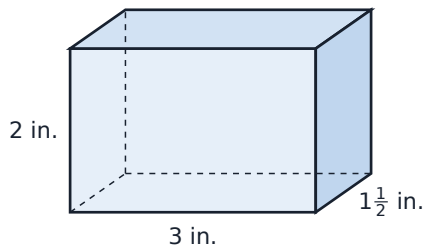
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q3/5

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q4/5

ID S47-F1-Q4

2033

QUESTION ID

S47-F1-Q4

TYPE

Short answer

CORRECT ANSWER

9 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q4/5

ID S47-F1-Q4

True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q5/5

2034

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q1/5

2036

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q1/5

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q2/5

2037

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q2/5

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet
- B.** 30 cubic feet
- C.** 120 cubic feet
- D.** 19 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q3/5

2038

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q3/5

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q4/5

2039

QUESTION ID

S47-F2-Q4

TYPE

Short answer

CORRECT ANSWER

108 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q4/5

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q5/5

2040

QUESTION ID

S47-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 47 • FORWARD 2 • Q5/5

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q1/5

2043

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

60 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q1/5

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q2/5

2044

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 12×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q2/5

True or false?

For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q3/5

2045

QUESTION ID

S48-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

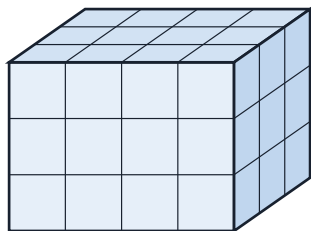
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q3/5

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q4/5

2046

QUESTION ID

S48-M-Q4

TYPE

Short answer

CORRECT ANSWER

36 cubic units

GRADING NOTE Base area $4 \times 3 = 12$; $12 \times 3 = 36$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q4/5

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q5/5

2047

QUESTION ID

S48-M-Q5

TYPE

Short answer

CORRECT ANSWER

72 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.a • Set 48 • MAIN • Q5/5

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q1/5

2049

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q1/5

True or false?

The area of a 5-by-2 rectangle is 10 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q2/5

2050

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q2/5

Which expression gives the area of a 6-by-3 rectangle?

- A. $6 - 3$
- B. $6 + 3$
- C. 6×3
- D. $2 \times (6 + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q3/5

2051

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6×3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q3/5

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q4/5

2052

QUESTION ID

S48-B1-Q4

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q4/5

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q5/5

2053

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.a • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q1/5

2055

QUESTION ID

S48-B2-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q1/5

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q2/5

2056

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q2/5

Which expression is equal to $4 \times 5 \times 2$?

- A. 9×2
- B. $20 + 2$
- C. $4 + 5 + 2$
- D. 4×10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q3/5

2057

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4×10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 2 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q4/5

2058

QUESTION ID

S48-B2-Q4

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q4/5

2058

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q5/5

2059

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q1/5

2061

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q1/5

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q2/5

2062

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q2/5

A prism is packed with cubes that have edges $\frac{1}{2}$ unit long: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

- A.** $6 + 4 + 2 = 12$ cubic units
- B.** $3 \times 2 \times 1 = 6$ cubic units
- C.** $6 \times 4 \times 2 = 48$ cubic units
- D.** $\frac{1}{2} \times 48 = 24$ cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q3/5

2063

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

**B. $3 \times 2 \times 1 = 6$
cubic units**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q3/5

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q4/5

2064

QUESTION ID

S48-F1-Q4

TYPE

Short answer

CORRECT ANSWER

27/64 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q4/5

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q5/5

2065

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q1/5

2067

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q1/5

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q2/5

2068

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

70 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q2/5

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

- A.** 1,008 m **B.** 7 m
C. 72 m **D.** 96 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q3/5

2069

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q3/5

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q4/5

2070

QUESTION ID

S48-F2-Q4

TYPE

Short answer

CORRECT ANSWER

100 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q4/5

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q5/5

2071

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 48 • FORWARD 2 • Q5/5

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Multiplying multi-digit numbers	4.NBT.B.5
BACKWARD 2	The area of the base	4.MD.A.3
FORWARD 1	Real-world volume with fractional edge lengths	6.G.A.2
FORWARD 2	Real-world volume problems with right prisms	7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q1/5

2074

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

4,320 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q1/5

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q2/5

2075

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

60 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q2/5

A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A. 90 cubic cm
- B. 36 cubic cm
- C. 180 cubic cm
- D. 18 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q3/5

2076

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 180 cubic cm

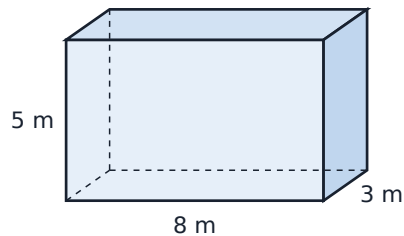
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q3/5

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q4/5

2077

QUESTION ID

S49-M-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q4/5

A cube-shaped box has edges that are 7 inches long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q5/5

2078

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

343 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 49 • MAIN • Q5/5

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$24 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q1/5

2080

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q1/5

2080

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$288 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q2/5

2081

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q2/5

2081

True or false?

$$36 \times 5 = 150$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q3/5

2082

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q3/5

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q4/5

2083

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 343

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$15 \times 24$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q5/5

2084

QUESTION ID

S49-B1-Q5

TYPE

Short answer

CORRECT ANSWER

360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 49 • BACKWARD 1 • Q5/5

2084

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q1/5

2086

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

288 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q1/5

True or false?

A 9-by-4 rectangle has an area of 36 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q2/5

2087

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q2/5

The base of a box is 8 m by 3 m.

What is the area of the base?

- A. 11 square meters
- B. 24 square meters
- C. 22 square meters
- D. 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q3/5

2088

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q3/5

A square has sides that are 7 inches long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q4/5

2089

QUESTION ID

S49-B2-Q4

TYPE

Short answer

CORRECT ANSWER

49 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q4/5

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q5/5

2090

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q1/5

2092

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1 $\frac{1}{8}$ cubic feet

GRADING NOTE $\frac{9}{8}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q1/5

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q2/5

2093

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

37 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q2/5

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

- A.** $6\frac{1}{4}$ cubic cm **B.** 15 cubic cm
C. $7\frac{1}{2}$ cubic cm **D.** $15\frac{5}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q3/5

2094

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15 $\frac{5}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q3/5

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q4/5

2095

QUESTION ID

S49-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1 1/2 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q4/5

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q5/5

2096

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world volume problems with right prisms • Grade 7 • Geometry • 7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q1/5

2098

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q1/5

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q2/5

2099

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

75 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q2/5

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

- A.** 3 inches
- B.** 30 inches
- C.** 5 inches
- D.** 60 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q3/5

2100

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q3/5

A rectangular prism is 4.5 cm by 2 cm by 3 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q4/5

2101

QUESTION ID

S49-F2-Q4

TYPE

Short answer

CORRECT ANSWER

27 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q4/5

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q5/5

2102

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • FORWARD 2 • Q5/5

Find a missing dimension when the volume is known

MAIN

Grade 5 main question

5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number

4.NBT.B.6

BACKWARD 2

Finding an unknown factor

3.OA.A.4

FORWARD 1

Solve one-step equations of the form $px = q$

6.EE.B.7

FORWARD 2

Solve two-step equations

7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q1/5

2105

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q1/5

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q2/5

2106

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

25 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q2/5

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A. 6 feet
- B. 30 feet
- C. 40 feet
- D. 227 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q3/5

2107

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q3/5

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q4/5

2108

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q4/5

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q5/5

2109

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 50 • MAIN • Q5/5

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q1/5

2111

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$150 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q2/5

2112

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q2/5

2112

True or false?

$$240 \div 8 = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q3/5

2113

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q3/5

$$84 \div 7 = ?$$

- A.** 12
- B.** 77
- C.** 11
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q4/5

2114

QUESTION ID

S50-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$60 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q5/5

2115

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 50 • BACKWARD 1 • Q5/5

2115

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q1/5

2117

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q1/5

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q2/5

2118

QUESTION ID

S50-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q2/5

What number goes in the box?

$$4 \times \square = 36$$

- A. 9
- B. 8
- C. 32
- D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q3/5

2119

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q3/5

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q4/5

2120

QUESTION ID

S50-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q4/5

True or false?

If $8 \times \square = 56$, then $\square = 6$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q5/5

2121

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q1/5

2123

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q1/5

Solve for b.

$$6b = 150$$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q2/5

2124

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q2/5

Solve for w.

$$28w = 84$$

- A.** 56
- B.** 2,352
- C.** 112
- D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q3/5

2125

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q3/5

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q4/5

2126

QUESTION ID

S50-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$15h = 52.5; h = 3.5$
cm**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q4/5

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q5/5

2127

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q1/5

2129

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE $20(x + 2) = 120$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q1/5

Solve for x.

$$6(x + 3) = 54$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q2/5

2130

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q2/5

Solve for h.

$$12h + 8 = 68$$

- A.** 5
- B.** 6.3
- C.** 60
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q3/5

2131

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q3/5

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q4/5

2132

QUESTION ID

S50-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$12h + 40 = 100;$$

$$h = 5 \text{ feet}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q4/5

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q5/5

2133

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 2 • Q5/5

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 3	The area formula for rectangles	4.MD.A.3
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q1/5

2136

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

112 cubic feet

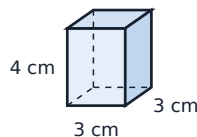
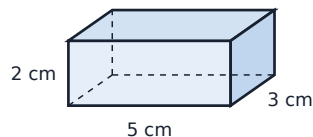
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q1/5

A figure is made of the two prisms shown, which do not overlap.

What is the total volume?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q2/5

2137

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

66 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q2/5

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units **B.** 8 cubic units
C. 10 cubic units **D.** 16 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q3/5

2138

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q3/5

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q4/5

2139

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

576 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q4/5

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q5/5

2140

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 51 • MAIN • Q5/5

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is split into two rectangles that do not overlap.
Their areas are 15 square units and 8 square units.
What is the area of the figure?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q1/5

2142

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

23 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q1/5

True or false?

The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q2/5

2143

QUESTION ID

S51-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

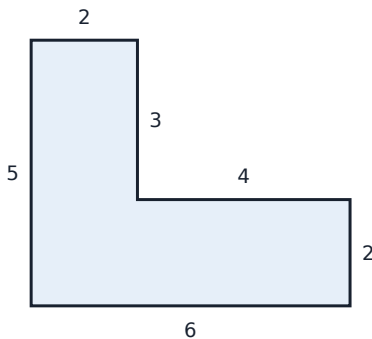
ID S51-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q2/5

The figure is split into a 6-by-2 rectangle along the bottom and a rectangle above it.

What is the area of the 6-by-2 rectangle?

- A.** 12 square units **B.** 16 square units
C. 6 square units **D.** 8 square units



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q3/5

2144

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q3/5

A figure is made of two squares that do not overlap.
Their areas are 9 square units and 25 square units.
To find the area of the figure, what do you do with the two areas?

- A.** Divide them
- B.** Multiply them
- C.** Subtract them
- D.** Add them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q4/5

2145

QUESTION ID

S51-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Add them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q4/5

A square with an area of 4 square units is cut out of a rectangle with an area of 48 square units.

What is the area of the part that is left?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q5/5

2146

QUESTION ID

S51-B1-Q5

TYPE

Short answer

CORRECT ANSWER

44 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 5.MD.C.5.c

BACKWARD 3

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 6 units by 4 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q1/5

2148

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q1/5

True or false?

A rectangle with an area of 32 square units and a length of 8 units has a width of 4 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q2/5

2149

QUESTION ID

S51-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q2/5

A rectangle is 10 m by 8 m.

What is its area?

- A.** 36 square meters
- B.** 800 square meters
- C.** 80 square meters
- D.** 18 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q3/5

2150

QUESTION ID

S51-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 80 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q3/5

A rectangle has an area of 45 square feet and a width of 5 feet.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q4/5

2151

QUESTION ID

S51-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q4/5

True or false?

A 4-by-8 rectangle has an area of 24 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q5/5

2152

QUESTION ID

S51-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q1/5

2154

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q1/5

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q2/5

2155

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q2/5

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

- A.** 2 cubic units **B.** $\frac{1}{2}$ cubic unit
- C.** 4 cubic units **D.** $1\frac{1}{2}$ cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q3/5

2156

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q3/5

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q4/5

2157

QUESTION ID

S51-F1-Q4

TYPE

Short answer

CORRECT ANSWER

27 1/2 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q4/5

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q5/5

2158

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q1/5

2160

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

96 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q1/5

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q2/5

2161

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3 cubic meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q2/5

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units **B.** 800 cubic units
C. 1,000 cubic units **D.** 40 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q3/5

2162

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 960 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q3/5

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q4/5

2163

QUESTION ID

S51-F2-Q4

TYPE

Short answer

CORRECT ANSWER

36 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q4/5

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q5/5

2164

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 51 • FORWARD 2 • Q5/5

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q1/5

2167

QUESTION ID

S52-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q1/5

In the ordered pair (7, 2), what does the 7 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q2/5

2168

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

**Start at the origin
and move 7 units
along the x-axis (to
the right).**

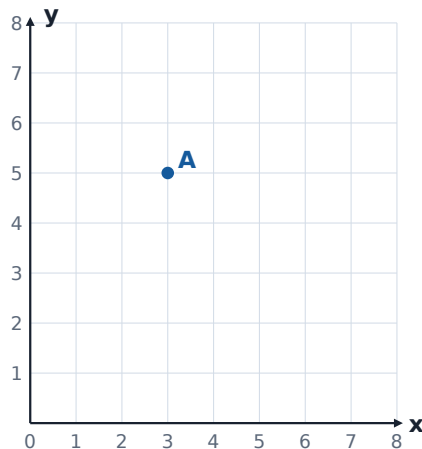
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q2/5

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q3/5

2169

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q3/5

True or false?

To reach (0, 6), you start at the origin and move 6 units to the right along the x-axis.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q4/5

2170

QUESTION ID

S52-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

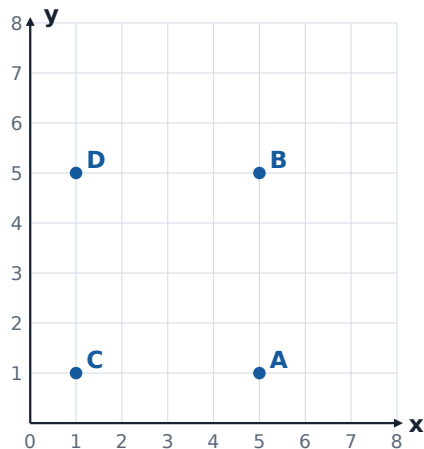
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q4/5

Which point is at (5, 1)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q5/5

2171

QUESTION ID

S52-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Point A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 52 • MAIN • Q5/5

SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q1/5

2173

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q1/5

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q2/5

2174

QUESTION ID

S52-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q2/5

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q3/5

2175

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q3/5

How many units is it from 0 to 9 on a number line?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q4/5

2176

QUESTION ID

S52-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q4/5

True or false?

On a number line, 3 is farther from 0 than 7 is.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q5/5

2177

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q1/5

2179

QUESTION ID

S52-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q1/5

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q2/5

2180

QUESTION ID

S52-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q2/5

How many degrees are in a right angle?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q3/5

2181

QUESTION ID

S52-B2-Q3

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q3/5

True or false?

Parallel lines never meet.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q4/5

2182

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q4/5

Which letter is made of two perpendicular line segments?

- A.** S
- B.** L
- C.** O
- D.** C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q5/5

2183

QUESTION ID

S52-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 52 • BACKWARD 2 • Q5/5

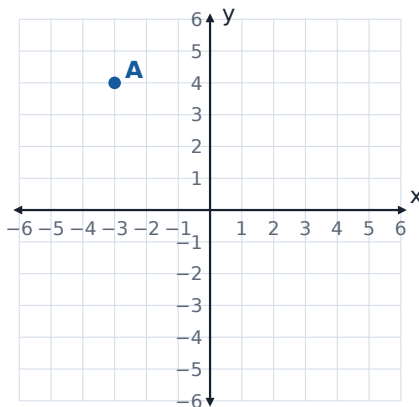
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q1/5

2185

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(-3, 4)

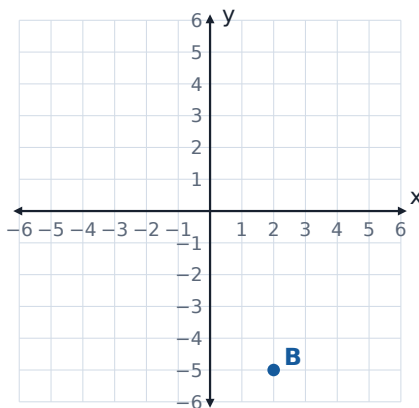
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q2/5

2186

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, -5)

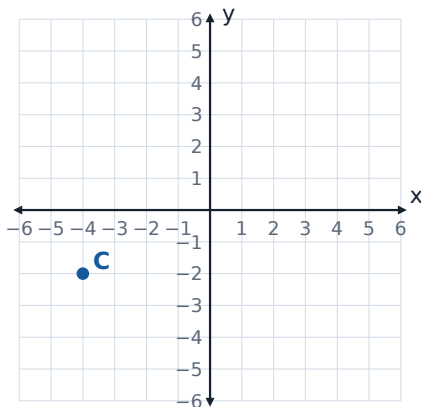
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q2/5

Which ordered pair names point C?

- A. $(-4, -2)$
- B. $(-2, -4)$
- C. $(4, 2)$
- D. $(-4, 2)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q3/5

2187

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(-4, -2)$

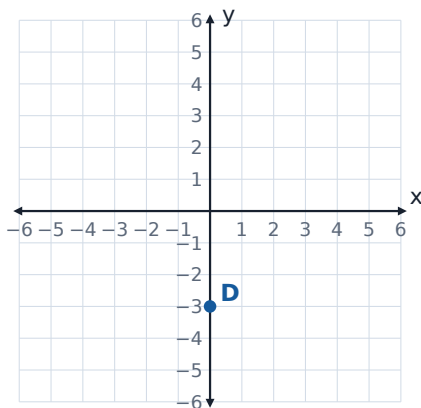
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q3/5

What are the coordinates of point D?

D = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q4/5

2188

QUESTION ID

S52-F1-Q4

TYPE

Short answer

CORRECT ANSWER

(0, -3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

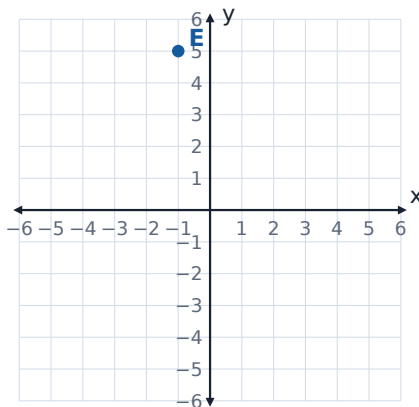
ID S52-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q4/5

True or false?

Point E is located at $(5, -1)$.

- A. True
- B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q5/5

2189

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 52 • FORWARD 1 • Q5/5

SET 52 • 5.G.A.1

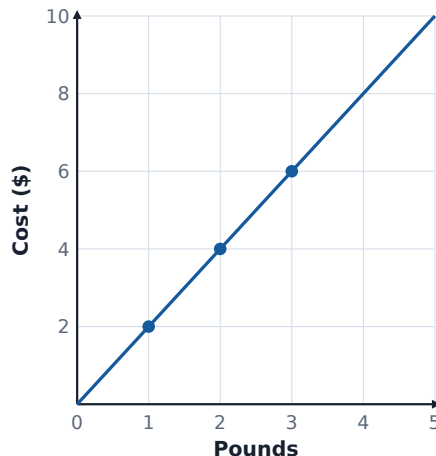
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q1/5

2191

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**3 pounds of pears
cost \$6.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

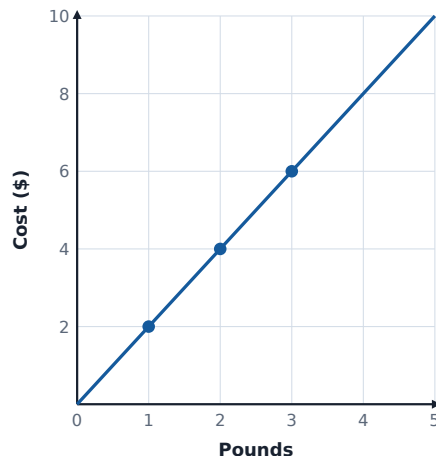
ID S52-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q1/5

The graph shows the cost of pears.

What does the point (1, 2) represent?

- A. The cost starts at \$2.
- B. 2 pounds cost \$1.
- C. \$1 buys 2 pounds.
- D. 1 pound costs \$2.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q2/5

2192

QUESTION ID

S52-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 1 pound costs \$2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q2/5

True or false?

On the graph of a proportional relationship between pounds and cost, the point (0, 0) means 0 pounds cost \$0.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q3/5

2193

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q3/5

The graph of a proportional relationship passes through (1, r) and (4, 20).

What is r ?

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q4/5

2194

QUESTION ID

S52-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q4/5

A graph shows a proportional relationship between the time x , in hours, that a bus drives and the distance y , in miles. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q5/5

2195

QUESTION ID

S52-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. In 2 hours, the distance is 90 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 52 • FORWARD 2 • Q5/5

Graph points in the first quadrant

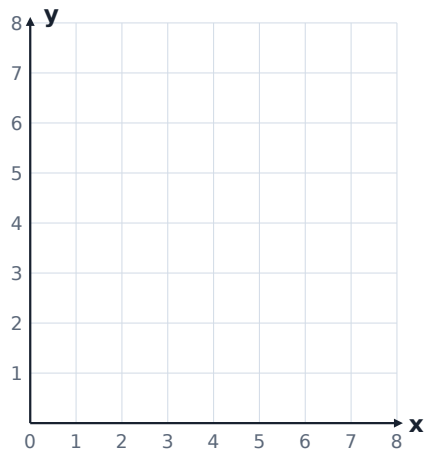
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Read a scaled bar graph	3.MD.B.3
BACKWARD 3	Fractions and whole numbers on a number line	3.NF.A.2
FORWARD 1	Reflect points across the axes	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points A(2, 7), B(6, 3), and C(0, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

2198

QUESTION ID

S53-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

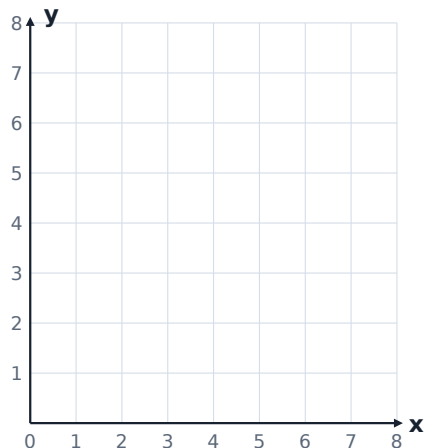
A plotted at (2, 7), B at (6, 3), and C at (0, 5) on the y-axis, each labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q2/5

2199

QUESTION ID

S53-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

D plotted at (4, 4), E at (7, 0) on the x-axis, and F at (1, 6), each labeled

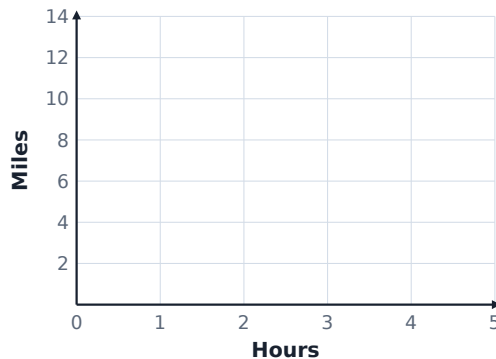
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q2/5

Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q3/5

2200

QUESTION ID

S53-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

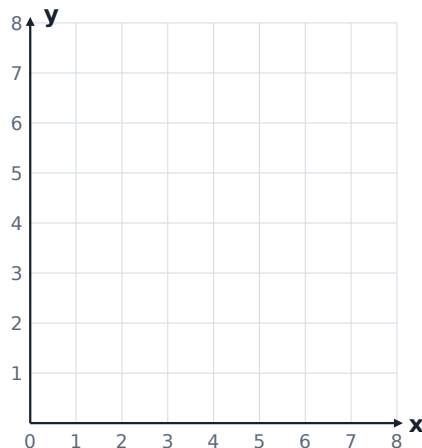
Points plotted at (1, 3), (2, 6), (3, 9), (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q3/5

Plot and label the points G(3, $2\frac{1}{2}$) and H($5\frac{1}{2}$, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

2201

QUESTION ID

S53-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

G plotted at (3, $2\frac{1}{2}$), halfway between 2 and 3 on the y-axis scale; H plotted at ($5\frac{1}{2}$, 4); each labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

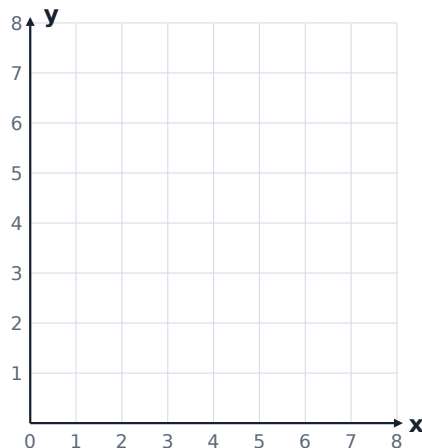
ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

Plot P(2, 3) and Q(2, 7).

How far apart are the two points?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q5/5

ID S53-M-Q5

2202

QUESTION ID

S53-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: P plotted at (2, 3) and Q plotted at (2, 7)

Written: 4 units

GRADING NOTE Grade both: both points at the correct places, and the distance 4 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q5/5

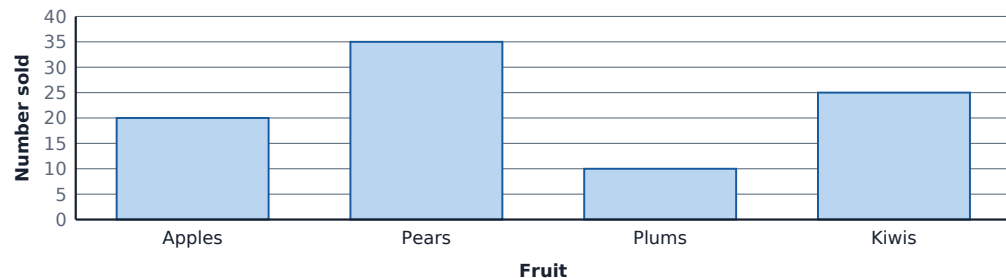
ID S53-M-Q5

SET 53 • 5.G.A.2

BACKWARD 2

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

What does each grid line on the scale stand for?



- A.** 10 **B.** 20 **C.** 5 **D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q1/5

2204

QUESTION ID

S53-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

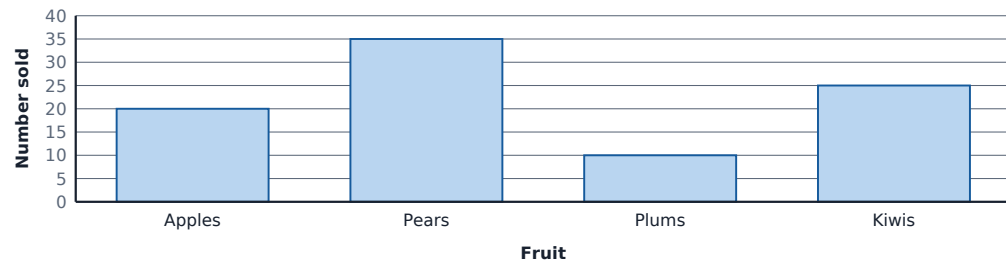
C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q1/5

How many pears were sold?



Pears: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q2/5

2205

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

35

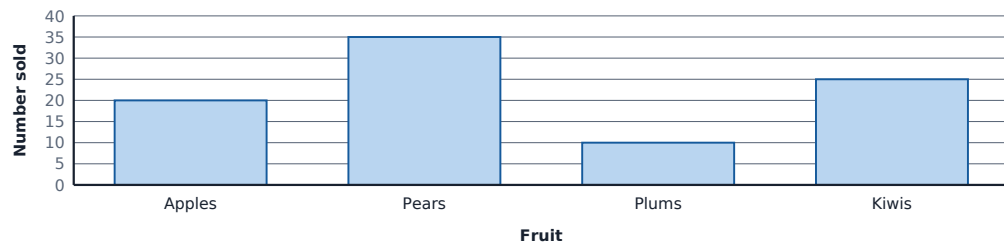
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q2/5

True or false?

More kiwis than apples were sold.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q3/5

2206

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

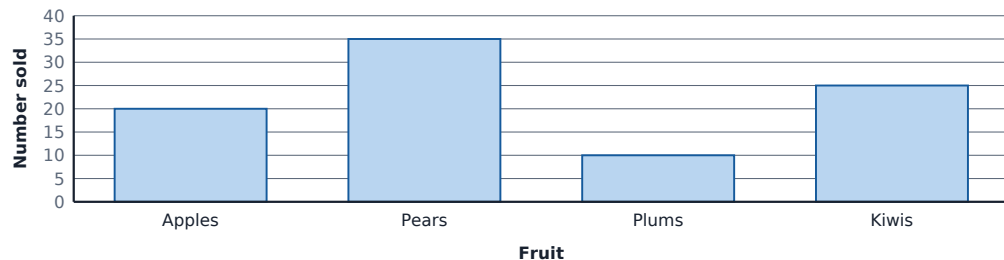
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q3/5

How many fewer plums than pears were sold?



Plums: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q4/5

2207

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

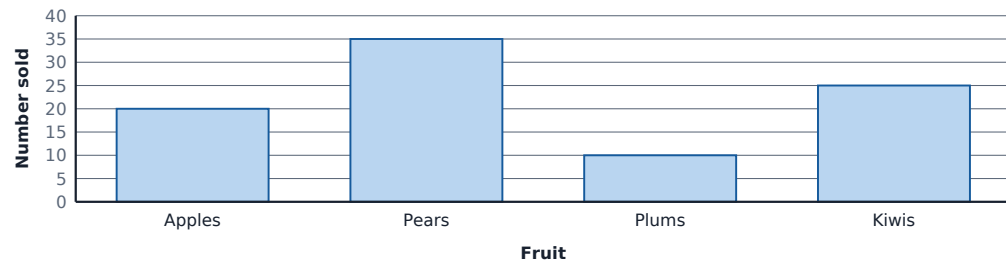
25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q4/5

How many apples and kiwis were sold in all?



Fruit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q5/5

2208

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

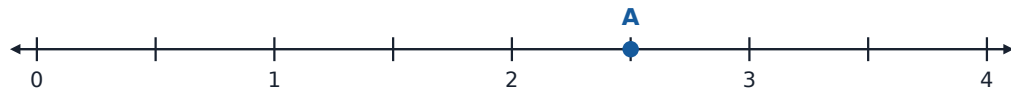
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 5.G.A.2

BACKWARD 3

Fractions and whole numbers on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q1/5

2210

QUESTION ID

S53-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/2

GRADING NOTE 5/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q1/5

True or false?

$2\frac{1}{2}$ is halfway between 2 and 3 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q2/5

2211

QUESTION ID

S53-B3-Q2

TYPE

True or false

CORRECT ANSWER

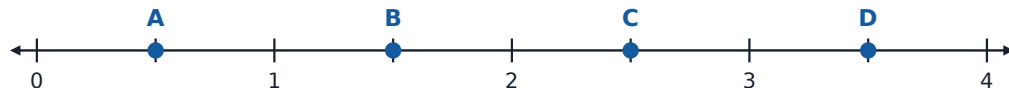
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q2/5

Which point is at $1\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q3/5

2212

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q4/5

2213

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q4/5

True or false?

On a number line, $5\frac{1}{2}$ is between 6 and 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q5/5

2214

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

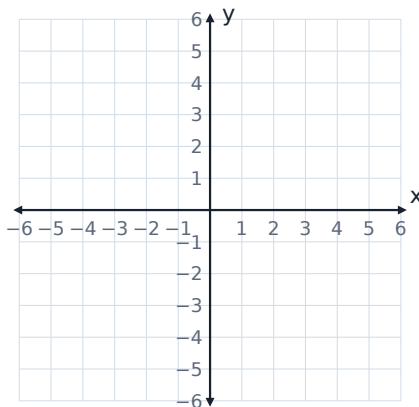
Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • 5.G.A.2

FORWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Plot and label A(2, 4). Then plot its reflection across the x-axis and label it B.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q1/5

2216

QUESTION ID

S53-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**A plotted at (2, 4)
and B at (2, -4),
each labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q1/5

The point $(5, 2)$ is reflected across the x-axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q2/5

2217

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(5, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q2/5

The point $(-3, 5)$ is reflected across the x-axis.

What are the coordinates of its image?

- A. $(3, -5)$
- B. $(3, 5)$
- C. $(-3, -5)$
- D. $(5, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q3/5

2218

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(-3, -5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q3/5

The point $(-4, 1)$ is reflected across the y -axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q4/5

2219

QUESTION ID

S53-F1-Q4

TYPE

Short answer

CORRECT ANSWER

(4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q4/5

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q5/5

2220

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 53 • FORWARD 1 • Q5/5

SET 53 • 5.G.A.2

FORWARD 2

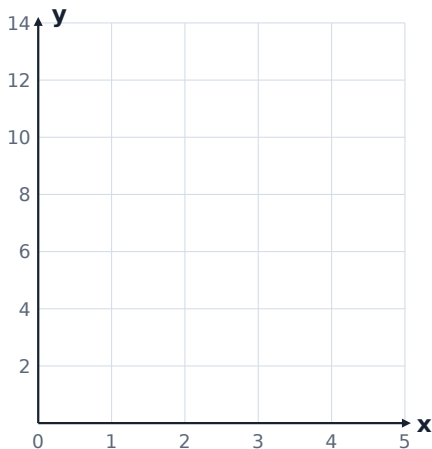
Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6), and (4, 12).

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q1/5

2222

QUESTION ID

S53-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at (1, 3), (2, 6) and (4, 12)

Written: Yes; the points lie on a straight line through (0, 0)

GRADING NOTE Grade both: all three points at the correct places, and "yes" with the reason (a straight line through the origin, or $y = 3x$ for every point).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

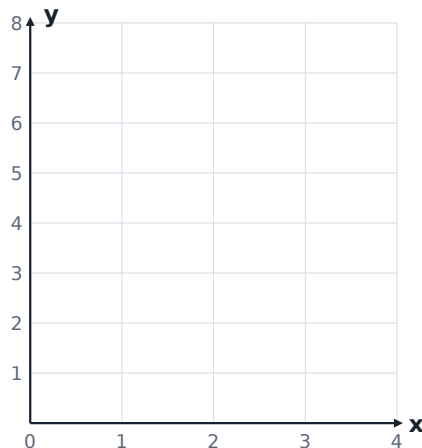
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q1/5

Graph the pairs $(0, 2)$, $(1, 4)$, and $(2, 6)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q2/5

2223

QUESTION ID

S53-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at $(0, 2)$, $(1, 4)$ and $(2, 6)$

Written: No; the points are on a line, but the line does not pass through $(0, 0)$

GRADING NOTE Grade both: all three points at the correct places, and "no" with the reason (the line does not go through the origin; when x is 0, y is 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q2/5

Which graph shows a proportional relationship?

- A. A curve through (0, 0)
- B. Points that are not on a line
- C. A straight line through (0, 3)
- D. A straight line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q3/5

2224

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. A straight line
through (0, 0)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The points (2, 1), (4, 2), and (6, 3) show a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q4/5

2225

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q4/5

Graph the pairs from the table.

Is the relationship proportional? Explain using your graph.

x	1	2	3
y	4	8	12



Answer: _____

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q5/5

2226

QUESTION ID

S53-F2-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points plotted at (1, 4), (2, 8) and (3, 12)

Written: Yes; the points lie on a straight line through (0, 0)

GRADING NOTE Grade both: all three points at the correct places, and "yes" with the reason (a straight line through the origin, or $y = 4x$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 53 • FORWARD 2 • Q5/5

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
BACKWARD 3	Solve problems with a scaled bar graph	3.MD.B.3
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

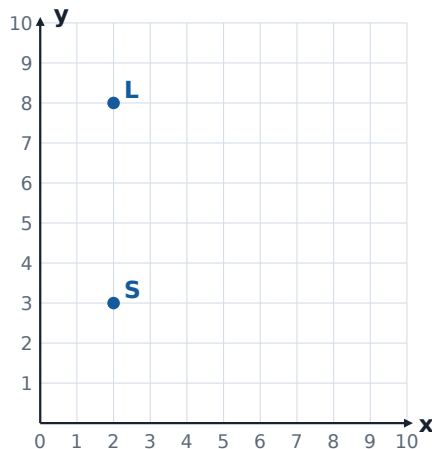
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q1/5

2229

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

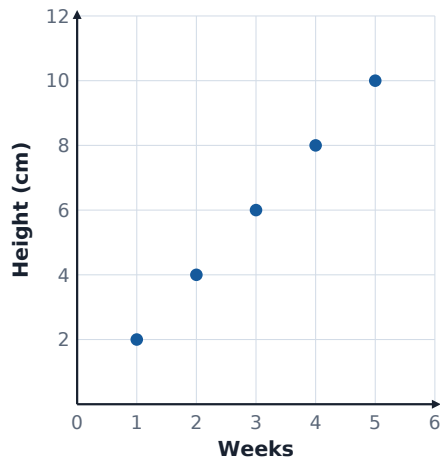
ID S54-M-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q1/5

The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q2/5

2230

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q2/5

The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

What does the point mean?

- A.** Working 3 hours earns \$12. **B.** Working 12 hours earns \$3.
C. The pay starts at \$12. **D.** Each hour earns \$3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q3/5

2231

QUESTION ID

S54-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. Working 3 hours
earns \$12.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

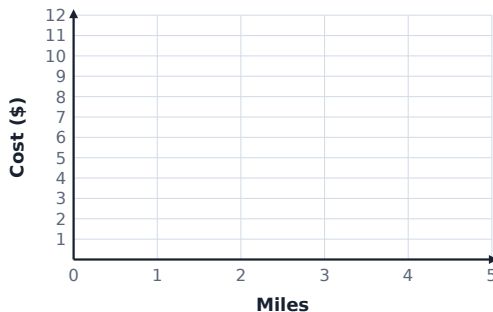
ID S54-M-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q3/5

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q4/5

2232

QUESTION ID

S54-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 5), (2, 7), (3, 9), (4, 11)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q4/5

On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q5/5

2233

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

600 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 54 • MAIN • Q5/5

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q1/5

2235

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q1/5

True or false?

The distance from 3 to 10 is 13 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q2/5

2236

QUESTION ID

S54-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q2/5

How many units apart are 1 and 8 on a number line?

- A.** 9
- B.** 8
- C.** 1
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q3/5

2237

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q3/5

What is the distance from 6 to 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q4/5

2238

QUESTION ID

S54-B2-Q4

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q4/5

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q5/5

2239

QUESTION ID

S54-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

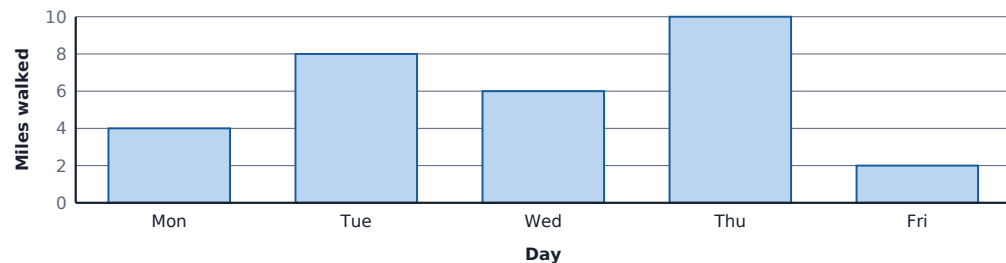
Grade 2 • Measurement & Data • 2.MD.B.6 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 5.G.A.2

BACKWARD 3

Solve problems with a scaled bar graph • Grade 3 • Measurement & Data •
3.MD.B.3

How many miles were walked on Thursday?



Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q1/5

2241

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

10

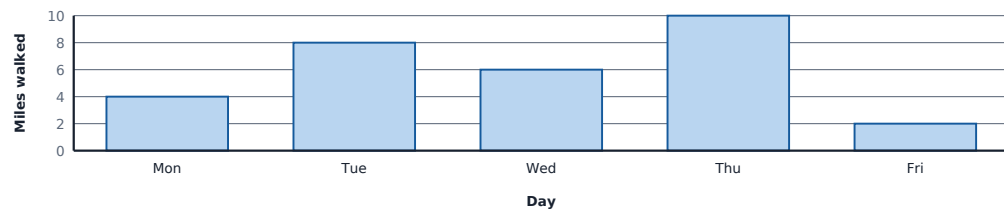
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q1/5

True or false?

The same number of miles were walked on Monday and Wednesday.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q2/5

2242

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

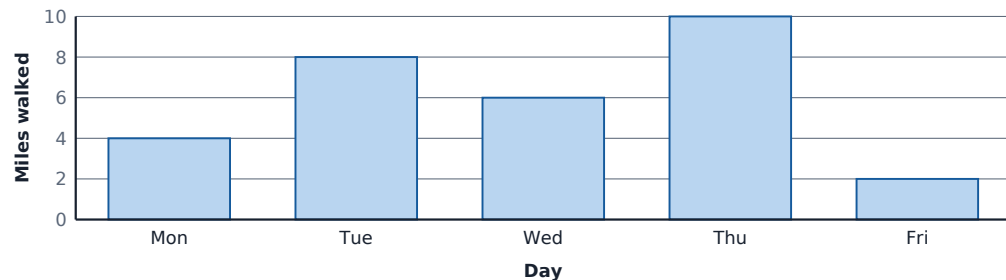
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q2/5

On which day were 8 miles walked?



A. Thursday B. Friday C. Monday D. Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

2243

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

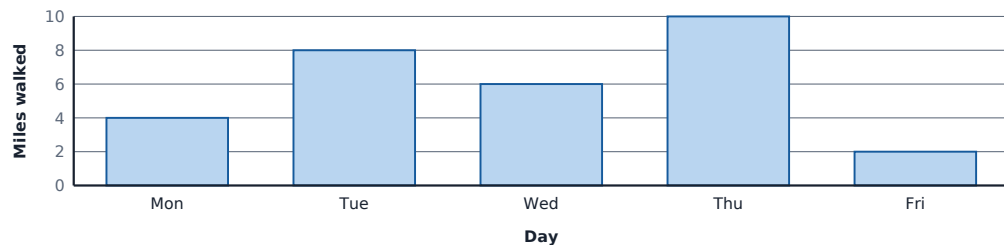
D. Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

How many more miles were walked on Thursday than on Friday?



Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q4/5

2244

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

8

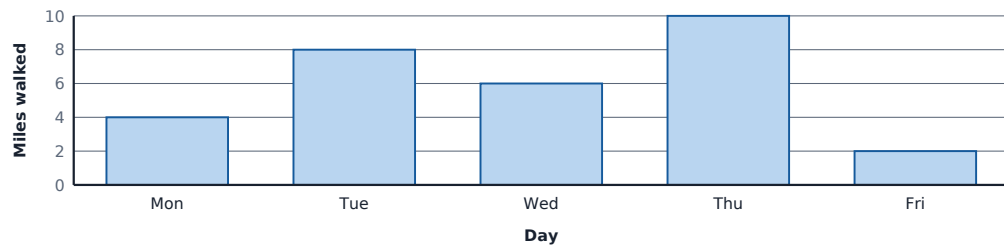
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q4/5

True or false?

Each grid line on the scale stands for 2 miles.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q5/5

2245

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q1/5

2247

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q1/5

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q2/5

2248

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q2/5

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

- A.** 7 blocks **B.** 1 block
C. 4 blocks **D.** 6 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q3/5

2249

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q3/5

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q4/5

2250

QUESTION ID

S54-F1-Q4

TYPE

Short answer

CORRECT ANSWER

16 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q4/5

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q5/5

2251

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q1/5

2253

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q1/5

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q2/5

2254

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q2/5

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

- A.** 12 m **B.** 35 m
C. 2 m **D.** 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q3/5

2255

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q3/5

A garden plan has corners at (0, 0), (6, 0), (6, 4), and (0, 4). Each unit is 3 feet.

What are the real length and width?

Length and width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q4/5

2256

QUESTION ID

S54-F2-Q4

TYPE

Short answer

CORRECT ANSWER

18 feet by 12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q5/5

2257

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 54 • FORWARD 2 • Q5/5

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Draw shapes with given conditions	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

Every parallelogram has opposite sides of equal length. A rhombus is a parallelogram, so every rhombus has opposite sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q1/5

2260

QUESTION ID

S55-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q1/5

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A. Kite
- B. Pentagon
- C. Triangle
- D. Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q2/5

2261

QUESTION ID

S55-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q2/5

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q3/5

ID S55-M-Q3

2262

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE A square is a quadrilateral with 4 sides of equal length, so it has every attribute of a rhombus.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q3/5

ID S55-M-Q3

True or false?

All quadrilaterals have 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q4/5

2263

QUESTION ID

S55-M-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q4/5

Which statement is true?

- A.** Every parallelogram is a square.
- B.** Every square is a rectangle.
- C.** Every quadrilateral is a rectangle.
- D.** Every rectangle is a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q5/5

2264

QUESTION ID

S55-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Every square is a rectangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 55 • MAIN • Q5/5

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q1/5

2266

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q1/5

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q2/5

2267

QUESTION ID

S55-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q2/5

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q3/5

2268

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q3/5

True or false?

A square has 4 sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q4/5

2269

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q4/5

Name a quadrilateral that always has 4 right angles.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q5/5

2270

QUESTION ID

S55-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Rectangle

GRADING NOTE Square is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q1/5

2272

QUESTION ID

S55-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q1/5

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q2/5

2273

QUESTION ID

S55-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q2/5

What do we call two lines that meet to form right angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q3/5

2274

QUESTION ID

S55-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Perpendicular lines

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q3/5

True or false?

An obtuse angle is smaller than a right angle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q4/5

2275

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q4/5

How many pairs of parallel sides does a rectangle have?

- A.** 4
- B.** 0
- C.** 2
- D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q5/5

2276

QUESTION ID

S55-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q1/5

2278

QUESTION ID

S55-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q1/5

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q2/5

2279

QUESTION ID

S55-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. A right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q2/5

How many right angles does a square have?

Right angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q3/5

2280

QUESTION ID

S55-B3-Q3

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q3/5

True or false?

A rhombus always has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q4/5

2281

QUESTION ID

S55-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q4/5

Which shape always has two pairs of parallel sides?

- A.** Rectangle
- B.** Kite
- C.** Triangle
- D.** Pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q5/5

2282

QUESTION ID

S55-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • 5.G.B.3

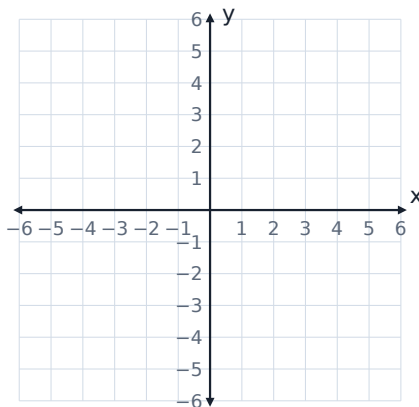
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot A(−2, 1), B(3, 1), C(3, −3), and D(−2, −3). Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q1/5

2284

QUESTION ID

S55-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A, B, C and D plotted and joined in order A-B-C-D-A

Written: Rectangle

GRADING NOTE Grade both: the four points and the four connecting segments, and the name. The sides are 5 and 4 units with four right angles, so "rectangle" is the most specific name. "Parallelogram" or "quadrilateral" are true but not the most specific; "square" is wrong.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

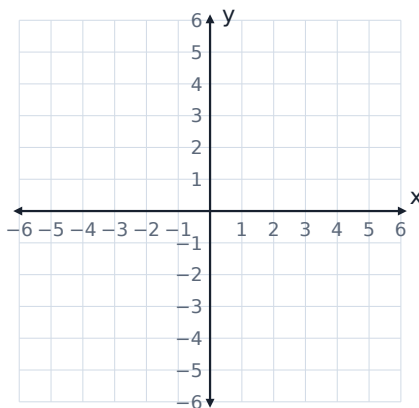
ID S55-F1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q1/5

Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q2/5

2285

QUESTION ID

S55-F1-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: E, F, G and H plotted and joined in order E-F-G-H-E

Written: Square

GRADING NOTE Grade both: the four points and the four connecting segments, and the name. All four sides are equal and all four angles are right angles, so "square" is the most specific name. "Rhombus", "rectangle", "parallelogram" and "quadrilateral" are true but not the most specific.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q2/5

The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

- A.** $(1, -1)$ **B.** $(-1, -1)$
C. $(-1, 4)$ **D.** $(4, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q3/5

2286

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(-1, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q3/5

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q4/5

2287

QUESTION ID

S55-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q4/5

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q5/5

2288

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 55 • FORWARD 1 • Q5/5

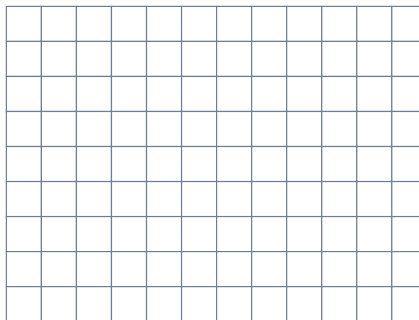
SET 55 • 5.G.B.3

FORWARD 2

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a triangle with sides of 3 units, 4 units, and 5 units. Put the 3-unit and 4-unit sides along grid lines.

Classify the triangle by its angles.



Kind of triangle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q1/5

2290

QUESTION ID

S55-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A triangle with a 3-unit side and a 4-unit side along grid lines meeting at a right angle, and the third side (5 units) joining their ends

Written: Right triangle

GRADING NOTE Grade both: the drawn side lengths 3 and 4 along grid lines with the third side joining them, and "right triangle".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q1/5

Can a triangle have sides of 2 cm, 2 cm, and 5 cm?
Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q2/5

2291

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE $2 + 2 = 4$ is less than 5, so the two short sides cannot meet to close the triangle.

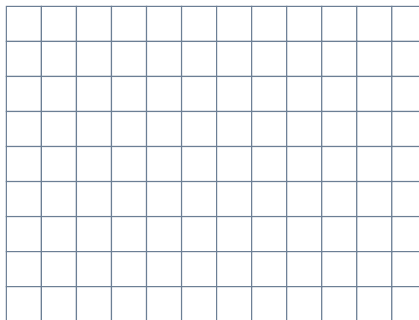
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q2/5

Draw a triangle with angles of 90° , 45° , and 45° . Make its two shorter sides 4 units long, along grid lines.

Could a triangle with these same angles be drawn in a different size?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q3/5

2292

QUESTION ID

S55-F2-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A right triangle with two 4-unit legs along grid lines meeting at a right angle, and the third side joining their ends

Written: Yes

GRADING NOTE Grade both: the drawing (two 4-unit legs at a right angle) and "yes": three angles do not fix the size, so larger or smaller triangles with the same angles can be drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q3/5

Can a triangle have angles of 100° , 50° , and 40° ?
Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q4/5

2293

QUESTION ID

S55-F2-Q4

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE The angles of a triangle add to 180° , but $100 + 50 + 40 = 190$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

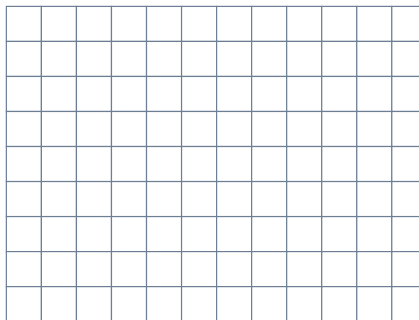
Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q4/5

2293

Draw a quadrilateral that has exactly one pair of parallel sides.

What is the name of this kind of quadrilateral?

Name: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q5/5

2294

QUESTION ID

S55-F2-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A quadrilateral with one pair of parallel sides (for example a horizontal top and bottom of different lengths) and one pair of sides that are not parallel

Written: Trapezoid

GRADING NOTE Grade both: exactly one pair of parallel sides in the drawing, and the name "trapezoid".

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 55 • FORWARD 2 • Q5/5

Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Draw quadrilaterals and triangles with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A. Square, rectangle, parallelogram, quadrilateral
- B. Rectangle, square, quadrilateral, parallelogram
- C. Quadrilateral, parallelogram, rectangle, square
- D. Parallelogram, quadrilateral, square, rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q1/5

2297

QUESTION ID

S56-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**C. Quadrilateral,
parallelogram,
rectangle, square**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q1/5

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q2/5

2298

QUESTION ID

S56-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q2/5

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q3/5

2299

QUESTION ID

S56-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Quadrilateral and
parallelogram**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q3/5

A shape has 4 sides of equal length and 4 right angles.

What is the most specific name for it?

- A. Rectangle
- B. Rhombus
- C. Square
- D. Quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q4/5

2300

QUESTION ID

S56-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q4/5

True or false?

Every parallelogram is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q5/5

2301

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 56 • MAIN • Q5/5

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q1/5

2303

QUESTION ID

S56-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q1/5

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q2/5

2304

QUESTION ID

S56-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q2/5

How many angles does a quadrilateral have?

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q3/5

2305

QUESTION ID

S56-B1-Q3

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

Every quadrilateral is a square.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q4/5

2306

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q4/5

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q5/5

2307

QUESTION ID

S56-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A trapezoid with exactly one pair of parallel sides

GRADING NOTE Any quadrilateral with fewer than two pairs of parallel sides is correct, for example a trapezoid with exactly one pair of parallel sides or a kite with no parallel sides. A rhombus, rectangle or square is wrong because each is a parallelogram.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

2309

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q1/5

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

2310

QUESTION ID

S56-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q2/5

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

2311

QUESTION ID

S56-B2-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

The sides of a square that meet at a corner are perpendicular.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

2312

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q4/5

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

2313

QUESTION ID

S56-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 4 sides of equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q1/5

2315

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

60 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q1/5

A rhombus has a base of 7 in. and a height of 5 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q2/5

2316

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q2/5

A trapezoid has bases of 4 m and 10 m and a height of 6 m. It is cut into a 4-m by 6-m rectangle and two triangles, each with a base of 3 m and a height of 6 m.

What is the area of the trapezoid?

- A.** 84 square meters **B.** 42 square meters
C. 60 square meters **D.** 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q3/5

2317

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 42 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q3/5

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q4/5

2318

QUESTION ID

S56-F1-Q4

TYPE

Short answer

CORRECT ANSWER

32 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q4/5

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q5/5

2319

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 5.G.B.4

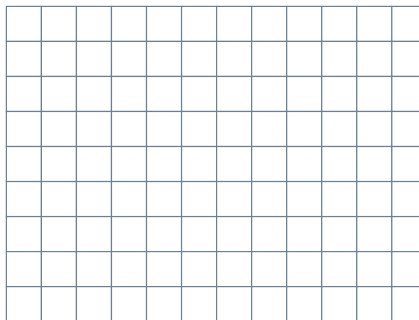
FORWARD 2

Draw quadrilaterals and triangles with given conditions • Grade 7 • Geometry •
7.G.A.2

Draw a rectangle that is not a square.

Write its side lengths.

Side lengths: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q1/5

ID S56-F2-Q1

2321

QUESTION ID

S56-F2-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A rectangle drawn along grid lines with two different side lengths

Written: Two different lengths that match the drawing, for example 6 units by 3 units

GRADING NOTE Grade both: four right angles with unequal adjacent sides in the drawing, and side lengths that match it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q1/5

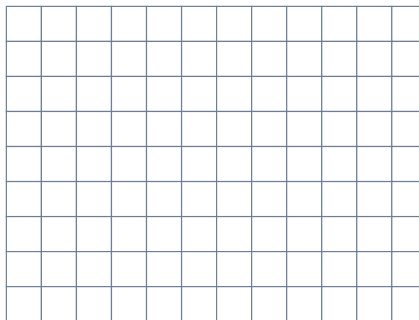
ID S56-F2-Q1

Draw a rhombus that is not a square, with every side 5 units long.

(Hint: a side can go 3 units across and 4 units up.)

How do you know it is not a square?

Reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q2/5

ID S56-F2-Q2

2322

QUESTION ID

S56-F2-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A quadrilateral with four 5-unit sides whose angles are not right angles, for example with vertices (0, 0), (5, 0), (8, 4) and (3, 4)

Written: Its angles are not right angles.

GRADING NOTE Grade both: four equal sides that are not perpendicular, and a reason that names the angles (not right angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q2/5

ID S56-F2-Q2

Can a quadrilateral have exactly 3 right angles? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q3/5

2323

QUESTION ID

S56-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE The angles of a quadrilateral add to 360° . If three are 90° , the fourth is $360 - 270 = 90^\circ$, so it has 4 right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

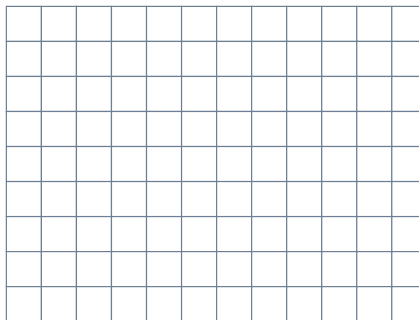
ID S56-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q3/5

Draw a parallelogram that is not a rectangle, with a base of 6 units and a height of 3 units.

What is its area?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q4/5

ID S56-F2-Q4

2324

QUESTION ID

S56-F2-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A parallelogram with a 6-unit base along a grid line and the opposite side 3 units above it, shifted sideways so the angles are not right angles

Written: 18 square units

GRADING NOTE Grade both: the drawing meets all three conditions, and the area 18 square units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q4/5

ID S56-F2-Q4

A triangle has sides of 6 cm, 6 cm, and 6 cm. Can triangles of different shapes be drawn with these sides? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q5/5

2325

QUESTION ID

S56-F2-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Three side lengths determine one unique triangle (here an equilateral triangle); every copy is the same shape and size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 56 • FORWARD 2 • Q5/5

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Right, acute, and obtuse triangles	4.G.A.2
BACKWARD 2	Kinds of angles	4.G.A.1
BACKWARD 3	Triangles	2.G.A.1
FORWARD 1	Area of triangles	6.G.A.1
FORWARD 2	Write and solve equations for unknown angles	7.G.B.5

SET 57 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A. Obtuse
- B. Scalene
- C. Equilateral
- D. Right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q1/5

2328

QUESTION ID

S57-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Equilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q1/5

A triangle has angles of 90° , 50° , and 40° .

Classify it by its angles.

Classification: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q2/5

2329

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q2/5

True or false?

An isosceles triangle has at least two sides of equal length. Every equilateral triangle is also an isosceles triangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q3/5

2330

QUESTION ID

S57-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q3/5

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

- A.** Acute equilateral
- B.** Right scalene
- C.** Right isosceles
- D.** Obtuse scalene

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q4/5

2331

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Obtuse scalene

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q4/5

A triangle has angles of 70° , 70° , and 40° .

Classify it by its angles and by its sides.

By angles: _____

By sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q5/5

2332

QUESTION ID

S57-M-Q5

TYPE

Short answer

CORRECT ANSWER

Acute; isosceles

GRADING NOTE Two equal angles mean two equal sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 57 • MAIN • Q5/5

SET 57 • 5.G.B.4

BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q1/5

2334

QUESTION ID

S57-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q1/5

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q2/5

2335

QUESTION ID

S57-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q2/5

A triangle has one angle greater than 90° .

What kind of triangle is it?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q3/5

2336

QUESTION ID

S57-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Obtuse triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q3/5

True or false?

In an acute triangle, all three angles are less than 90° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q4/5

2337

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q4/5

Which angle measure is acute?

- A.** 100°
- B.** 180°
- C.** 60°
- D.** 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q5/5

2338

QUESTION ID

S57-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 5.G.B.4

BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q1/5

2340

QUESTION ID

S57-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q1/5

What kind of angle measures 35° ?

- A.** Right
- B.** Straight
- C.** Obtuse
- D.** Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q2/5

2341

QUESTION ID

S57-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q2/5

What kind of angle measures 150° ?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q3/5

2342

QUESTION ID

S57-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q3/5

True or false?

Two right angles together make 360° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q4/5

2343

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q4/5

What kind of angle measures exactly 90° ?

Kind: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q5/5

2344

QUESTION ID

S57-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 5.G.B.4

BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q1/5

2346

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q1/5

True or false?

A triangle has 3 angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q2/5

2347

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q2/5

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q3/5

2348

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q3/5

True or false?

All triangles have sides of equal length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q4/5

2349

QUESTION ID

S57-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q4/5

How many vertices (corners) does a triangle have?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q5/5

2350

QUESTION ID

S57-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q1/5

2352

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

24 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q1/5

A triangle has a base of 10 in. and a height of 7 in.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q2/5

2353

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q2/5

An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters **B.** 40 square meters
C. 10 square meters **D.** 9 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q3/5

2354

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10 square meters

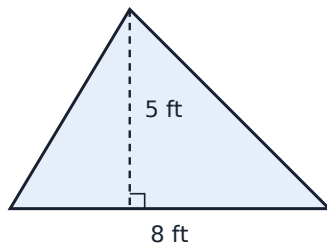
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q3/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q4/5

2355

QUESTION ID

S57-F1-Q4

TYPE

Short answer

CORRECT ANSWER

20 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q4/5

True or false?

Two triangles with the same base and the same height have the same area.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q5/5

2356

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 5.G.B.4

FORWARD 2

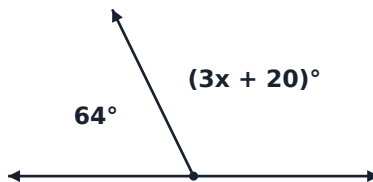
Write and solve equations for unknown angles • Grade 7 • Geometry • 7.G.B.5

The two angles form a straight line.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q1/5

2358

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$(3x + 20) + 64 = 180; x = 32$$

GRADING NOTE Any equivalent equation is correct (for example $3x + 84 = 180$). Both the equation and $x = 32$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

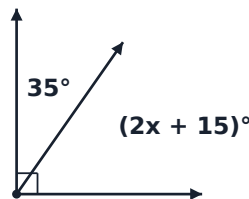
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q1/5

The two angles together form a right angle.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q2/5

2359

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$$(2x + 15) + 35 = 90;$$
$$x = 20$$

GRADING NOTE Any equivalent equation is correct (for example $2x + 50 = 90$). Both the equation and $x = 20$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

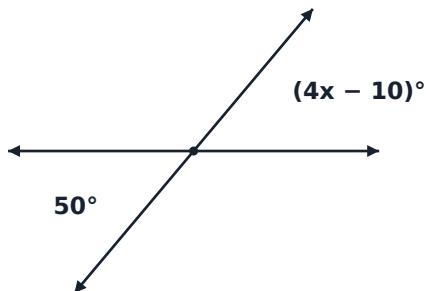
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q2/5

Two lines cross. The marked angles are vertical angles.

Write an equation and solve for x .

Equation: _____

$x =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q3/5

ID S57-F2-Q3

2360

QUESTION ID

S57-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$4x - 10 = 50; x = 15$

GRADING NOTE Vertical angles are equal. Any equivalent equation is correct. Both the equation and $x = 15$ are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

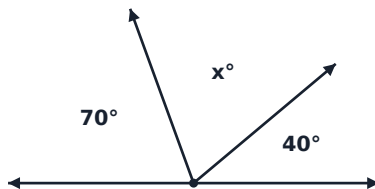
Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q3/5

ID S57-F2-Q3

Three angles together form a straight line.

Which equation can be used to find x ?

- A. $x = 40 \times 70$
- B. $40 + x + 70 = 180$
- C. $40 + x = 70$
- D. $40 + x + 70 = 90$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q4/5

2361

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $40 + x + 70 = 180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q4/5

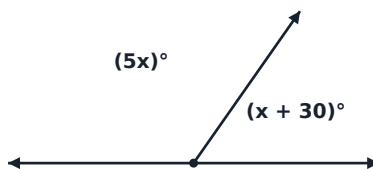
Angle ABC and angle CBD together form a straight line. Angle ABC is $(5x)^\circ$ and angle CBD is $(x + 30)^\circ$.

Write an equation, solve for x , and find both angles.

Equation: _____

$x =$ _____

Angles: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q5/5

ID S57-F2-Q5

2362

QUESTION ID

S57-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**$5x + (x + 30) = 180;$
 $x = 25;$ angle
ABC = 125° , angle
CBD = 55°**

GRADING NOTE Any equivalent equation is correct ($6x + 30 = 180$). All three parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 7 • Geometry • 7.G.B.5 • Set 57 • FORWARD 2 • Q5/5

ID S57-F2-Q5

Place triangles in a hierarchy of categories and subcategories

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Classify triangles by their angles	4.G.A.2
BACKWARD 2	Shapes in different categories share attributes	3.G.A.1
FORWARD 1	Area of special triangles	6.G.A.1
FORWARD 2	Decide whether conditions give one triangle, more than one, or no...	7.G.A.2

SET 71 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A. Triangle, isosceles triangle, equilateral triangle
- B. Isosceles triangle, triangle, equilateral triangle
- C. Triangle, equilateral triangle, isosceles triangle
- D. Equilateral triangle, isosceles triangle, triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q1/5

2365

QUESTION ID

S71-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Triangle, isosceles triangle, equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q1/5

True or false?

Every right isosceles triangle belongs to both the category of right triangles and the category of isosceles triangles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q2/5

2366

QUESTION ID

S71-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q2/5

Isosceles means at least two sides of equal length.

A triangle has sides of 7, 7, and 7 units. Which of these categories does it belong to?

triangle, isosceles, equilateral, scalene

Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q3/5

2367

QUESTION ID

S71-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Triangle, isosceles,
and equilateral**

GRADING NOTE All three are required. Scalene is wrong because a scalene triangle has no sides of equal length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q3/5

2367

Isosceles means at least two sides of equal length.

Which statement is true?

- A. Some isosceles triangles are equilateral.
- B. No right triangle is isosceles.
- C. Every isosceles triangle is equilateral.
- D. Some scalene triangles are isosceles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q4/5

2368

QUESTION ID

S71-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Some isosceles triangles are equilateral.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q4/5

Explain why a triangle cannot be both a right triangle and an obtuse triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q5/5

2369

QUESTION ID

S71-M-Q5

TYPE

Short answer

CORRECT ANSWER

A right triangle has a 90° angle and an obtuse triangle has an angle greater than 90° . The three angles of a triangle add to 180° , so these two angles together would already be more than 180° .

GRADING NOTE Accept the 180° angle-sum reason, or that a triangle can have only one angle of 90° or more because the two sides would never meet. The categories "right" and "obtuse" do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 71 • MAIN • Q5/5

SET 71 • 5.G.B.4

BACKWARD 1

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A triangle with angles of 30° , 60° , and 90° is a right triangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q1/5

2371

QUESTION ID

S71-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q1/5

A triangle has angles of 110° , 40° , and 30° .

What kind of triangle is it?

- A.** Equilateral
- B.** Acute
- C.** Right
- D.** Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q2/5

2372

QUESTION ID

S71-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Obtuse

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q2/5

A triangle has angles of 50° , 60° , and 70° .

Is it acute, right, or obtuse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q3/5

2373

QUESTION ID

S71-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Acute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q3/5

True or false?

A triangle can have two obtuse angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q4/5

2374

QUESTION ID

S71-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q4/5

Which three angles could be the angles of an acute triangle?

- A. 100° , 50° , 30°
- B. 120° , 30° , 30°
- C. 80° , 60° , 40°
- D. 90° , 45° , 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q5/5

2375

QUESTION ID

S71-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 80° , 60° , 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • 5.G.B.4

BACKWARD 2

Shapes in different categories share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

Rhombuses, rectangles, and squares are all quadrilaterals.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q1/5

2377

QUESTION ID

S71-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q1/5

What do all quadrilaterals have in common?

- A.** 4 right angles
- B.** 4 sides
- C.** 2 pairs of parallel sides
- D.** 4 sides of equal length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q2/5

2378

QUESTION ID

S71-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 4 sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q2/5

Name a quadrilateral that is not a rectangle.

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q3/5

2379

QUESTION ID

S71-B2-Q3

TYPE

Short answer

CORRECT ANSWER

**A rhombus that is
not a square**

GRADING NOTE Any quadrilateral that is not a rectangle is correct, for example a trapezoid or a rhombus that is not a square.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q3/5

True or false?

Every quadrilateral is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q4/5

2380

QUESTION ID

S71-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q4/5

A square is also a ____.

- A.** rectangle
- B.** pentagon
- C.** hexagon
- D.** triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q5/5

2381

QUESTION ID

S71-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • 5.G.B.4

FORWARD 1

Area of special triangles • Grade 6 • Geometry • 6.G.A.1

A right isosceles triangle has legs that are 6 cm long.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q1/5

2383

QUESTION ID

S71-F1-Q1

TYPE

Short answer

CORRECT ANSWER

18 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q1/5

True or false?

A triangle with a base of 4 in. and a height of $3\frac{1}{2}$ in. has an area of 7 square inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q2/5

2384

QUESTION ID

S71-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q2/5

A square with sides of 8 m is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 64 square meters **B.** 8 square meters
C. 32 square meters **D.** 16 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q3/5

2385

QUESTION ID

S71-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 32 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q3/5

An isosceles triangle has a base of 10 ft and a height of 12 ft.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q4/5

2386

QUESTION ID

S71-F1-Q4

TYPE

Short answer

CORRECT ANSWER

60 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q4/5

True or false?

A right triangle with legs of 5 units and 9 units has an area of 45 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q5/5

2387

QUESTION ID

S71-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 71 • FORWARD 1 • Q5/5

SET 71 • 5.G.B.4

FORWARD 2

Decide whether conditions give one triangle, more than one, or none • Grade 7
• Geometry • 7.G.A.2

How many differently shaped triangles can be drawn with sides of 5 cm, 5 cm, and 8 cm?

- A. Exactly two
- B. More than one
- C. Exactly one
- D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q1/5

2389

QUESTION ID

S71-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Exactly one

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q1/5

True or false?

Triangles of many different sizes can be drawn with angles of 60° , 60° , and 60° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q2/5

2390

QUESTION ID

S71-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q2/5

Can a triangle have sides of 3 m, 4 m, and 8 m? Explain.

Answer: _____

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q3/5

2391

QUESTION ID

S71-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE $3 + 4 = 7$ is less than 8, so the two shorter sides cannot meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q3/5

True or false?

A triangle can have angles of 90° , 60° , and 40° .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q4/5

2392

QUESTION ID

S71-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q4/5

Can a right triangle also be isosceles? If it can, give its angle measures.

Answer: _____

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q5/5

2393

QUESTION ID

S71-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Yes; 90° , 45° , 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 71 • FORWARD 2 • Q5/5